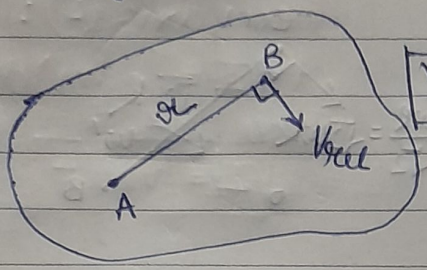
Rotational Motion

Rigid Body : Body whose shape, size and geometry remains unchanged.

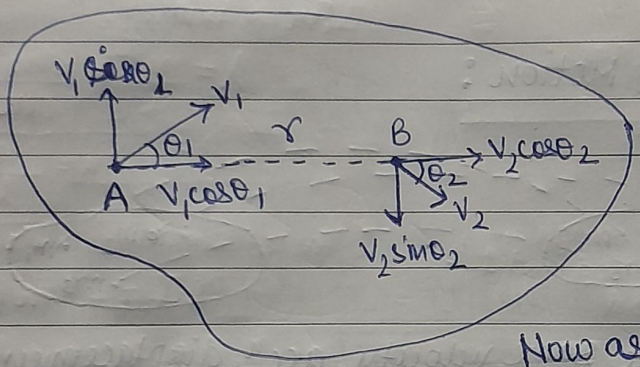
★ Separation between any pair of points on a rigid body remains unchanged.



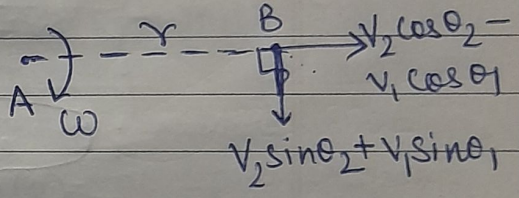
r is fix

$$\boxed{V_{app} = V_{sep} = 0}$$

★ In a rigid body, a point should be at rest wrt other or it should perform circular motion about other.



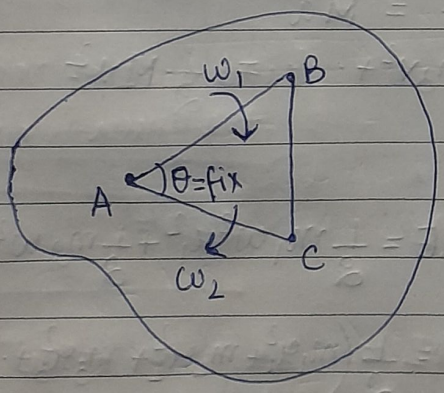
Now V_{BA} :-



Now as $r \to$ fix $\Rightarrow V_2\cos\theta_2 - V_1\cos\theta_1 = 0$

$$\boxed{V_2\cos\theta_2 = V_1\cos\theta_1}$$

$$\boxed{\omega = \frac{(V_{BA})_\perp}{r} = \frac{V_2\sin\theta_2 + V_1\sin\theta_1}{r}}$$

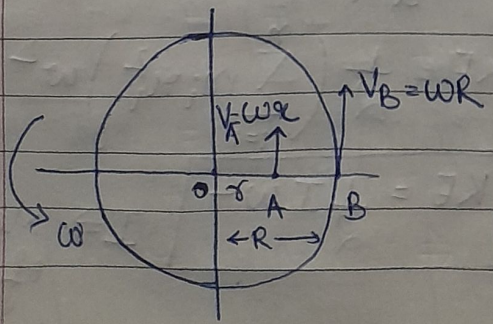


$\frac{d\theta}{dt} = \omega_2 - \omega_1$ As θ is fix $\Rightarrow \frac{d\theta}{dt} = 0$

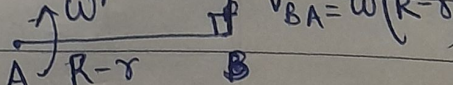
$\therefore \boxed{\omega_2 = \omega_1}$

★ Every particle will be rotating with same ω wrt other in a rigid body.

Ex



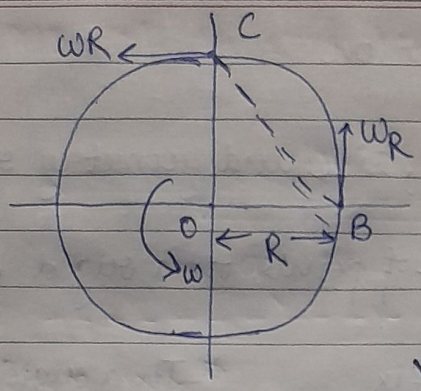
V_{BA} :



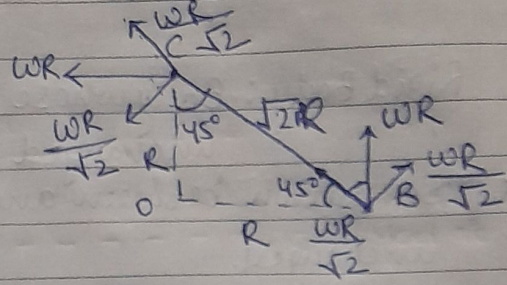
$$\omega' = \frac{V_{BA}}{R-r} = \frac{\omega(R-r)}{R-r} = \omega$$

$$\therefore \boxed{\omega' = \omega}$$

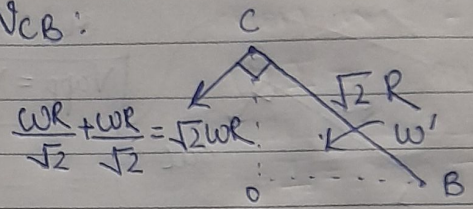
Q



Find V_{CB} and ω_{CB}



V_{CB}:

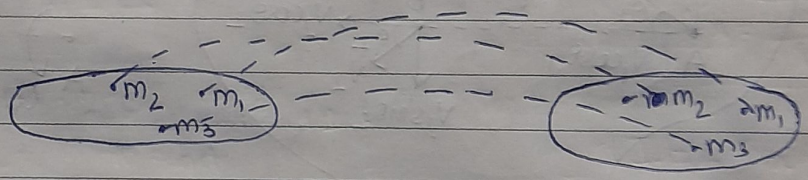


$\therefore V_{CB} = \sqrt{2}\omega R$

$\omega' = \frac{\sqrt{2}\omega R}{\sqrt{2}R}$

$\therefore \boxed{\omega' = \omega}$

Pure Translational Motion:



All particles will have same velocity, acc^n, displacement and distance travelled.

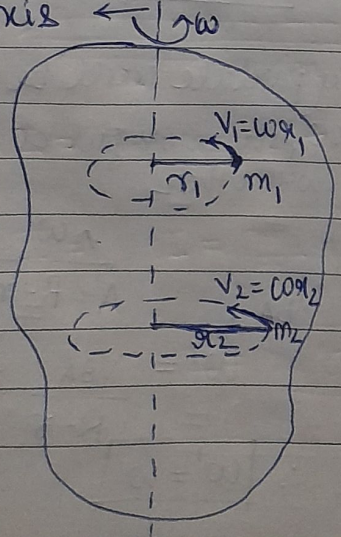
$$\vec{P} = m_1\vec{v} + m_2\vec{v} + m_3\vec{v} + \cdots = M\vec{v}$$

$$\vec{F}_{net} = m_1\vec{a} + m_2\vec{a} + m_3\vec{a} \cdots = M\vec{a}$$

$$KE = \frac{1}{2}m_1v^2 + \frac{1}{2}m_2v^2 + \frac{1}{2}m_3v^2 + \cdots = \frac{1}{2}Mv^2$$

Pure Rotational Motion:

All points on axis remain at rest.



$$KE = \frac{1}{2}m_1\omega^2r_1^2 + \frac{1}{2}m_2\omega^2r_2^2 + \frac{1}{2}m_3\omega^2r_3^2 \cdots$$

$$KE = \frac{1}{2}\left(m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + \cdots\right)\omega^2$$

$$KE = \frac{1}{2}\left(\sum_{i=1}^{n} m_ir_i^2\right)\omega^2$$

$$\boxed{KE = \frac{1}{2}I\omega^2}$$

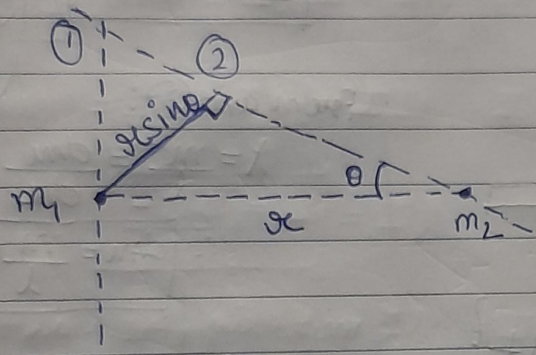
$I \Rightarrow$ Moment of Inertia

Moment of Inertia:

$$I = \sum_{i=1}^{n} m_i r_i^2 \qquad I = \int dm\, r^2$$

→ Unit = Kg-m²
→ Dimension = $[ML^2T^0]$
→ Scalar
→ Either zero or positive
→ Depends on mass and how the mass is distributed wrt axis.
→ Depends on mass, shape, size, geometry and density.
→ Depends on axis.

Q



Find I from the given axes.

① From axis ① ⇒ $I = \sum m_i r_i^2$

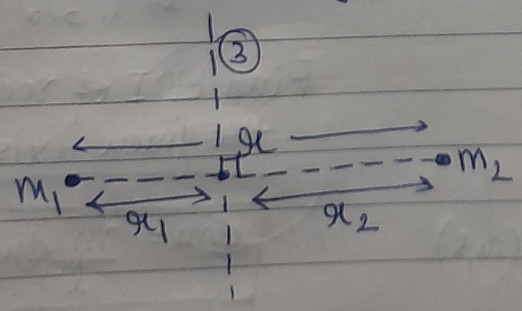
$I = m_1(0) + m_2(r)^2$

$\boxed{I = m_2 r^2}$

② From axis ② ⇒ $I = m_1(r\sin\theta)^2 + m_2(0)$

$\boxed{I = m_1 r^2 \sin^2\theta}$

③ Axis ③ is passing through COM and ⊥ to line joining the masses.



$r_1 = \frac{m_2 r}{m_1+m_2}$, $r_2 = \frac{m_1 r}{m_1+m_2}$

$I = m_1 r_1^2 + m_2 r_2^2$

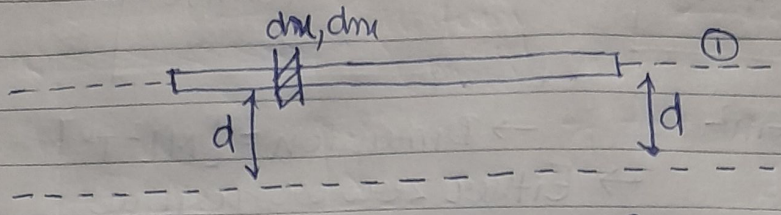
$I = m_1\left(\frac{m_2 r}{m_1+m_2}\right)^2 + m_2\left(\frac{m_1 r}{m_1+m_2}\right)^2$

$I = m_1\left(\frac{m_2^2 r^2}{(m_1+m_2)^2}\right) + m_2\left(\frac{m_1^2 r^2}{(m_1+m_2)^2}\right)$

$I = \frac{r^2}{(m_1+m_2)^2}\left(m_1 m_2 (m_2 + m_1)\right)$

$\boxed{I = \frac{m_1 m_2 r^2}{m_1+m_2}}$ $\boxed{I = \mu r^2}$ (μ = reduced mass $= \frac{m_1 m_2}{m_1+m_2}$)

① MOI of Uniform Rod (m, l):



From axis ①:

$$\boxed{I = 0}$$

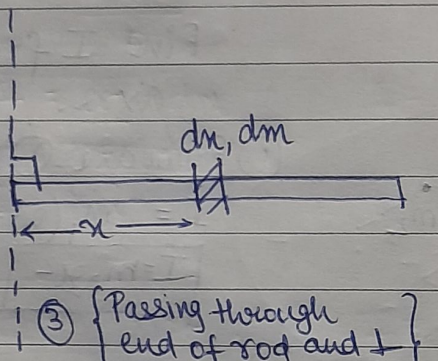
as rod lies on axis.

From axis ②:

$$\int dI = \int (dm)d^2$$

$$\boxed{I = md^2}$$

(Even if mass is non-uniformly distributed as d remains same)



From axis ③:

$$\lambda = \frac{m}{l} = \frac{dm}{dx}$$

$$\therefore \boxed{dm = \frac{m}{l}dx}$$

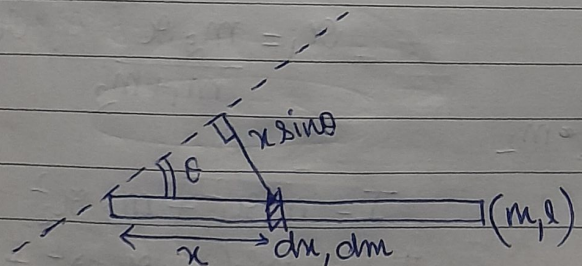
$$dI = dm(x)^2$$

$$\int dI = \frac{m}{l}\int_0^l x^2 dx$$

$$I = \frac{m}{l}\left(\frac{x^3}{3}\right)_0^l = \frac{ml^2}{3}$$

$$\star \boxed{I = \frac{ml^2}{3}}$$

Q



Find MOI of rod from the given axis.

$$\boxed{dm = \frac{m}{l}dx}$$

$$dI = dm(x\sin\theta)^2 \Rightarrow dI = \frac{m}{l}\sin^2\theta\, x^2 dx$$

$$I = \int dI = \frac{m}{l}\sin^2\theta \int_0^l x^2 dx = \frac{m}{l}\sin^2\theta \left(\frac{x^3}{3}\right)_0^l$$

$$\boxed{I = \frac{ml^2\sin^2\theta}{3}}$$

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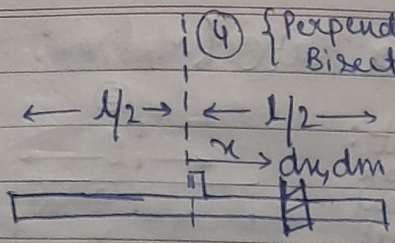


From axis ④:

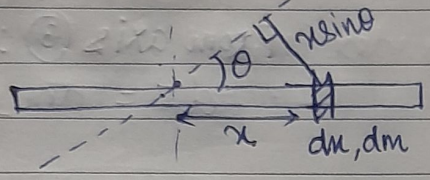
$dI = dm(x)^2$

$dI = \frac{m}{l} x^2 dx$

$I = \int dI = \frac{m}{l} \int_{-l/2}^{l/2} x^2 dx = \frac{m}{l}\left(\frac{x^3}{3}\right)$

$dm = \frac{m}{l} dx$

$\star \boxed{I = \frac{ml^2}{12}}$

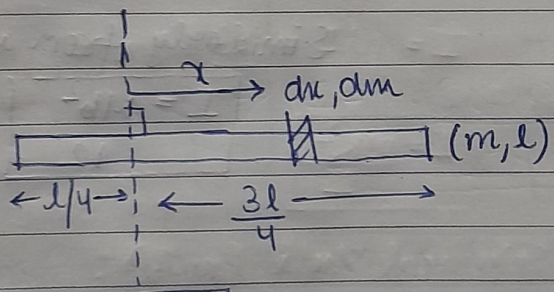


$dI = \frac{m}{l}(x\sin\theta)^2 dx$

$I = \int dI = \frac{m\sin^2\theta}{l} \int_{-l/2}^{l/2} x^2 dx$

$\boxed{I = \frac{ml^2\sin^2\theta}{12}}$

Q



Find MOI of rod from the given axis.

$dI = (dm)x^2$

$dI = \frac{m}{l} x^2 dx$

$dm = \frac{m}{l} dx$

$I = \int dI = \frac{m}{l} \int_{-l/4}^{3l/4} x^2 dx = \frac{m}{l}\left(\frac{x^3}{3}\right)_{-l/4}^{3l/4}$

$I = \frac{m}{3l}\left(\frac{27l^3}{64} + \frac{l^3}{64}\right) = \frac{m}{3l} \times \frac{28}{64} l^3$

$\boxed{I = \frac{7ml^2}{48}}$

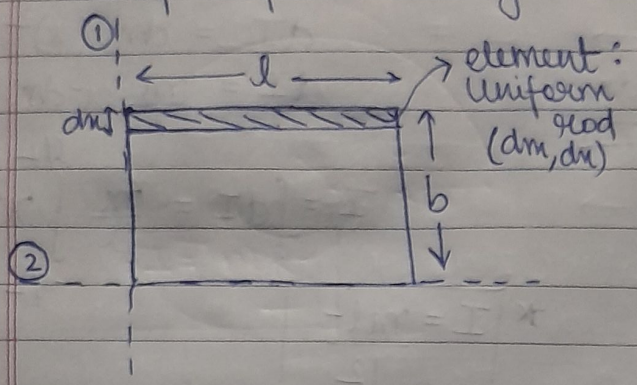
Alternate: Consider the axis as ends of two rods of length $l/4$ & $3l/4$.

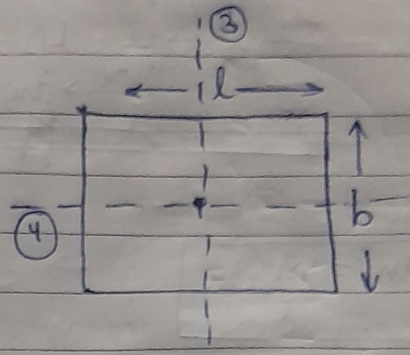
$I = \frac{\left(\frac{3m}{4}\right)\left(\frac{3l}{4}\right)^2}{3} + \frac{\left(\frac{m}{4}\right)\left(\frac{l}{4}\right)^2}{3} = \frac{\frac{3m}{4} \times \frac{9l^2}{16} + \frac{m}{4} \times \frac{l^2}{16}}{3} = \frac{28ml^2}{3 \times 64}$

$\boxed{I = \frac{7ml^2}{48}}$

② MOI of Uniform Rectangular Plate (M, l, b):

$\sigma = \frac{M}{lb}$





From axis ①:

$$dI = \frac{dm\,l^2}{3}$$

$$I = \int dI = \frac{l^2}{3}\int dm$$

$$\therefore \boxed{I = \frac{Ml^2}{3}}$$

From axis ③:

$$dI = dm\,\frac{l^2}{12}$$

$$I = \int dI = \frac{l^2}{12} = \int dm$$

$$\boxed{I = \frac{Ml^2}{12}}$$

Similarly from axis ②:

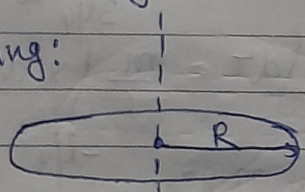
$$\boxed{I = \frac{Mb^2}{3}}$$

Similarly from axis ④:

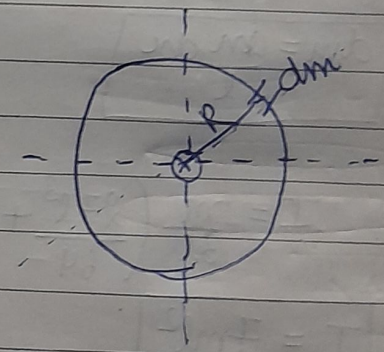
$$\boxed{I = \frac{Mb^2}{12}}$$

③ MOI of Ring (m, R):

(I) Full Ring:



{Passing through centre of Ring and ⊥ to plane of Ring} ①



From axis ①: $dI = dm\,R^2$

$$I = \int dI = R^2 \int dm$$

$$\boxed{I = mR^2}$$

↳ Even if mass is non-uniformly distributed.

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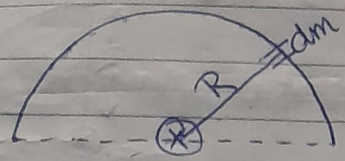
(II) Semi-Circular Ring (m, R):

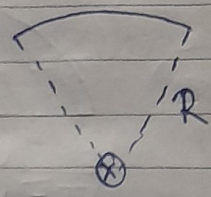


$dI = dm R^2$

$I = \int dI = R^2 \int dm$

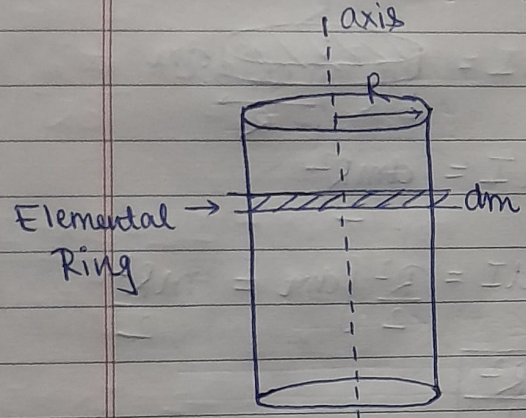
$\boxed{I = mR^2}$

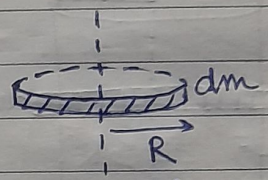
(III) Circular arc (m, R):



$\boxed{I = mR^2}$

(4) Moment of Inertia of Hollow Cylinder (m, R):





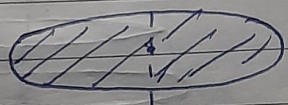
$dI = dm R^2$

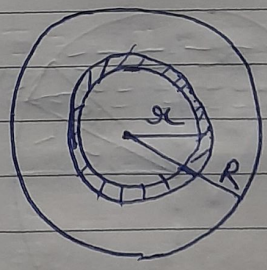
$I = \int dI = R^2 \int dm$

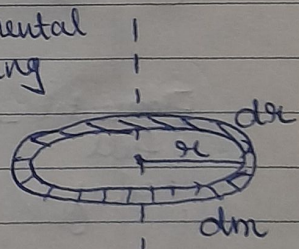
$\boxed{I = mR^2}$ {Even if mass is non-uniformly distributed}

(5) MOI of Uniform Disc (m, R):

(I)







{Passing through centre of disc & ⊥ to plane of disc} axis

$dm = \sigma dA$

$dm = \frac{m}{\pi R^2}(2\pi x\, dx)$

$\boxed{dm = \frac{2mx\,dx}{R^2}}$

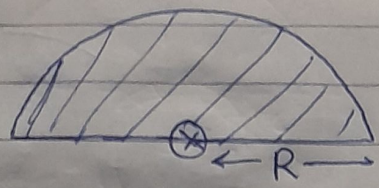
$dI = dm\, x^2$

$I = \int dI = \int \frac{2mx\,dx}{R^2} x^2$

$I = \frac{2m}{R^2}\int_0^R x^3 dx = \frac{2m}{4R^2}\left(x^4\right)_0^R$

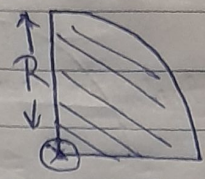
$\boxed{I = \frac{mR^2}{2}}$

(II) Semicircular disc (m, R):



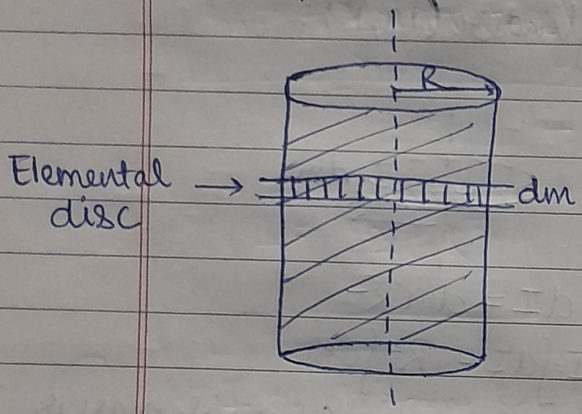
$$I = \frac{mR^2}{2}$$

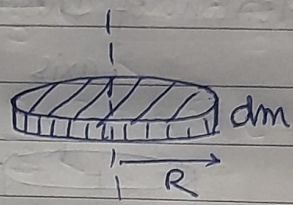
(III) Quater Ring (m, R)



$$I = \frac{mR^2}{2}$$

(6) MOI of Uniform Solid Cylinder (m, R):



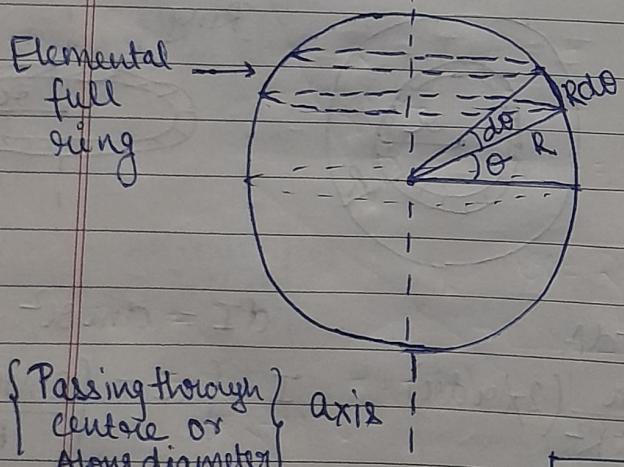


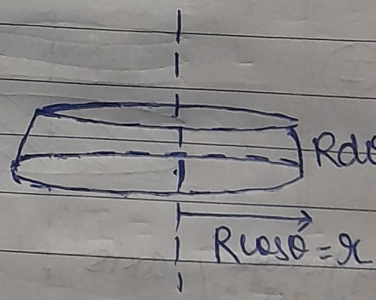
$$dI = \frac{dmR^2}{2}$$

$$I = \int dI = \frac{R^2}{2}\int dm = \frac{mR^2}{2}$$

$$\boxed{I = \frac{mR^2}{2}}$$

(7) MOI of Hollow sphere | Spherical shell (m, R):





$$dI = dm(r^2)$$

$$dI = \frac{m}{2}\cos\theta d\theta (R^2\cos^2\theta)$$

$$\boxed{dm = \frac{m}{2}\cos\theta d\theta}$$

$$I = \int dI = \frac{mR^2}{2}\int_{-\pi/2}^{\pi/2} \cos^2\theta \cos\theta d\theta$$

$dm = \sigma dA$

$$dm = \frac{m}{2\cancel{4}\cancel{\pi}R^2}(2\cancel{\pi}R\cos\theta \times R d\theta) = \frac{m}{2}\cos\theta d\theta$$

$$I = \frac{mR^2}{2}\int_{-\pi/2}^{\pi/2} \cos^2\theta \cos\theta \, d\theta$$

$$= \frac{mR^2}{2}\left(\sin\theta - \frac{\sin^3\theta}{3}\right)_{-\pi/2}^{\pi/2}$$

$$= \frac{mR^2}{2}\left\{\left(\frac{2}{3}\right) - \left(\frac{-2}{3}\right)\right\} = \frac{mR^2}{2}\left(\frac{4}{3}\right)$$

$$\boxed{I = \frac{2}{3}mR^2}$$

$$\int \cos^2\theta \cos\theta \, d\theta$$

$$= \int (1 - \sin^2\theta)\cos\theta \, d\theta$$

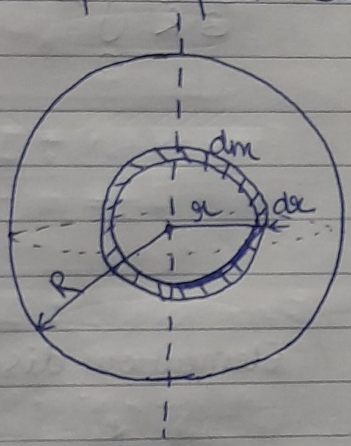
Let $\sin\theta = t$

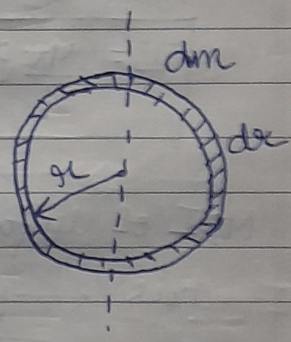
$\cos\theta \, d\theta = dt$

$$= \int (1 - t^2)\, dt = t - \frac{t^3}{3}$$

(8) MOI of Uniform Solid Sphere (m, R):

Elemental spherical shell





$dm = \rho dV$

$$dm = \frac{3m}{4\pi R^3} \times 4\pi x^2 dx$$

$$\boxed{dm = \frac{3m}{R^3} x^2 dx}$$

$$dI = \frac{2}{3} dm \, x^2$$

$$dI = \frac{2}{3}\left(\frac{3m}{R^3} x^2 dx\right) x^2$$

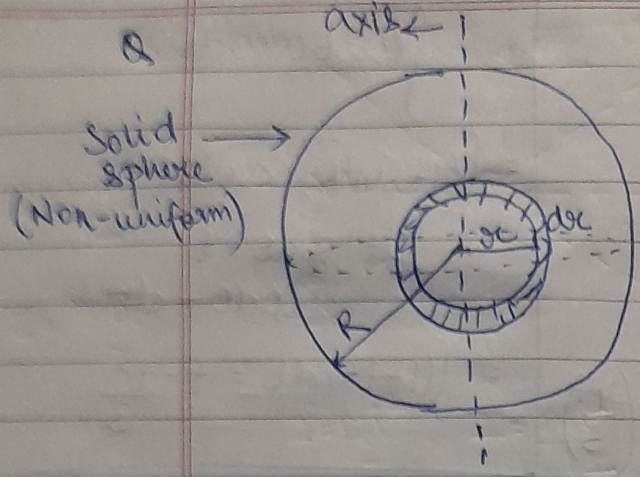
$$I = \int dI = \int \frac{2m}{R^3} x^4 dx = \frac{2m}{R^3}\int_0^R x^4 dx$$

$$I = \frac{2m}{R^3}\left(\frac{x^5}{5}\right)_0^R$$

$$\boxed{I = \frac{2}{5}mR^2}$$

★ Note: $\boxed{I_{\text{solid sphere}} < I_{\text{hollow sphere}}}$ provided m, R are same

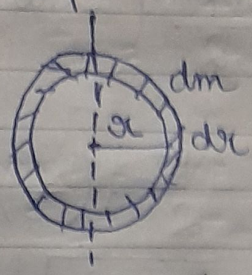
Q



Given ρ varies as $\rho = \rho_0 \frac{r}{R}$

Find MOI about given axis

Elemental Spherical shell



$dm = \rho dV$

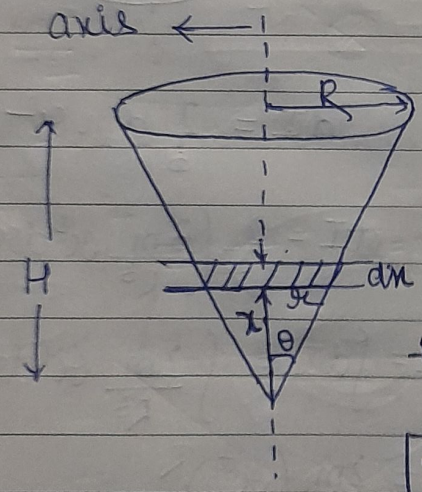
$$\boxed{dm = \rho_0 \frac{r}{R}\left(4\pi x^2 dx\right)}$$

$$dI = \frac{2}{3}(dm)x^2 = \frac{2}{3}\left(\rho_0 \frac{r}{R}\left(4\pi x^2 dx\right)\right)x^2$$

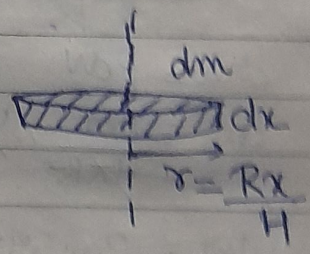
$$I = \int dI = \frac{8\pi\rho_0}{3R}\int_0^R x^5 dx = \frac{8\pi\rho_0}{3R}\left(\frac{R^6}{6}\right)$$

$$\boxed{I = \frac{4\pi}{9}\rho_0 R^5}$$

(9) MOI of Uniform Solid Cone:



Elementary disc



$$\tan\theta = \frac{x}{x} = \frac{R}{H}$$

$$\boxed{x = \frac{Rx}{H}}$$

$dm = \rho dV$

$$= \frac{3m}{\pi R^2 H}\left(\pi x^2 dx\right)$$

$$= \frac{3m}{R^2 H}\left(\frac{R^2 x^2}{H^2} dx\right)$$

$$\boxed{dm = \frac{3m}{H^3} x^2 dx}$$

$$dI = \frac{dm\, x^2}{2}$$

$$dI = \frac{1}{2}\left(\frac{3m}{H^3} x^2 dx\right)\left(\frac{R^2 x^2}{H^2}\right)$$

$$I = \int dI = \frac{3R^2 m}{2H^5}\int_0^H x^4 dx$$

$$I = \frac{3R^2 m}{2H^5}\left(\frac{H^5}{5}\right) = \frac{3}{10} mR^2$$

$$\boxed{I = \frac{3}{10} mR^2}$$

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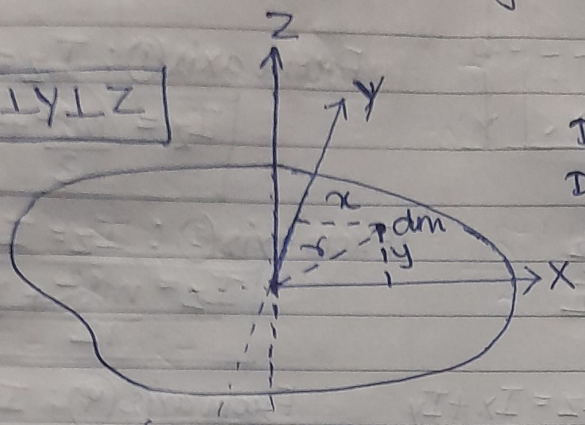
Date:

Perpendicular Axis Theorem:

→ Applicable only for laminar objects (planar, 1-D, 2-D).



$X \perp Y \perp Z$

$r = \sqrt{x^2+y^2}$

Distance of dm from X axis = y
Distance of dm from Y axis = x
Distance of dm from Z axis = r

$I_x = \int dm \cdot y^2$

$I_y = \int dm\, x^2$

$I_z = \int dm(\sqrt{x^2+y^2})^2$

$= \int dm(x^2+y^2)$

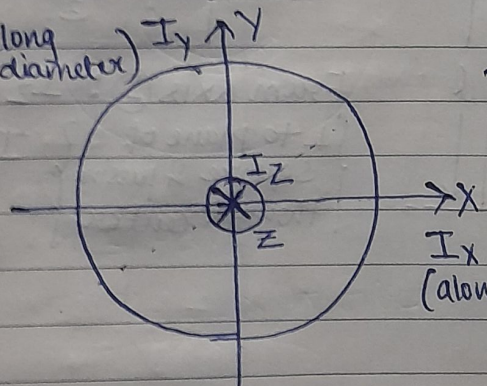
$= \int dm\, x^2 + \int dm\, y^2$

$\therefore \boxed{I_z = I_y + I_x}$

→ X & Y are in same plane as object.

→ Z is ⊥ to the plane of object and passing through intersection of X & Y.

Ex



$m, R \Rightarrow$ Uniform Ring.

We know, MOI about Z ⇒ $I_z = mR^2$

By ⊥ axis theorem:

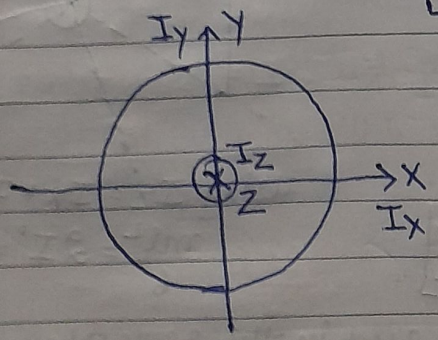
$I_z = I_x + I_y$ — ①

By symmetry:

$I_x = I_y$ — ②

From ① and ② ⇒ $\boxed{I_x = I_y = \frac{mR^2}{2}}$

Q



$m, R \Rightarrow$ Uniform Disc

$I_z = \frac{mR^2}{2}$

By ⊥ axis Theorem:

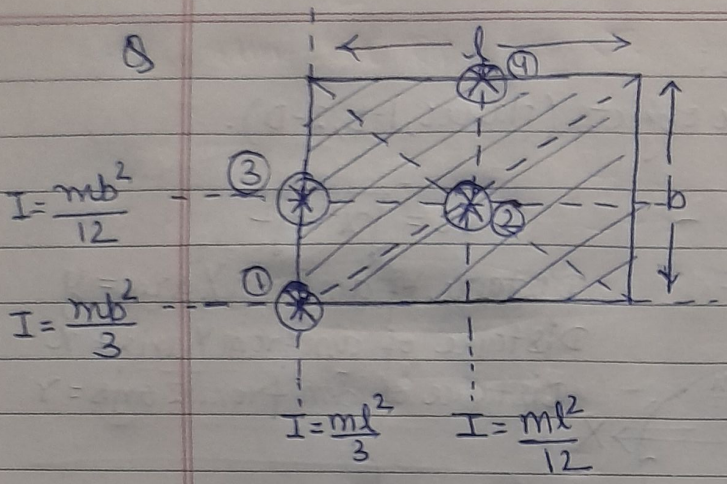
$I_z = I_x + I_y$ — ①

By symmetry:

$I_x = I_y$ — ②

From ① and ② ⇒ $\boxed{I_x = I_y = \frac{mR^2}{4}}$

Q



Find MOI of the Rectangular Plate by the given axes.

From axis ①: $I_Z = I_X + I_Y$

$$\boxed{I_Z = \frac{ml^2}{3} + \frac{mb^2}{3}}$$

From axis ②: $I_Z = I_X + I_Y$

$$\boxed{I_Z = \frac{ml^2}{12} + \frac{mb^2}{12}}$$

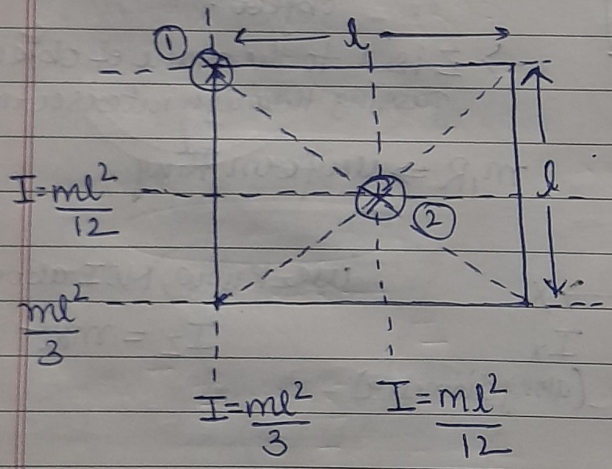
From axis ③: $I_Z = I_X + I_Y$

$$\boxed{I_Z = \frac{ml^2}{3} + \frac{mb^2}{12}}$$

From axis ④: $I_Z = I_X + I_Y$

$$\boxed{I_Z = \frac{ml^2}{12} + \frac{mb^2}{3}}$$

Ex Square Plate (m, l):



From axis ①: $I_Z = I_X + I_Y$

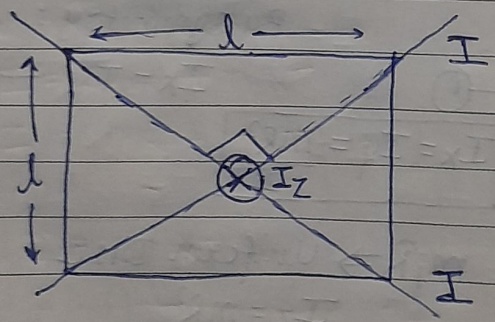
$$\boxed{I_Z = \frac{ml^2}{3} + \frac{ml^2}{3} = \frac{2ml^2}{3}}$$

From axis ②: $I_Z = I_X + I_Y$

{⊥ to plane of square & passing through center}

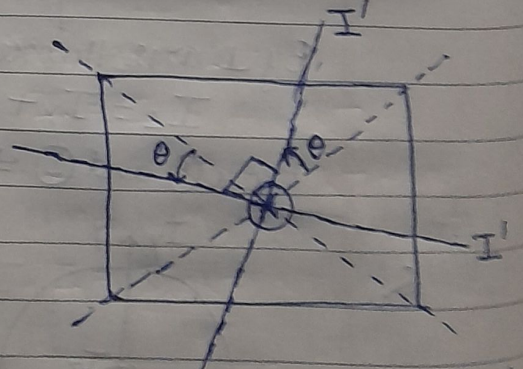
$$I_Z = \frac{ml^2}{12} + \frac{ml^2}{12}$$

$$\boxed{I_Z = \frac{ml^2}{6}}$$



Using $I_Z = I_X + I_Y \Rightarrow I_Z = 2I$

$$\frac{ml^2}{6} = 2I \Rightarrow \boxed{I = \frac{ml^2}{12}}$$

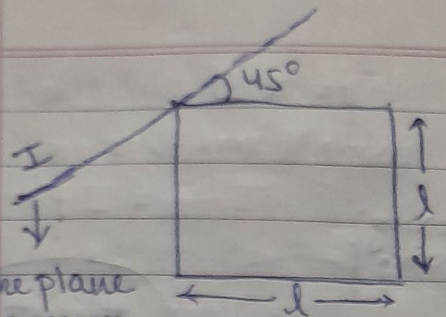


Using $I_Z = I_X + I_Y \Rightarrow I_Z = 2I'$

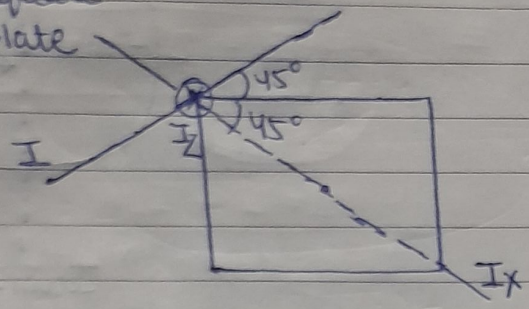
$$\frac{ml^2}{6} = 2I' \Rightarrow \boxed{I' = \frac{ml^2}{12}}$$

★ Any axis passing through centre and in the same plane as square plate, MOI will be $ml^2/12$.

Q



Find MOI about the given axis.



Using ⊥ axis theorem:

$$I_z = I_x + I_y$$

$$= \frac{2ml^2}{3} = \frac{ml^2}{12} + I$$

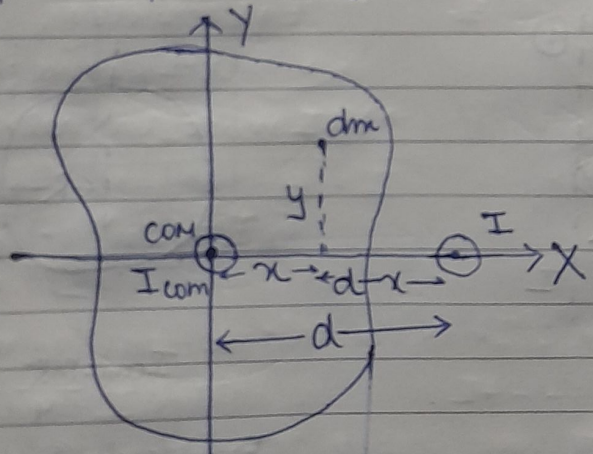
$$\Rightarrow I = \frac{2ml^2}{3} - \frac{ml^2}{12}$$

$$I = ml^2\left(\frac{2}{3} - \frac{1}{12}\right)$$

$$\boxed{I = \frac{7ml^2}{12}}$$

Parallel Axis Theorem:

→ Applicable for all objects (3-D, 2-D and 1-D).



$$I_{com} = \int dm\,(x^2+y^2)$$

$$I = \int dm\,((d-x)^2+y^2)$$

$$= \int dm\,((d^2+x^2-2xd)+y^2)$$

$$= \int dm\,(x^2+y^2) + \int dm\,d^2 - 2d\int dm\cdot x$$

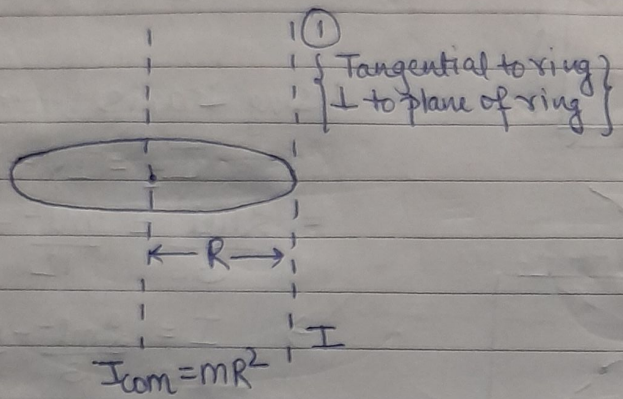
$$I = \int dm(x^2+y^2) + d^2\int dm - 2d\int dm\,x$$

(Sum of mass moments from COM is zero (Refer COM))

$$\therefore \boxed{I = I_{com} + md^2}$$

Note: One axis must pass through COM and the other axis about which MOI needs to be found must be parallel to this axis.

Uniform Ring (m, R):

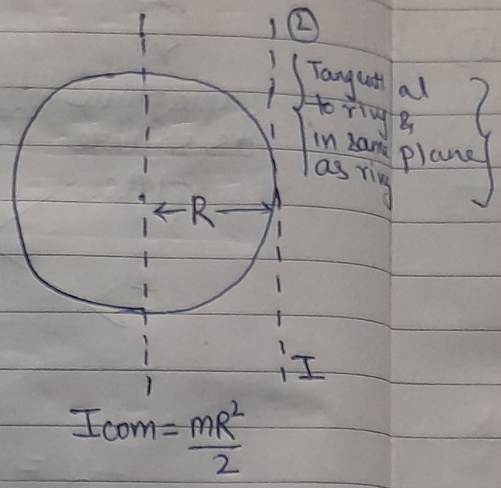


From axis ①:

$I = I_{com} + mR^2$

$I = mR^2 + mR^2$

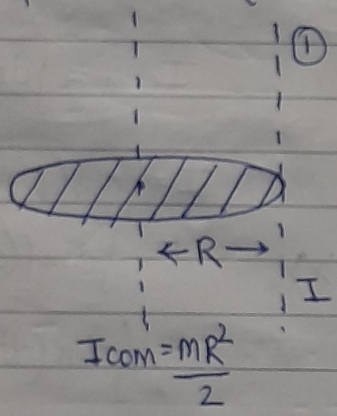
$\boxed{I = 2mR^2}$



From axis ②: $I = I_{com} + mR^2$

$\boxed{I = \frac{3}{2}mR^2}$

Uniform Disc (m, R):

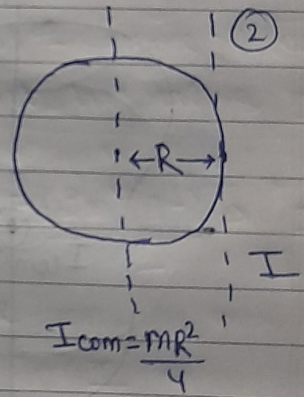


From axis ①:

$I = I_{com} + mR^2$

$I = \frac{mR^2}{2} + mR^2$

$\boxed{I = \frac{3mR^2}{2}}$



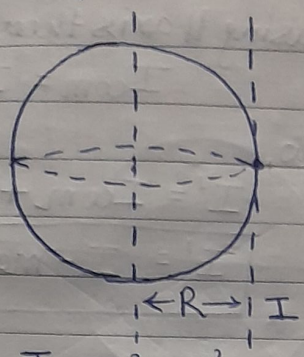
From axis ②:

$I = I_{com} + mR^2$

$I = \frac{mR^2}{4} + mR^2$

$\boxed{I = \frac{5mR^2}{4}}$

Hollow Sphere (m, R):

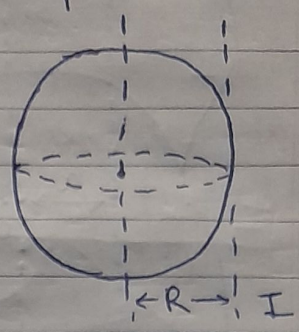


$$I_{com} = \frac{2}{3}mR^2$$

$$I = \frac{2}{3}mR^2 + mR^2$$

$$\boxed{I = \frac{5}{3}mR^2}$$

Solid sphere (m, R):

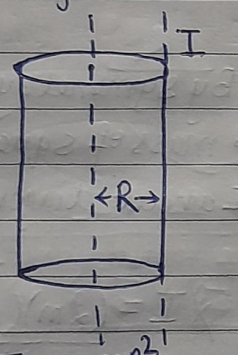


$$I_{com} = \frac{2}{5}mR^2$$

$$I = \frac{2}{5}mR^2 + mR^2$$

$$\boxed{I = \frac{7}{5}mR^2}$$

Hollow Cylinder (m, R):

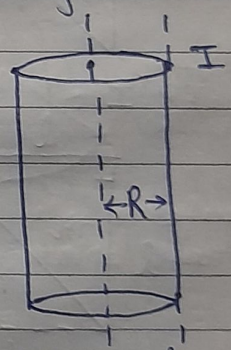


$$I_{com} = mR^2$$

$$I = mR^2 + mR^2$$

$$\boxed{I = 2mR^2}$$

Solid Cylinder (m, R):

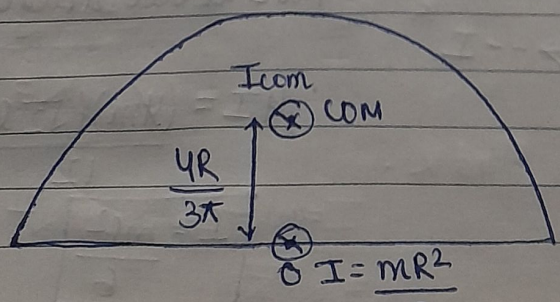


$$I_{com} = \frac{mR^2}{2}$$

$$I = \frac{mR^2}{2} + mR^2$$

$$\boxed{I = \frac{3mR^2}{2}}$$

Q Given Uniform Semi-circular disc (m, R). Find its MOI about an axis passing through its COM and ⊥ to the plane of disc.



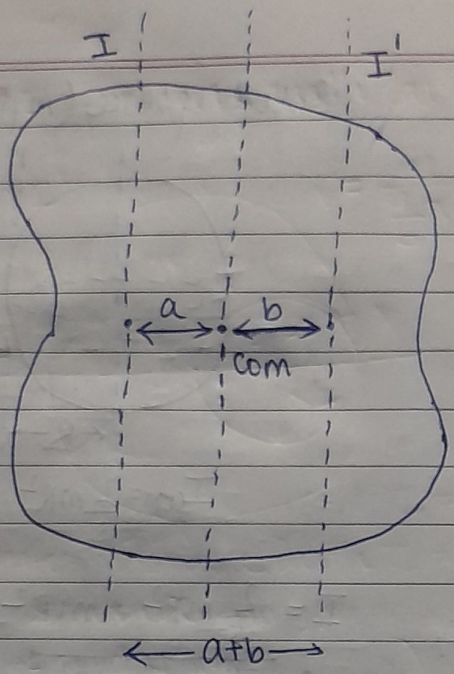
$$I = I_{com} + md^2$$

$$I_{com} = I - md^2$$

$$I_{com} = \frac{mR^2}{2} - m\frac{16R^2}{9\pi^2}$$

$$\boxed{I_{com} = mR^2\left(\frac{1}{2} - \frac{16}{9\pi^2}\right)}$$

Q



Find I'.

Using ∥ axis theorem:

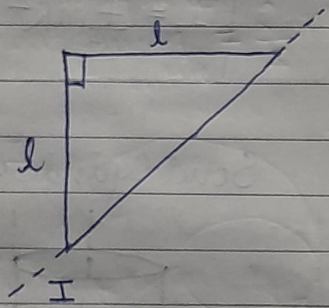
$$I_{com} = I - ma^2 \quad \text{—— (1)}$$

Now,

$$I' = I_{com} + mb^2$$

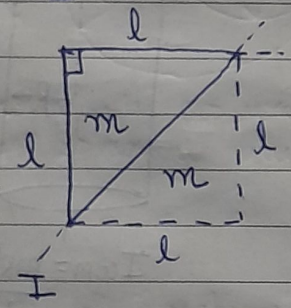
$$\boxed{I' = I - ma^2 + mb^2}$$

Q



Given isoceles uniform triangular plate of mass m

Find I.



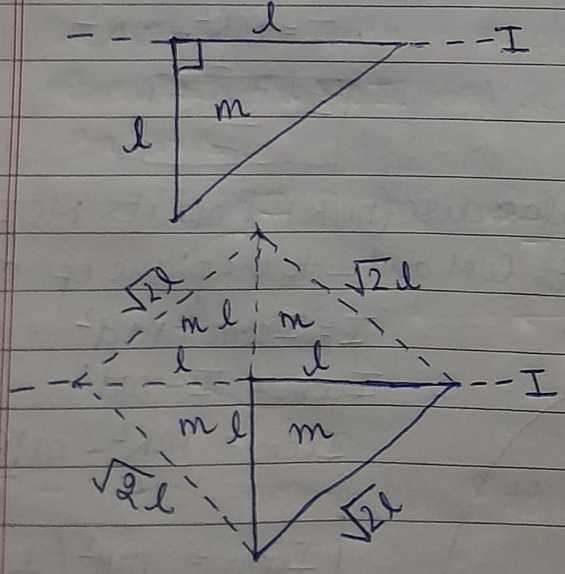
By symmetry,

→ mass of square plate = 2m

→ MOI about diagonal is $\frac{ml^2}{12}$

$$2I = \frac{(2m)(l^2)}{12} \Rightarrow \boxed{I = \frac{ml^2}{12}}$$

Q



Find I

By symmetry,

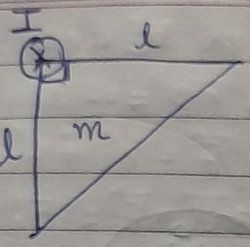
→ mass of square plate = 4m

→ Length of square plate = $\sqrt{2}l$

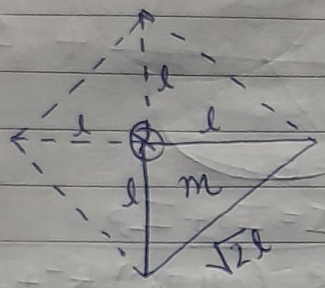
→ MOI about diagonal is $\frac{ml^2}{12}$

$$4I = \frac{(4m)(\sqrt{2}l)^2}{12}$$

$$\boxed{I = \frac{ml^2}{6}}$$

Q 

Find I.



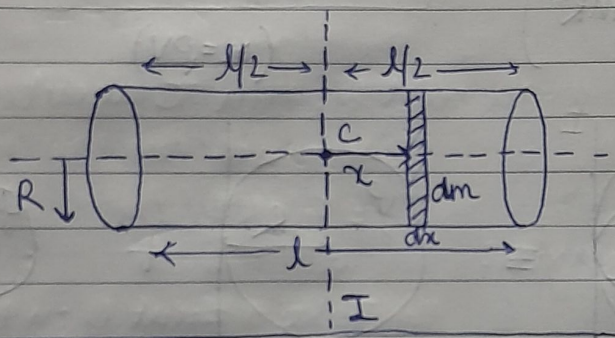
By symmetry,

→ mass of square plate = 4m

→ length of square plate = $\sqrt{2}l$

→ MOI about axis passing through center and ⊥ to plane = $\frac{ml^2}{6}$

$$\Rightarrow \frac{1}{4}I = \frac{(\frac{1}{4}m)(\sqrt{2}l)^2}{6} \Rightarrow \boxed{I = \frac{ml^2}{3}}$$

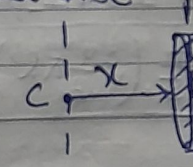
Q 

Given solid cylinder of mass m and R.

Find I

$$\boxed{dm = \frac{m}{l}dx}$$

Elemental Disc



$$dI_{com} = \frac{dm R^2}{4}$$

$$dI = dI_{com} + dm x^2$$

$$I = \int dI = \int dI_{com} + \int dm x^2$$

$$I = \int \frac{dm R^2}{4} + \int dm x^2$$

$$I = \int \frac{m\,dx R^2}{l 4} + \int \frac{m\,dx}{l} x^2 \qquad I = \int \frac{dm R^2}{4} + \int \frac{m}{l} dx\, x^2$$

$$I = \frac{R^2}{4}\int dm + \frac{m}{l}\int_{-l/2}^{l/2} x^2 dx$$

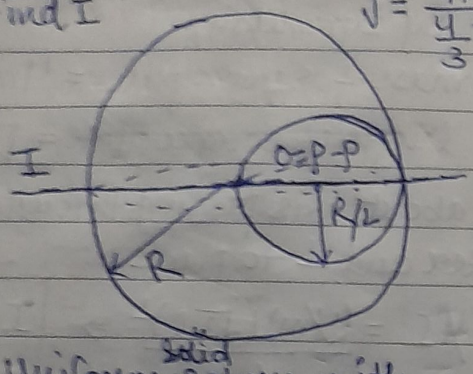
$$I = \frac{mR^2}{4} + \frac{m}{3l}\left(x^3\right)_{-l/2}^{l/2} = \frac{mR^2}{4} + \frac{m}{3l}\left(\frac{l^3}{4}\right)$$

$$\boxed{I = \frac{mR^2}{4} + \frac{ml^2}{12}}$$

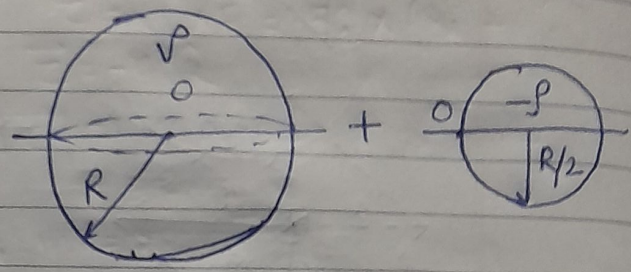
MOI of objects with cavity:

Ex Find I

$\rho = \frac{m}{\frac{4}{3}\pi\left(R^3 - \frac{R^3}{8}\right)}$



$\equiv$



Uniform solid sphere with spherical cavity.

(I of solid sphere without cavity about centre $= \frac{2}{5}mR^2$)

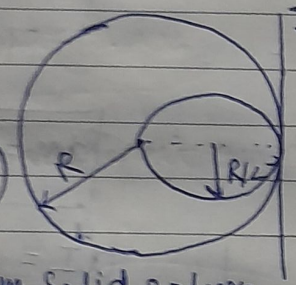
$I = I_\rho + I_{-\rho}$

$$I = \frac{2}{5}\left(\rho\frac{4}{3}\pi R^3\right)R^2 + \frac{2}{5}\left(-\rho\frac{4}{3}\pi\frac{R^3}{8}\right)\left(\frac{R}{2}\right)^2$$

$(m = \rho V)$

Ex Find I

$\rho = \frac{m}{\frac{4}{3}\pi\left(R^3 - \frac{R^3}{8}\right)}$



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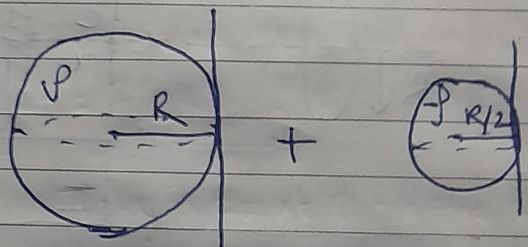
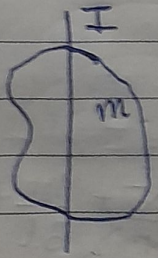


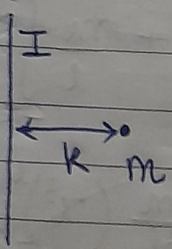
Uniform solid sphere with spherical cavity

$I = I_\rho + I_{-\rho}$

$$I = \frac{2}{5}\left(\rho\frac{4}{3}\pi R^3\right)R^2 + \frac{2}{5}\left(-\rho\frac{4}{3}\pi\frac{R^3}{8}\right)\left(\frac{R}{2}\right)^2$$

Radius of Gyration: It is the distance from an axis at which the mass of a body may be assumed to be concentrated and at which the MOI will be equal to MOI of the actual mass about the axis.



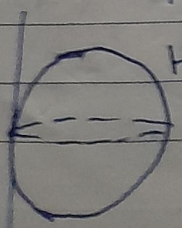


$K \rightarrow$ radius of gyration

$$I = mK^2$$

$$K = \sqrt{\frac{I}{m}}$$

Ex



Find K.

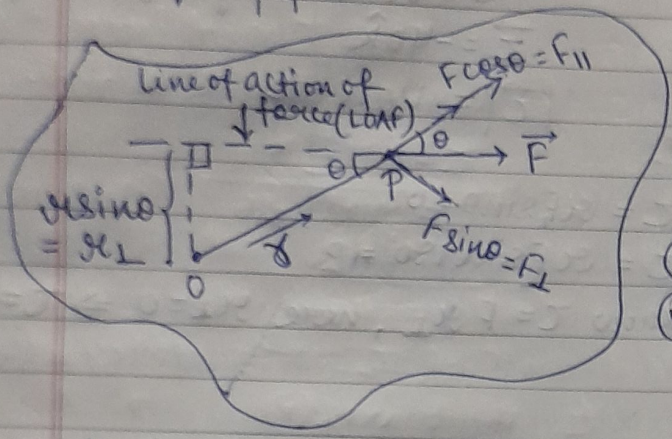
$I = mK^2 \Rightarrow \frac{5}{3}mR^2 = mK^2$

$$K = R\sqrt{\frac{5}{3}}$$

Torque: It is rotating effect of a force.

→ Depends on force as well as where the force is applied.

Torque of a point:



① O is the point about which torque is to be found.

② P is point of application of force

③ $\overrightarrow{OP} = \vec{r}$

④ θ is angle b/w $\vec{r}$ and $\vec{F}$

→ $r_\perp$ is also called lever arm.

$$\boxed{\vec{\tau} = \vec{r} \times \vec{F}}$$

If $\theta = 0$, $\tau = 0$

$$\boxed{\tau = rF\sin\theta}$$

$$\boxed{\tau = r(F\sin\theta) = rF_\perp}$$

$$\boxed{\tau = F(r\sin\theta) = Fr_\perp}$$

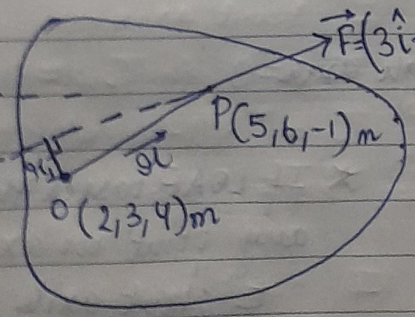
→ Direction can be determined using right hand rule.

⊗ → clockwise ⊙ → anticlockwise.

∴ Torque is vector quantity, an axial vector.

Unit = N-m Dimension = $[ML^2T^{-2}]$

Ex



① Find τ of the force about point O.

$\overrightarrow{OP} = \vec{r} = (5\hat{i}+6\hat{j}-\hat{k}) - (2\hat{i}+3\hat{j}+4\hat{k})$

$\vec{r} = 3\hat{i}+3\hat{j}-5\hat{k}$

$\vec{F} = 3\hat{i}+4\hat{j}-2\hat{k}$

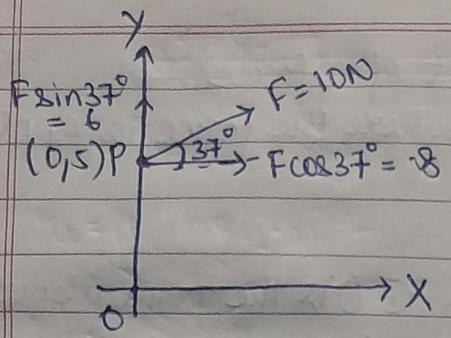
$\vec{\tau} = \vec{r} \times \vec{F}$

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 3 & -5 \\ 3 & 4 & -2 \end{vmatrix} = 14\hat{i} - 9\hat{j} + 3\hat{k} \text{ N-m}$$

$$|\vec{\tau}| = \sqrt{14^2+9^2+3^2} = \sqrt{196+81+9} = \sqrt{286} \text{ N-m}$$

② Find length of lever arm ⇒ $\tau = Fr_\perp \Rightarrow r_\perp = \frac{\tau}{F} = \frac{\sqrt{286}}{\sqrt{29}} = \sqrt{\frac{286}{29}}$ m

Q



Find $\vec{\tau}$ about point O.

$\vec{r} = \vec{OP} = 5\hat{j}$

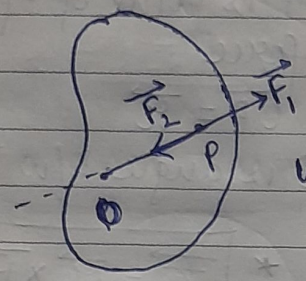
$\vec{F} = 8\hat{i} + 6\hat{j}$

$\vec{\tau} = \vec{r} \times \vec{F} = 5\hat{j} \times (8\hat{i} + 6\hat{j})$

$\vec{\tau} = -40\hat{k}$

$|\vec{\tau}| = 40\ N\text{-}m$

Note:



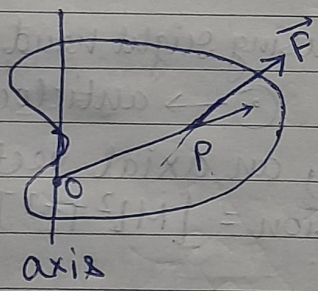
$\tau = r F_1 \sin 0 = 0$

$\tau = r F_2 \sin 180° = 0$

we know $\tau = F r_\perp$, here $r_\perp = 0 \Rightarrow \tau = 0$.

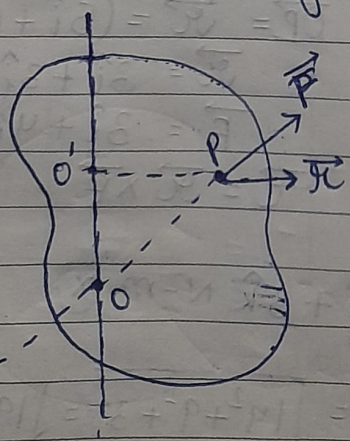
* If LOAF passes through point O, then τ about point O will be zero.

Torque about an axis:



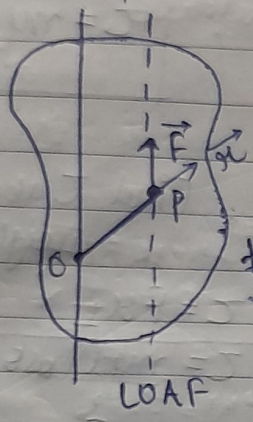
Take a point O on the axis and find torque about point O. The component of this torque along axis is defined as torque about axis.

① Case 1: LOAF is passing through axis.



* If LOAF passes through any point on axis then τ about axis will be zero.

② Case 2: LOAF parallel to axis



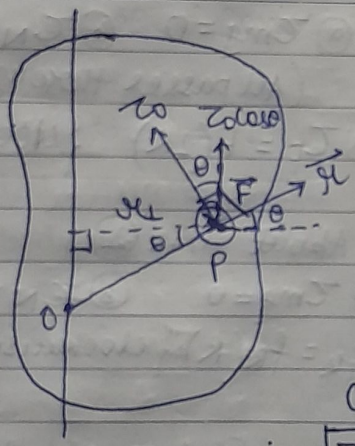
★ If a force parallel to axis is applied on object, then its torque about axis is zero.

Here, $\vec{\tau} = \vec{r} \times \vec{F} \Rightarrow \vec{\tau} = |\vec{\tau}|\hat{k}$

As $\vec{\tau}$ is outward there will be no component of $\vec{\tau}$ along the axis and hence τ about axis will be zero.

③ Case ③: Axis and LOAF are skew lines.

Skew lines → Neither intersecting nor parallel.



$\vec{F} = F(-\hat{k})$

$\tau_0 = rF\sin 90°$ (As r and F are in different planes)

$\boxed{\tau_0 = rF}$

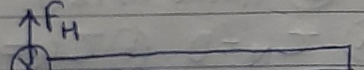
Also $\vec{\tau_0} = \vec{r} \times \vec{F}$

Use right hand rule for direction.

Component of τ_0 along axis is τ_{axis}

$\therefore \boxed{\tau_{axis} = \tau_0 \cos\theta = rF\cos\theta = F(r\cos\theta) = Fr_\perp}$

★ $\boxed{\tau_{axis} = Fr_\perp}$

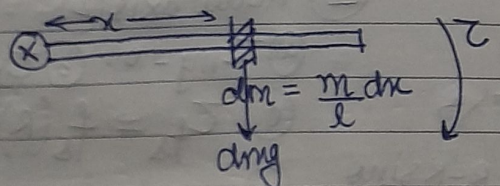
Q 

(m, l, uniform rod)

Smooth hinge

Find τ about hinge
① due to hinge
② due to weight.

① $\tau = 0$ as f_H passes through the axis, ie hinge.

② 

$d\tau = dmg\,x = \frac{mg}{l}x\,dx$

$\tau = \int d\tau = \frac{mg}{l}\int_0^l x\,dx = \frac{mg}{l}\left(\frac{l^2}{2}\right)$

$\boxed{\tau = \frac{mgl}{2}}$

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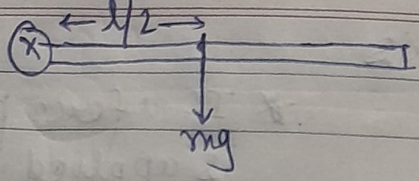
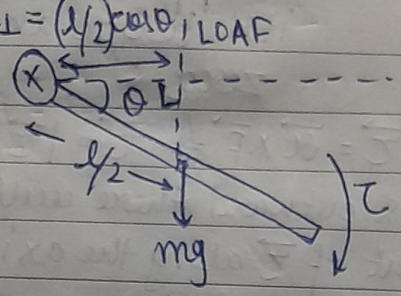
Alternate:



Assume weight at COM

$$\tau = mg\frac{l}{2}$$

Q

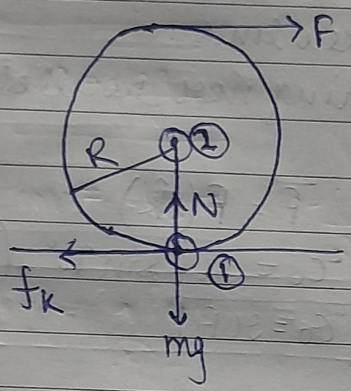


Find τ about hinge due to weight of rod.

$\tau = F r_\perp$

$$\tau = \frac{mgl}{2}\cos\theta \quad \text{(clockwise } @x\text{)}$$

Q



Find τ due to all forces.

(I) About axis (1)

(a) $\tau_{mg} = 0$ (b) $\tau_N = 0$ (c) $\tau_{fk} = 0$

(All passes through axis)

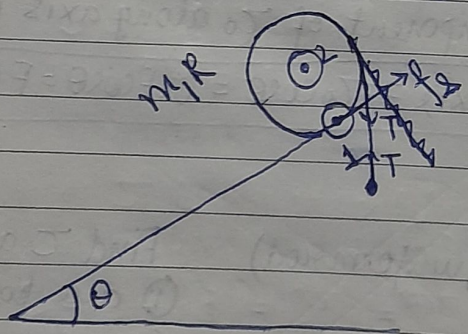
(d) $\tau_F = F(2R)$ {clockwise}

(II) About axis (2)

(a) $\tau_{mg} = 0$ (b) $\tau_N = 0$

(c) $\tau_{fk} = f_k(R)$ {clockwise} (d) $\tau_F = F(2R)$ {clockwise}

* Q



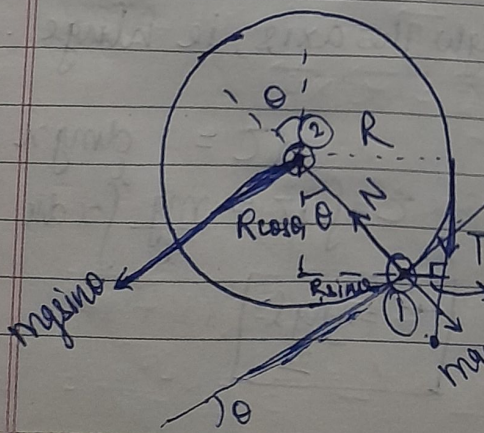
Find τ about axis (1) and (2) for all forces.

(I) About axis (1)

(a) $\tau_N = 0$ (b) $\tau_{f_s} = 0$

(c) $\tau_T = TR(1-\sin\theta)$ {clockwise}

(d) $\tau_{mg} = mg\sin\theta(R)$ {anticlockwise}



(II) About axis (2)

(a) $\tau_{mg} = 0$ (b) $\tau_N = 0$

(c) $\tau_{f_s} = f_s(R)$ {anticlockwise}

(d) $\tau_T = TR$ {clockwise}

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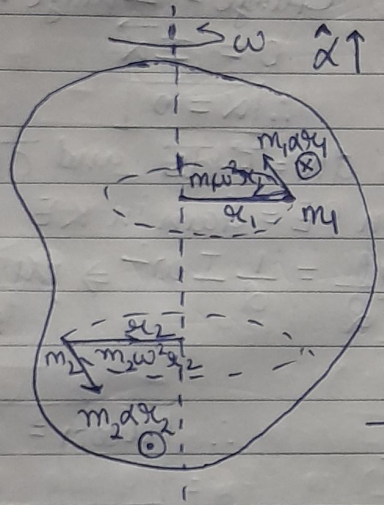
Newtons 2nd Law in Rotation :

$$\vec{\tau} = I\vec{\alpha}$$

$$\vec{\tau}_{Hinge} = I_{Hinge}\vec{\alpha} \qquad \vec{\tau}_{COM} = I_{com}\vec{\alpha}$$



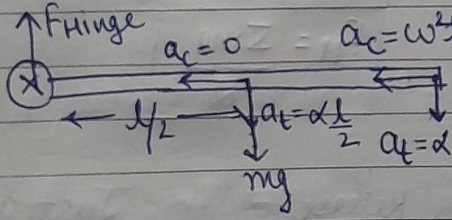
$$\vec{\tau} = (m_1\alpha R_1^2 + m_2\alpha R_2^2)\hat{\alpha}$$
$$\vec{\tau} = \{(m_1R_1^2 + m_2R_2^2)\alpha\}\hat{\alpha}$$
$$\vec{\tau} = I_{axis}\,\alpha\hat{\alpha}$$

$$\star\ \boxed{\vec{\tau} = I_{axis}\vec{\alpha}}$$

$\rightarrow \vec{\tau}$ is in direⁿ of $\vec{\alpha}$

Note: ① If body is hinged → write τ about hinge $\Rightarrow \vec{\tau}_{Hinge} = I_H\vec{\alpha}$
② If body is not hinged → write τ about COM $\Rightarrow \vec{\tau}_{com} = I_{com}\vec{\alpha}$

Q Vertical plane



Just after release
Find ① α ② ω ③ a_{end} ④ a_{com} ⑤ F_{Hinge}

① $\vec{\tau}_{Hinge} = I_{Hinge}\vec{\alpha}$

$$mg\frac{l}{2} = \frac{ml^2}{3}\alpha$$

$$\boxed{\alpha = \frac{3g}{2l}}\ \{\text{clockwise}\}$$

② Just after release $\omega = 0$

Alternate: $\frac{d\omega}{dt} = \frac{3g}{2l}$

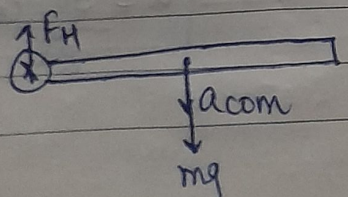
$$\int d\omega = \frac{3g}{2l}\int dt \Rightarrow \omega = \frac{3gt}{2l}$$

$\therefore$ At $t = 0$, $\omega = 0$.

③ $a_{end} = a_t = \alpha l = \frac{3g}{2}$

④ $a_{com} = a_t = \alpha\frac{l}{2} = \frac{3g}{4}$

⑤

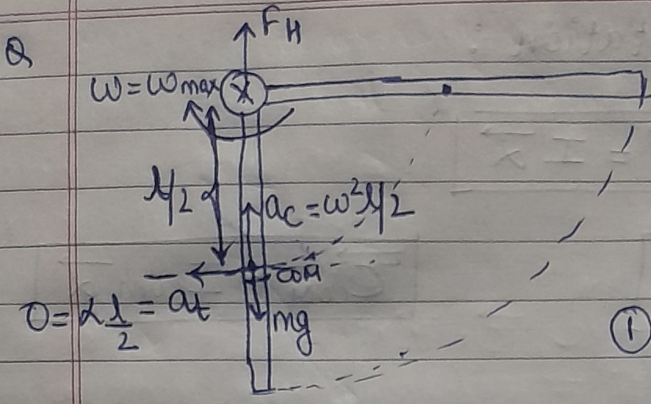


We know, $\vec{F}_{ext} = m\vec{a}_{com}$

$$mg - F_H = m\left(\frac{3g}{4}\right)$$

$$\boxed{F_H = \frac{mg}{4}}$$

Q



When rod becomes vertical, Find ① α ② ω ③ a_{com} ④ F_{Hinge}.

① $\vec{\tau}_{Hinge} = I_{Hinge}\vec{\alpha}$

$0 = I\vec{\alpha}$

$\therefore \boxed{\alpha = 0}$

② We know, W.d by $F_H = 0$ as $dW = \vec{F}\cdot d\vec{s}$ and $d\vec{s} = 0$.

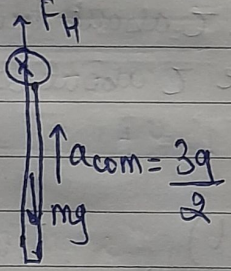
So using T.M.E.C $\Rightarrow$ Loss in PE = Gain in KE

$$mg\frac{l}{2} = \frac{1}{2} I\omega^2 \Rightarrow mg\frac{l}{2} = \frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2$$

$$\boxed{\omega = \sqrt{\frac{3g}{l}}}$$

③ $a_{com} = a_c = \omega^2 \frac{l}{2} = \frac{3g}{2}$ $\therefore \boxed{a_{com} = \frac{3g}{2}\ (\uparrow)}$

④

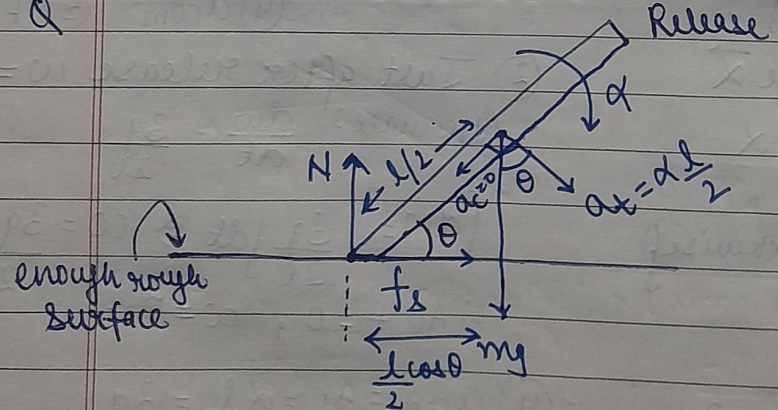


$F_{ext} = ma_{com}$

$F_H - mg = \frac{3mg}{2}$

$$\boxed{F_H = \frac{5mg}{2}}$$

Q



Just after release Find ① α ② ω ③ a_{com} ④ N ⑤ f_s

① $\vec{\tau} = I\vec{\alpha}$

$mg\frac{l}{2}\cos\theta = \frac{ml^2}{3}\alpha$

$$\boxed{\alpha = \frac{3g}{2l}\cos\theta}$$

② Just after release $\omega = 0$

③ $a_{com} = a_t = \alpha\frac{l}{2} = \frac{3g\cos\theta}{4}$

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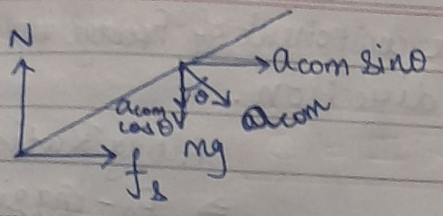
④



$F_{ext} = m a_{com}$

$mg - N = m\, a_{com}\cos\theta$

$mg - N = \frac{3mg\cos^2\theta}{4}$

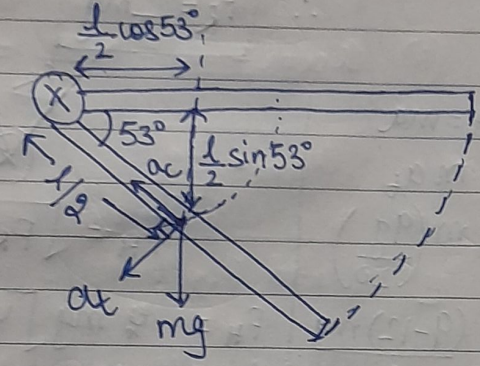
$$\boxed{N = mg\left(1 - \frac{3}{4}\cos^2\theta\right)}$$

⑤ $f_s = m a_{com}\sin\theta$

$f_s = m\left(\frac{3g\cos\theta}{4}\right)\sin\theta$

$$\boxed{f_s = \frac{3mg}{4}\cos\theta\sin\theta}$$

** Q



Release

When angular displacement of Rod is $\theta = 53°$. Find

① α ② ω ③ $(a_t)_{com}$ ④ $(a_c)_{com}$ ⑤ a_{com} ⑥ F_{hinge}.

① $\vec{\tau} = I\vec{\alpha}$

$mg\frac{l}{2}\times\frac{3}{5} = \frac{ml^2}{3}\alpha$

$$\boxed{\alpha = \frac{9g}{10l}}$$

② using TMEC $\Rightarrow$ Loss in PE = Gain in KE

$mg\frac{l}{2}\left(\frac{4}{5}\right) = \frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2$

$$\boxed{\omega = \sqrt{\frac{12g}{5l}}}$$

③ $a_t = \alpha\frac{l}{2} = \frac{9g}{10l}\left(\frac{l}{2}\right)$

$$\boxed{a_t = \frac{9g}{20}}$$

④ $a_c = \omega^2\frac{l}{2} = \frac{6}{5}\frac{l}{l}g\left(\frac{l}{2}\right)$

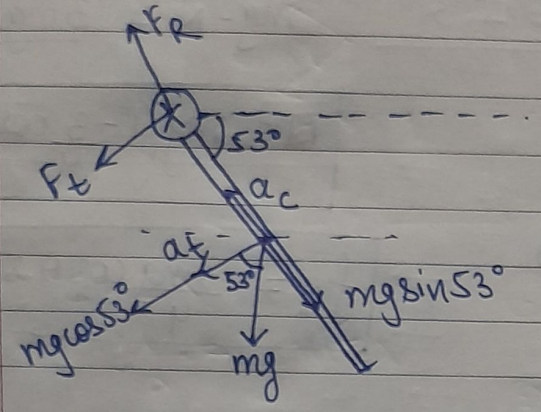
$$\boxed{a_c = \frac{6g}{5}}$$

⑤ $a_{com} = \sqrt{a_t^2 + a_c^2} = \sqrt{\left(\frac{9g}{20}\right)^2 + \left(\frac{6g}{5}\right)^2} = \frac{3g}{5}\sqrt{\left(\frac{3}{4}\right)^2 + (2)^2}$

$a_{com} = \frac{3g}{5}\sqrt{\frac{9}{16} + 4} = \frac{3g}{5}\sqrt{\frac{73}{16}} = \frac{3g\sqrt{73}}{20}$

$$\boxed{a_{com} = \frac{3g\sqrt{73}}{20}}$$

⑥ **Method 1:** Let us assume component of F_{Hinge} in radial (F_R) and tangential (F_t) direction.



Along Radial: $F_{ext} = ma$

$\Rightarrow F_R - mg\sin53° = ma_c$

$= F_R - mg\left(\frac{4}{5}\right) = m\left(\frac{6g}{5}\right)$

$= F_R = \frac{6mg}{5} + \frac{4mg}{5} = \frac{10mg}{5}$

$\therefore \boxed{F_R = 2mg}$

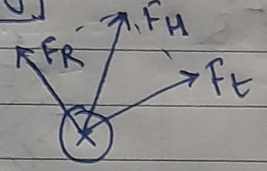
Along tangential: $F_{ext} = ma$

$= F_t + mg\cos53° = ma_t$

$= F_t + mg\left(\frac{3}{5}\right) = m\left(\frac{9g}{20}\right)$

$= F_t = \frac{9mg}{20} - \frac{3mg}{5} = \frac{(9-12)mg}{20}$

$\therefore \boxed{F_t = -\frac{3mg}{20}}$

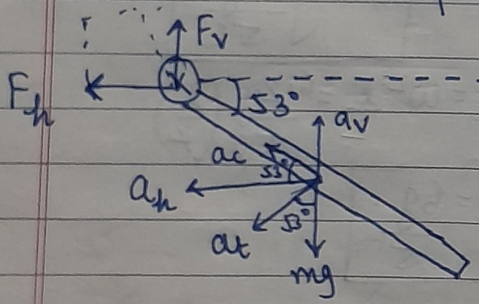


$F_H = \sqrt{F_R^2 + F_t^2}$

$F_H = 2mg\sqrt{4 + \frac{9}{400}} = mg\sqrt{\frac{1609}{400}}$

$F_H = \frac{mg}{20}\sqrt{1609} \approx 2.005mg$

Method 2: Let us assume component of F_{Hinge} in Horizontal (F_h) and vertical (F_v).



$a_h = a_c\cos53° + a_t\sin53°$

$= \frac{6g}{5}\times\frac{3}{5} + \frac{9g}{20}\times\frac{4}{5}$

$a_h = \frac{18g}{25} + \frac{9g}{25} = \frac{27g}{25}$

$a_v = a_c\sin53° - a_t\cos53°$

$= \frac{6g}{5}\left(\frac{4}{5}\right) - \frac{9g}{20}\left(\frac{3}{5}\right)$

$= \left(\frac{24}{25} - \frac{27}{100}\right)g$

$a_v = \frac{69g}{100}$

NL along Horizontal: $F_h = ma_h$

$\boxed{F_h = \frac{27mg}{25}}$

NL along vertical: $F_v - mg = m\left(\frac{69g}{100}\right)$

$\boxed{F_v = \frac{169mg}{100}}$

$F_{Hinge} = \sqrt{F_h^2 + F_v^2}$

$F_H = \frac{mg}{25}\sqrt{(27)^2 + \left(\frac{169}{4}\right)^2}$

$\therefore \boxed{F_H \approx 2.065mg}$

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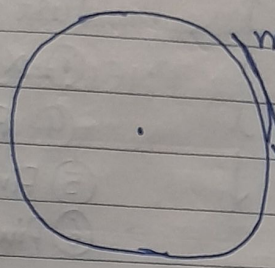
Q Horizontal Rough table (μ_K)

Top view

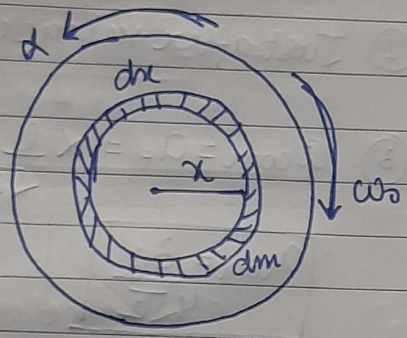


① Find angular acc^n of disc

② Find time taken by disc to stop

③ Find no. of rotations completed by disc before it stops.

① Due to τ of f_K disc will slow down.



$\sigma = \frac{m}{\pi R^2}$ $\quad \sigma = \frac{dm}{dA}$

$dm = \sigma\, dA = \frac{m}{\pi R^2}(2\pi x\, dx)$

$$\boxed{dm = \frac{2mx\,dx}{R^2}}$$

Now, $f_K = \mu_K N = \mu_K\, dm g$

$$\boxed{f_K = \mu_K \left(\frac{2mx\,dx}{R^2}\right) g}$$

Now, Torque to f_K on ring $= d\tau$, $\quad d\tau = f_K x$

$$\tau = \int d\tau = \frac{2m\mu_K g}{R^2}\int_0^R x^2\,dx \Rightarrow \boxed{\tau = \frac{2mg\mu_K R}{3}}$$

By Newtons 2nd law, $\tau = I\alpha$

$$\Rightarrow \frac{2mg\mu_K R}{3} = \frac{mR^2}{2}\alpha$$

$$\boxed{\alpha = \frac{4g\mu_K}{3R}}$$

② $\omega = \omega_0 + \alpha t$

$0 = \omega_0 - \frac{4g\mu_K}{3R}t$

$\frac{4g\mu_K}{3R}t = \omega_0$

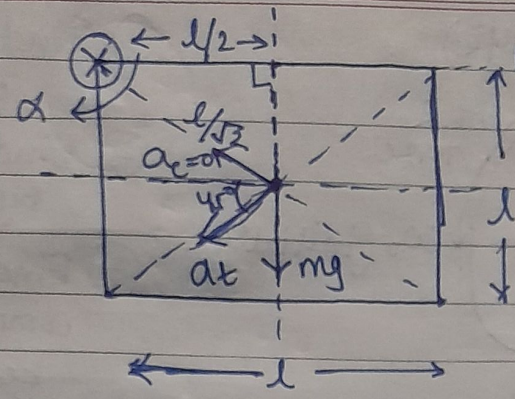
$$\boxed{t = \frac{3\omega_0 R}{4g\mu_K}}$$

③ $\omega^2 = \omega_0^2 + 2\alpha\theta$

$0 = \omega_0^2 - 2\left(\frac{4g\mu_K}{3R}\right)\theta$

$$\boxed{\theta = \frac{3\omega_0^2 R}{8\mu_K g}}$$

$$\boxed{\text{No of rotations} = \frac{\theta}{2\pi} = \frac{3\omega_0^2 R}{16\pi\mu_K g}}$$

Q 

Release Given uniform square plate
Just after release, find
① α ② ω ③ a_{com}
④ F_{Hinge}
⑤ Find ω_{max} of square plate
⑥ Find F_{Hinge} at this instant

① $\tau = I\alpha$

$$mg\frac{l}{2} = \frac{2}{3}ml^2\alpha$$

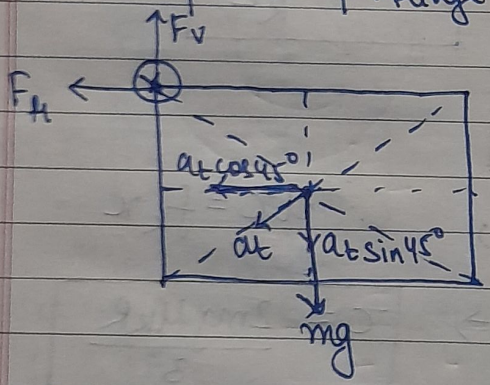
$$\boxed{\alpha = \frac{3g}{4l}}$$

② Just after release, $\omega = 0$

③ $a_{com} = a_t = \alpha\frac{l}{\sqrt{2}} = \frac{3g}{4l}\left(\frac{l}{\sqrt{2}}\right)$

$$\boxed{a_{com} = \frac{3g}{4\sqrt{2}}}$$

④ Let component of F_{Hinge} be F_H and F_V



NL along Hor^z:

$F_H = ma_t\cos45°$

$F_H = m\left(\frac{3g}{4\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right)$

$$\boxed{F_H = \frac{3mg}{8}}$$

NL along vert:

$mg - F_V = ma_t\sin45°$

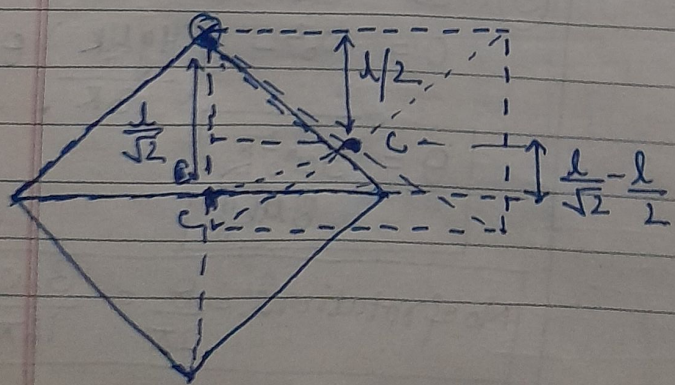
$mg - F_V = \frac{3mg}{8}$

$$\boxed{F_V = \frac{5mg}{8}}$$

$$F_{Hinge} = \sqrt{F_H^2 + F_V^2} = \frac{mg}{8}\sqrt{9+25}$$

$$\therefore \boxed{F_{Hinge} = \frac{mg}{8}\sqrt{34}}$$

⑤ ω_{max} will be when COM will be at the lowest point as loss in GPE will be more so gain in KE will also be more.



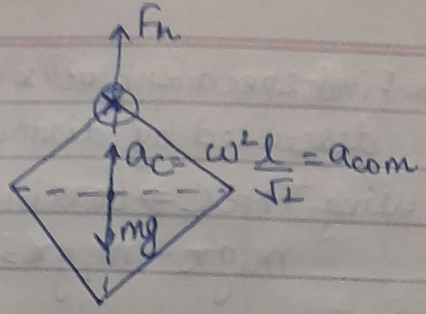
Using TMEC ⇒

Loss in GPE = Gain in KE

$$mg\left(\frac{l}{\sqrt{2}} - \frac{l}{2}\right) = \frac{1}{2}\left(\frac{2}{3}ml^2\right)\omega_{max}^2 - 0$$

$$\boxed{\omega_{max} = \sqrt{\frac{3g(2-\sqrt{2})}{2\sqrt{2}l}}}$$

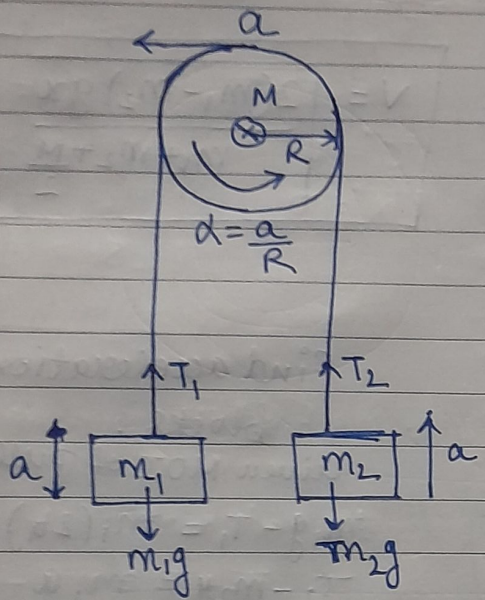
⑥



$$F_H - mg = m\omega^2 \frac{l}{\sqrt{2}}$$

$$\boxed{F_H = mg + \frac{3mg(2-\sqrt{2})}{4}}$$

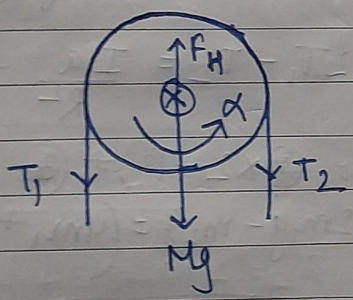
Q



Assume pulley to be a solid cylinder
There is no slipping b/w thread and pulley.
Find acceleration of blocks. ($m_1 > m_2$)

Note: There is no slipping b/w pulley and thread, that means there exists a f_s which will provide α. Due to this we can ~~write~~ say that ~~eqn~~ tension in both the sides of pulley is not same.
We consider pulley and thread as a system so f_s becomes Fint.

$$\boxed{F_H = Mg + T_1 + T_2}$$



Now, using NL on m_1:

$$m_1 g - T_1 = m_1 a \quad \text{—①}$$

using NL on m_2:

$$T_2 - m_2 g = m_2 a \quad \text{—②}$$

Using NL on pulley + thread; we write τ about hinge:

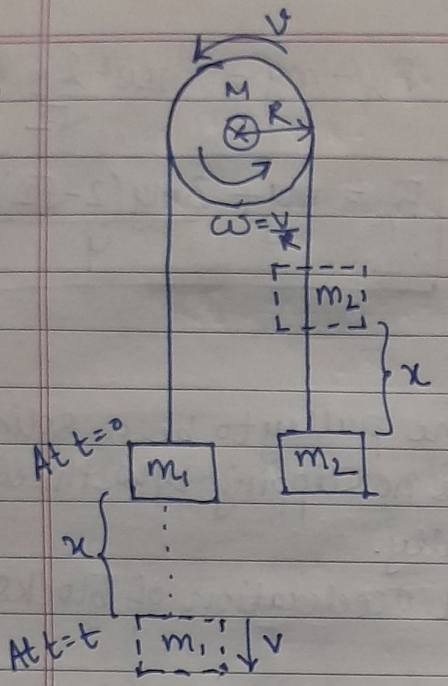
$$T_1 R - T_2 R = \frac{MR^2}{2}\left(\frac{a}{R}\right)$$

$$T_1 - T_2 = \frac{Ma}{2} \quad \text{—③}$$

Adding ① + ② + ③ $\Rightarrow$ $m_1 g - m_2 g = \left(m_1 + m_2 + \frac{M}{2}\right) a$

$$\boxed{a = \frac{(m_1 - m_2)g}{\left(m_1 + m_2 + \frac{M}{2}\right)}}$$

Q



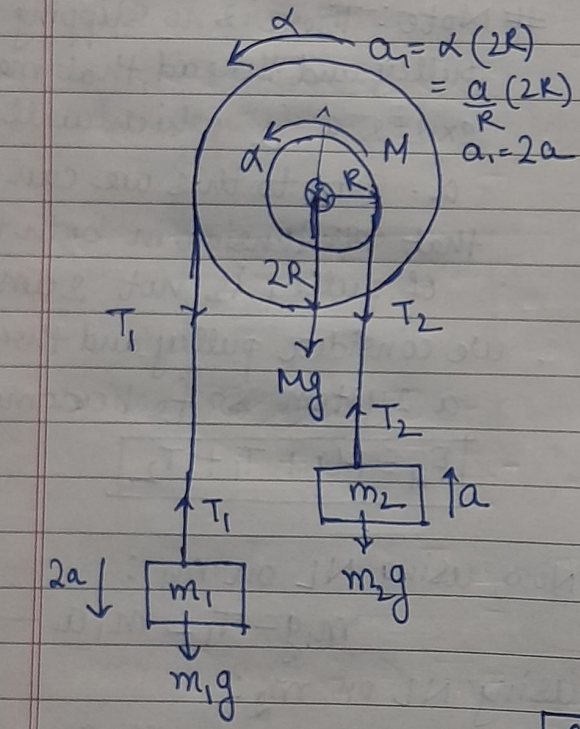
Find speed of blocks when m_1 has descended by distance x.

Using TMEC ⇒ Loss in PE = Gain in KE

$$m_1gx - m_2gx = \frac{1}{2}(m_1+m_2)v^2 + \frac{1}{2}\frac{MR^2}{2}\cdot\frac{v^2}{R^2}$$

$$\boxed{v = \sqrt{\frac{2(m_1-m_2)gx}{m_1+m_2+\frac{M}{2}}}}$$

Q



$a_1 = \alpha(2R)$

$= \frac{a}{R}(2R)$

$a_1 = 2a$

Find acceleration of blocks.

Given MOI of pulley = I

$$m_1g - T_1 = m_1(2a) \quad -(1)$$

$$T_2 - m_2g = m_1a \quad -(2)$$

$$T_1(2R) - T_2R = I\left(\frac{a}{R}\right)$$

$$2T_1 - T_2 = \frac{Ia}{R^2} \quad -(3)$$

Now, (1)×2 + (2) + (3)

$$2m_1g - m_2g = \left(4m_1 + m_2 + \frac{I}{R^2}\right)a$$

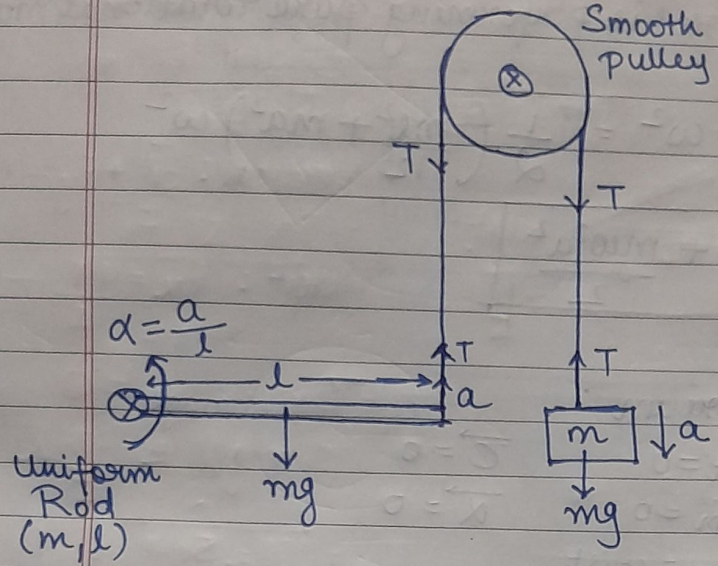
$$\boxed{a = \frac{(2m_1-m_2)g}{4m_1+m_2+\frac{I}{R^2}}}$$

Q Given Smooth pulley,

Just after release, Find

(1) Acc^n of block (2) angular acc^n of Rod (3) Tension in string

(4) F_{hinge}.



① $mg - T = ma$ — ①

$Tl - mg\frac{l}{2} = \frac{ml^2}{3}\left(\frac{a}{l}\right)$

$T - \frac{mg}{2} = \frac{ma}{3}$ — ②

Add ① and ②

$mg - \frac{mg}{2} = \left(m + \frac{m}{3}\right)a$

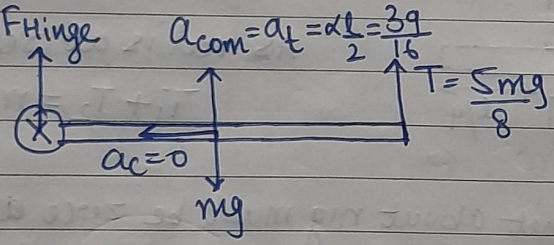
$\frac{mg}{2} = \frac{4m}{3}a$

$\boxed{a = \frac{3g}{8}}$

② $\alpha = \frac{a}{l}$

③ Put a in eqn ① $\Rightarrow mg - T = \frac{3mg}{8} \Rightarrow \boxed{T = \frac{5mg}{8}}$

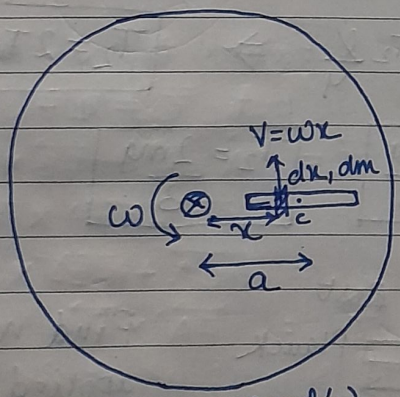
④



$F_H + \frac{5mg}{8} - mg = \frac{3mg}{16}$

$\boxed{F_{Hinge} = \frac{9mg}{16}}$

Q



Uniform Rod (m, l) is rotating on disc. Find KE of Rod.

$dKE = \frac{1}{2}dm\,v^2 = \frac{1}{2}\left(\frac{m}{l}dx\right)(\omega^2 x^2)$

$KE = \int dKE = \frac{m\omega^2}{2l}\int_{(a-\frac{l}{2})}^{(a+\frac{l}{2})} x^2\,dx$

$KE = \frac{m\omega^2}{6l}\left(x^3\right)_{(a-\frac{l}{2})}^{(a+l/2)} = \frac{m\omega^2}{6l}\left(\left(a+\frac{l}{2}\right)^3 - \left(a-\frac{l}{2}\right)^3\right)$

$KE = \frac{m\omega^2}{6l}\left(a^3 + \frac{l^3}{8} + 3a^2\frac{l}{2} + 3a\frac{l^2}{4} - a^3 + \frac{l^3}{8} + 3a^2\frac{l}{2} - 3a\frac{l^2}{4}\right) = \frac{m\omega^2 l^2}{24} + \frac{m\omega^2 a^2}{2}$

Alternate: we can say that rod is performing pure rotational motion about axis.

$$KE = \frac{1}{2} I_{axis}\, \omega^2 = \frac{1}{2}\left(\frac{ml^2}{12} + ma^2\right)\omega^2$$

$$\boxed{KE = \frac{m\omega^2 l^2}{24} + \frac{m\omega^2 a^2}{2}}$$

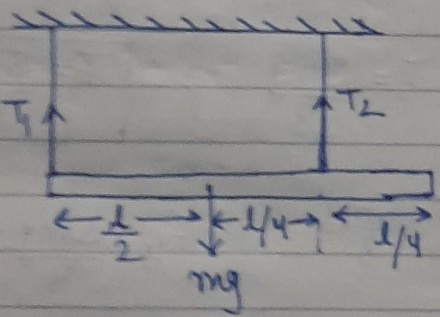
Mechanical Equilibrium:

$\vec{F}_{ext} = 0$ $\quad$ $\vec{\tau} = 0$

$\vec{a}_{com} = 0$ $\quad$ $\vec{\alpha} = 0$

$\vec{V}_{com} = \text{const}$

Q



Find Tension in strings

As rod is at rest,

$$T_1 + T_2 = mg \quad \text{—(1)}$$

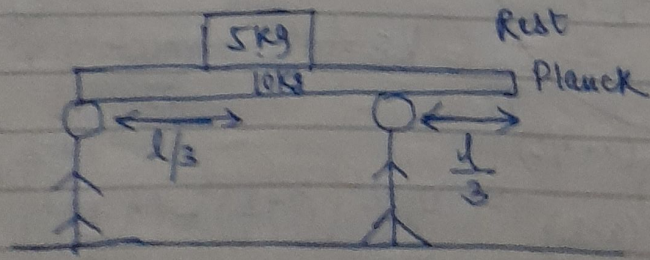
Also, T_{net} about mg must be zero as rod is at rest

$$T_1 \frac{l}{2} - T_2 \frac{l}{4} = 0$$

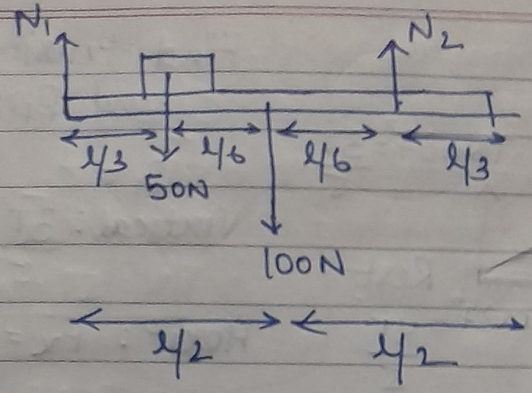
$$T_1 \frac{\cancel{l}}{2} = T_2 \frac{\cancel{l}}{4} \Rightarrow \boxed{T_2 = 2T_1} \quad \text{—(2)}$$

From (1) and (2) $\Rightarrow$ $\boxed{T_1 = \frac{mg}{3}}$ $\boxed{T_2 = \frac{2mg}{3}}$

Q



Find normal forces applied by the boys on planck.



$\vec{F}_{ext} = 0$

$\Rightarrow N_1 + N_2 = mg$ — (1)

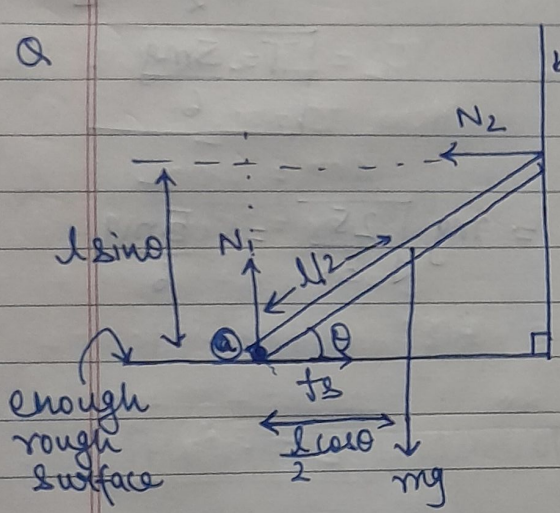
τ about N_2:

$100\frac{l}{6} + 50\frac{l}{3} = N_1\left(\frac{2l}{3}\right)$

$\frac{100}{3} = \frac{2N_1}{3} \Rightarrow \boxed{N_1 = 50N}$

$\boxed{N_2 = 100N}$

Q



Rod is at rest

(1) Draw FBD of rod and find all unknown forces on it.

Vertical: $N_1 = mg$

Horizontal: $N_2 = f_s$ — (1)

τ about (A) $= 0$

$\Rightarrow mg\frac{l}{2}\cos\theta = N_2 l \sin\theta$

$\boxed{N_2 = \frac{mg\cot\theta}{2}}$ $\boxed{f_s = \frac{mg\cot\theta}{2}}$

(2) Find μ b/w floor and rod so that Rod doesn't slip.

We know, $f_s \le f_{lim}$

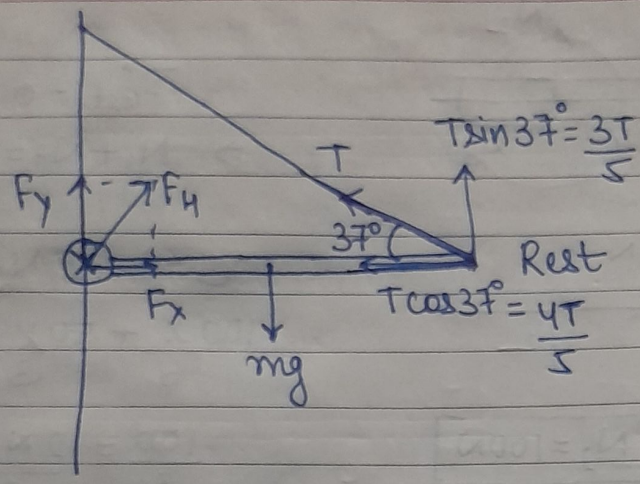
$= f_s \le \mu N_1$

$= \frac{mg\cot\theta}{2} \le \mu mg$

$\boxed{\mu \ge \frac{\cot\theta}{2}}$

$\boxed{\mu_{min} = \frac{\cot\theta}{2}}$

Q



Find tension in string and F_{Hinge}

Vertical: $\frac{3T}{5} + F_y = mg$ — (1)

Horiz: $F_x = \frac{4T}{5}$ — (2)

$\tau_{about\ hinge} = 0$

$mg\frac{\ell}{2} = \frac{3T}{5}\ell$

$$\boxed{T = \frac{5mg}{6}}$$

$$F_x = \frac{2}{3}\left(\frac{5mg}{6}\right) = \frac{2}{3}mg$$

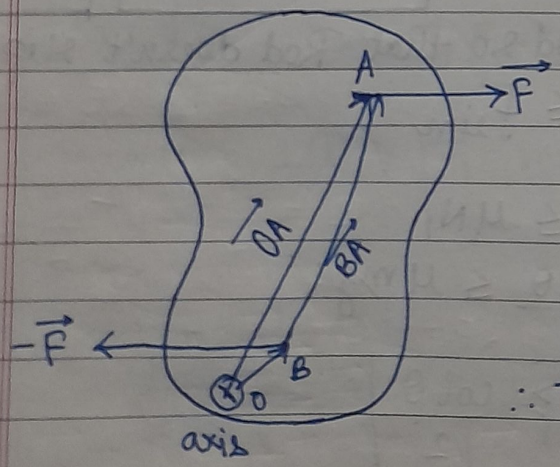
$$F_y = mg - \frac{3}{5}\left(\frac{5}{6}mg\right) = \frac{mg}{2}$$

$$F_h = \sqrt{F_x^2 + F_y^2} = mg\sqrt{\frac{4}{9} + \frac{1}{4}} = mg\sqrt{\frac{25}{36}} = \frac{5mg}{6}$$

$$\therefore \boxed{F_h = \frac{5mg}{6}}$$

Force Couple:

→ A pair of forces → Equal Magnitude → Opposite Direction



$\vec{F}_{net} = 0 \Rightarrow \vec{a}_{com} = 0 \Rightarrow \vec{V}_{com} = const$

But $\vec{\tau} \neq 0$

$$\vec{\tau} = \vec{OA} \times \vec{F} + \vec{OB} \times (-\vec{F})$$

$$= (\vec{OA} - \vec{OB}) \times \vec{F}$$

$$= (\vec{OA} + \vec{BO}) \times \vec{F}$$

$$\vec{\tau} = (\vec{BA}) \times \vec{F}$$

∴ It is independent of axis we choose

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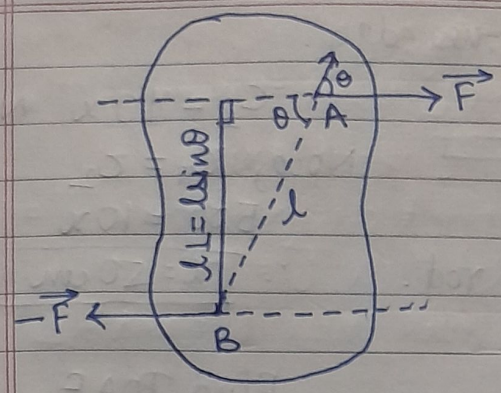


$\vec{\tau} = \vec{BA} \times \vec{F}$

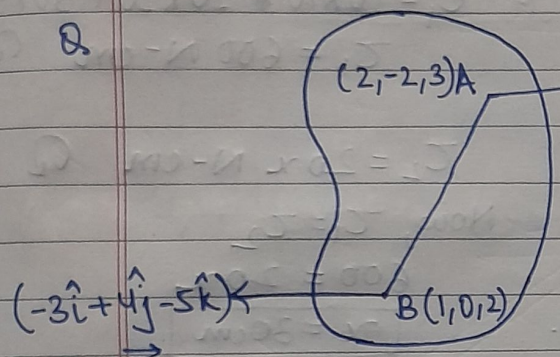
$\tau = lF\sin\theta$

$\tau = F(l\sin\theta)$

$\boxed{\tau = Fl_1}$

l_1 is separation b/w two LOAF

Q



$(3\hat{i}-4\hat{j}+5\hat{k})N = \vec{F}$

Find τ on this body.

$\vec{BA} = \hat{i}-2\hat{j}+\hat{k}$

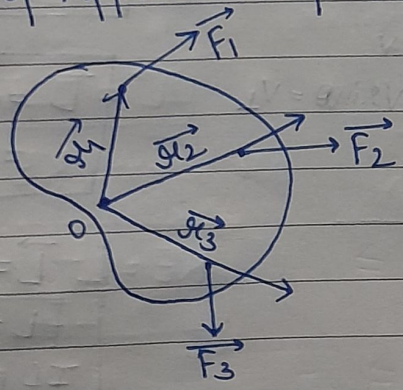
$\vec{\tau} = \vec{BA} \times \vec{F}$

$\vec{\tau} = (\hat{i}-2\hat{j}+\hat{k}) \times (3\hat{i}-4\hat{j}+5\hat{k})$

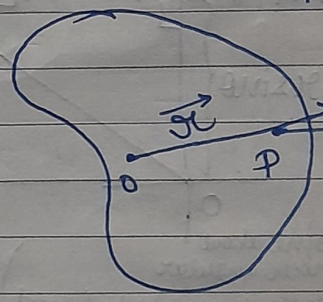
$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ 3 & -4 & 5 \end{vmatrix} = (-6\hat{i}-2\hat{j}+2\hat{k})\,N\text{-}m$$

$|\vec{\tau}| = \sqrt{44}$ N-m

Point of Application of Force:



P → POAF

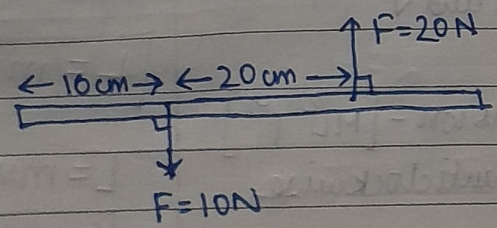


$\vec{\tau} = \vec{r_1} \times \vec{F_1} + \vec{r_2} \times \vec{F_2} + \vec{r_3} \times \vec{F_3}$

$\vec{\tau} = \vec{r} \times \vec{F_{net}}$

$\vec{\tau} = \vec{OP} \times (\vec{F_1}+\vec{F_2}+\vec{F_3})$

Q

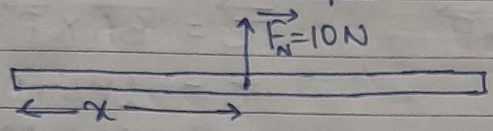


Find point of application of force.

$\tau_1 = 20\times30 - 10\times10 = 500$ N-cm ↺

(About end)

Let POAF be at x cm from one edge.



$\tau_2 = 10x$ N-cm ↺

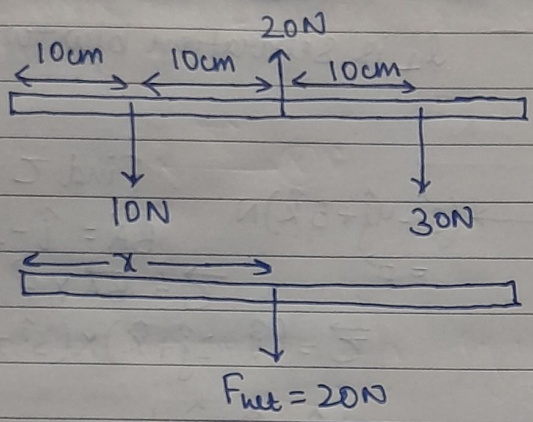
Now, $\tau_1 = \tau_2$

$500 = 10x$

$\therefore \boxed{x = 50\text{ cm}}$

$\therefore$ POAF is at 50cm from end of rod.

Q



Find POAF

$\tau_1 = 10\times10 + 20\times20 + 30\times30$

$\tau_1 = 600$ N-cm ↺

$\tau_2 = 20x$ N-cm ↻

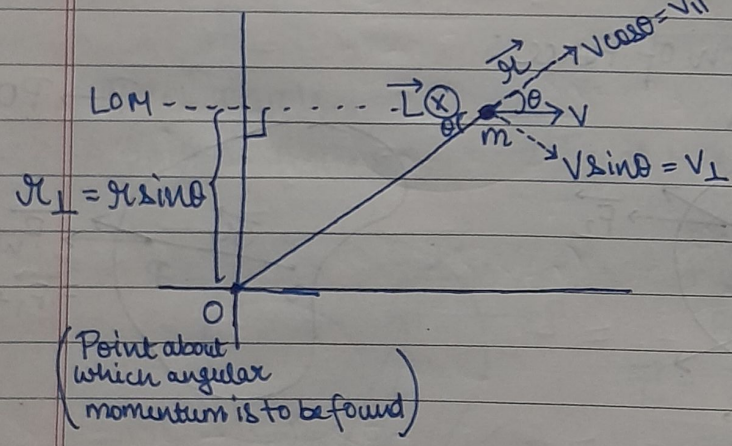
Now, $\tau_1 = \tau_2$

$600 = 20x$

$\boxed{x = 30\text{ cm}}$

$\therefore$ POAF is at 30cm from end of rod.

Angular Momentum about a point :



$\boxed{\vec{L} = \vec{r} \times \vec{p}}$

$\boxed{\vec{L} = \vec{r} \times m\vec{v} = m(\vec{r} \times \vec{v})}$

$\vec{L} = m(\vec{r} \times \vec{v})$

$\boxed{L = mrv\sin\theta}$

If $\theta = 0^\circ$ or $\theta = \pi$, $L = 0$

★ If line of motion passes through point O then $\vec{L}$ about Point O is zero.

$\boxed{L = mr(v\sin\theta) = mrv_{\perp}}$

$L = mv(r\sin\theta)$

★ $\boxed{L = mvr_{\perp}}$

→ Vector Quantity

→ Direction: $\vec{L} \perp \vec{v}$

$\vec{L} \perp \vec{r}$

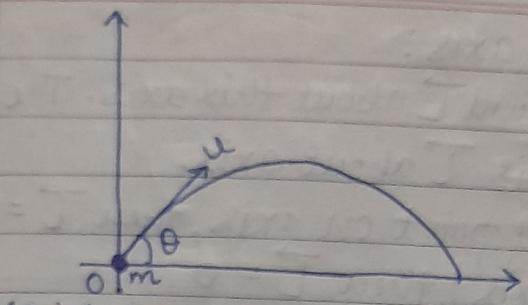
$\vec{L}$ will be ⊥ to plane of $\vec{r}$ and $\vec{v}$

→ axial vector

→ unit = $kg\,m^2/s$ → Dimension = $[ML^2T^{-1}]$

⊗ → clockwise ⊙ → anticlockwise

Q



(Point of projection)

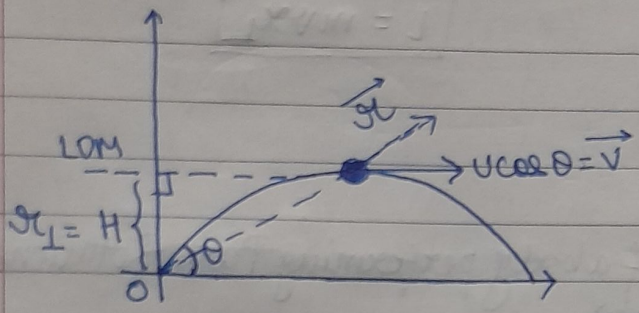
Find $\vec{L}$ about O when

① Particle is at highest point

② Particle is about to strike the ground

③ At any time t.

①



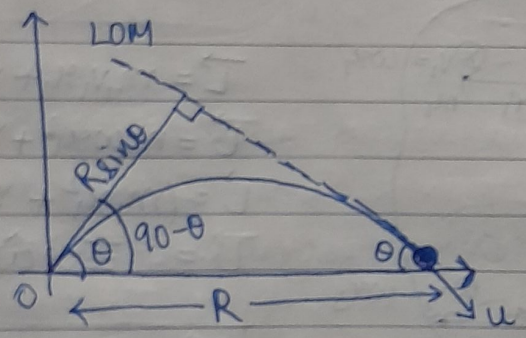
$\vec{L} = mvr_\perp$

$L = mu\cos\theta\,(H)$

$L = mu\cos\theta \dfrac{u^2\sin^2\theta}{2g}$

$$\boxed{\vec{L} = \frac{mu^3\sin^2\theta\cos\theta}{2g}(-\hat{k})}$$

②



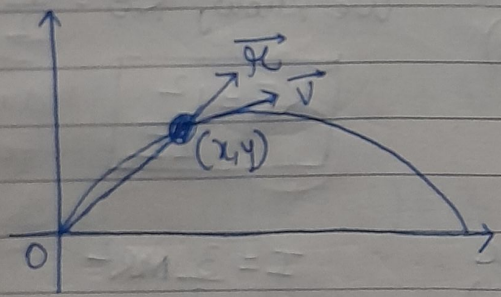
$L = mvr_\perp$

$L = mu R\sin\theta$

$L = mu\left(\dfrac{u^2\sin 2\theta}{g}\right)\sin\theta$

$$\boxed{\vec{L} = \frac{mu^3\sin 2\theta\sin\theta}{g}(-\hat{k})}$$

③



Let $\vec{r} = x\hat{i} + y\hat{j}$

$\vec{v} = v_x\hat{i} + v_y\hat{j}$

$\vec{L} = m(\vec{r} \times \vec{v})$

$\vec{L} = m\{(x\hat{i} + y\hat{j}) \times (v_x\hat{i} + v_y\hat{j})\}$

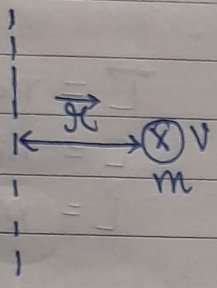
$\vec{L} = m(xv_y\hat{k} - yv_x\hat{k})$

$$\boxed{\vec{L} = m\left\{(u\cos\theta t)(u\sin\theta - gt) - \left(u\sin\theta t - \frac{1}{2}gt^2\right)(u\cos\theta)\right\}\hat{k}}$$

Angular Momentum about axis:

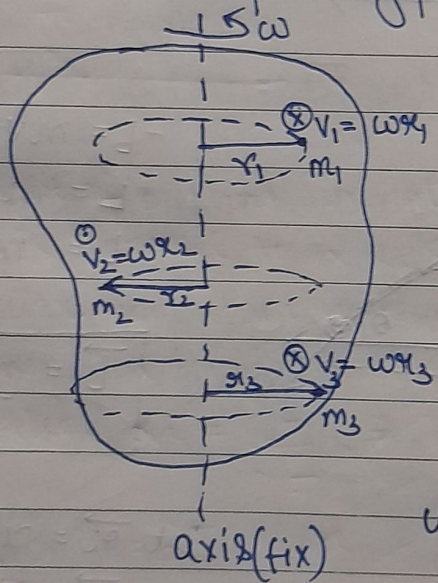
Take a point on axis and find $\vec{L}$ about this axis. The component of $\vec{L}$ along axis (parallel) is $\vec{L}$ about axis.

1. If LOM passes through any point on axis, then $\vec{L}=0$.
2. If LOM is parallel to axis, then $\vec{L}=0$
3. If axis and LOM are skew lines then $\vec{L}=mvx_\perp$



$$L = mvx_\perp$$

Angular Momentum of a body performing Pure Rotational Motion:



$$\vec{L} = m_1v_1x_1 + m_2v_2x_2 + m_3v_3r_3$$
$$\vec{L} = m_1\omega x_1^2 + m_2\omega x_2^2 + m_3\omega x_3^2$$
$$\vec{L} = (m_1x_1^2 + m_2x_2^2 + m_3r_3^2)\omega$$
$$\vec{L} = I\omega\hat{\omega}$$

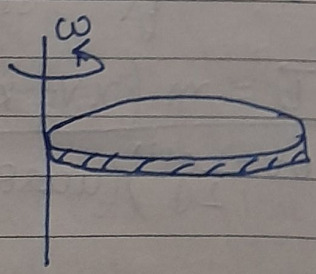
$$\boxed{\vec{L} = I\vec{\omega}}$$

We know, $KE = \frac{1}{2}I\omega^2 = \frac{(I\omega)^2}{2I}$

$$\boxed{KE = \frac{L^2}{2I}}$$

Q Find L and KE.

(1)

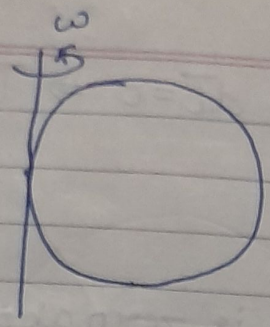


Disc (m, R)

$$I = \frac{3}{2}MR^2$$

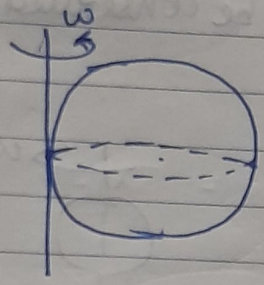
$$\boxed{L = I\omega = \frac{3}{2}comR^2}$$

$$\boxed{KE = \frac{3}{4}mR^2\omega^2}$$

② 

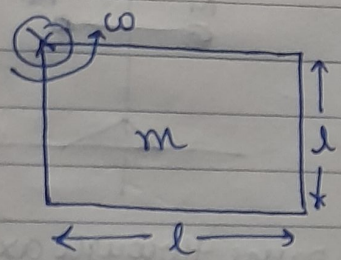
Disc (m, R) $\quad I = \frac{5}{4}mR^2$

$$L = \frac{5}{4}mR^2\omega \qquad KE = \frac{5}{8}mR^2\omega^2$$

③ 

Hollow sphere (m, R) $\quad I = \frac{5}{3}mR^2$

$$L = \frac{5}{3}mR^2\omega \qquad KE = \frac{5}{6}mR^2\omega^2$$

④ 

Square plate $\quad I = \frac{2}{3}ml^2$

$$L = \frac{2}{3}ml^2\omega \qquad KE = \frac{2}{6}ml^2\omega^2$$

Angular Momentum Conservation Theorem (AMCT):

$$\vec{L} = m(\vec{r} \times \vec{v})$$

$$\Rightarrow \frac{d\vec{L}}{dt} = m\left(\vec{r} \times \frac{d\vec{v}}{dt} + \frac{d\vec{r}}{dt} \times \vec{v}\right)$$

$$= \frac{d\vec{L}}{dt} = m\left(\vec{r} \times \vec{a} + \cancel{\vec{v} \times \vec{v}}^{\,0}\right)$$

$$= \frac{d\vec{L}}{dt} = m(\vec{r} \times \vec{a}) = \vec{r} \times m\vec{a}$$

$$= \frac{d\vec{L}}{dt} = \vec{r} \times \vec{F} = \vec{\tau}$$

$$\therefore \boxed{\frac{d\vec{L}}{dt} = \vec{\tau}}$$

$$\vec{L} = I\vec{\omega}$$

$$\Rightarrow \frac{d\vec{L}}{dt} = I\frac{d\vec{\omega}}{dt}$$

$$= \frac{d\vec{L}}{dt} = I\vec{\alpha} = \vec{\tau}$$

$$\therefore \boxed{\frac{d\vec{L}}{dt} = \vec{\tau}}$$

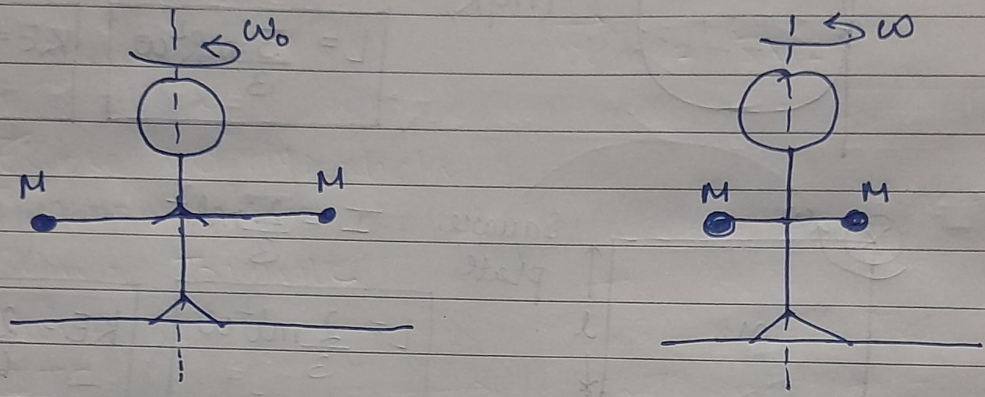
★ Rate of change of angular momentum is equal to torque applied.

If $\vec{\tau}_{ext} = 0 \Rightarrow \frac{d\vec{L}}{dt} = 0 \Rightarrow d\vec{L} = 0 \Rightarrow \vec{L} = \text{const}$

$$\therefore \boxed{\vec{L}_i = \vec{L}_f}$$

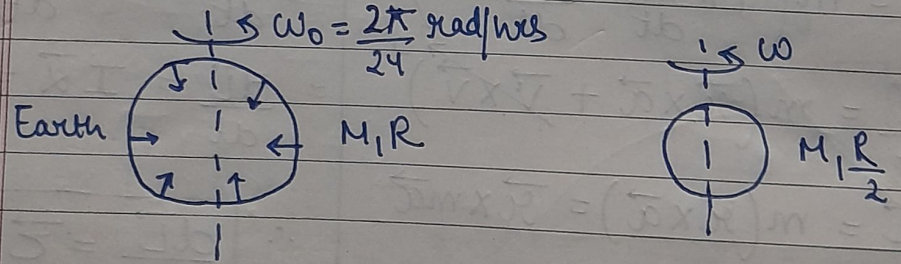
* If external torque on a system is zero about an axis, then its angular momentum will be conserved about that axis.

Ex



Here weight of boy and normal passes through axis. Also, weight of balls is parallel to axis. So no τ is produced. $\therefore \vec{\tau}_{ext} = 0$
So, angular momentum is conserved. $L = I\omega = \text{const}$.
When boy keeps his hand closer to body $I \downarrow$ and as $L = \text{const}$, $\omega \uparrow$. Also, $KE = \frac{L^2}{2I} \Rightarrow KE \propto \frac{1}{I}$. $\therefore$ KE also $\uparrow$.

Q



If earth shrinks due to internal forces to radius $\frac{R}{2}$. Find ω.

$\vec{\tau}_{ext} = 0$, using AMCT $\Rightarrow L_i = L_f$

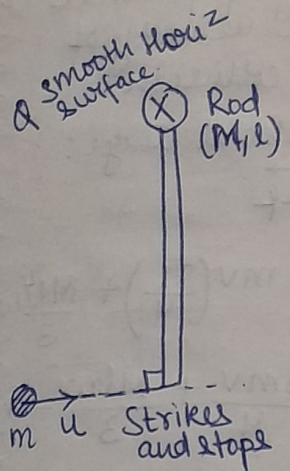
$$I_1\omega_1 = I_2\omega_2$$

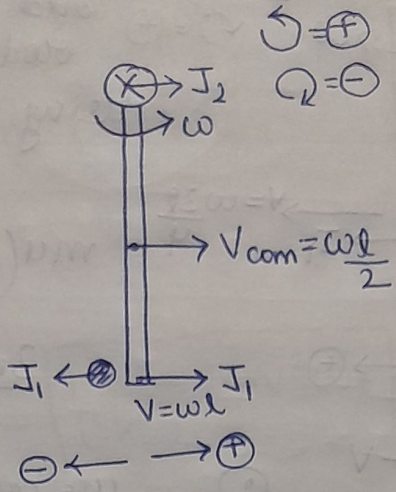
$$\frac{2}{5}MR^2\omega_0 = \frac{2}{5}M\left(\frac{R}{2}\right)^2\omega$$

$$\boxed{\omega = 4\omega_0}$$

Rotational Motion

Q Smooth Horiz surface





① Find ω of rod after collision.

Using AMCT on system about hinge:

$\vec{L_i} = \vec{L_f}$

$mul + 0 = 0 + \frac{Ml^2}{3}\omega$

$$\omega = \frac{3mu}{Ml}$$

② Find e.

$$e = \frac{v_{sep}}{v_{app}} = \frac{\omega l}{u} = \frac{\frac{3mu}{Ml} \times l}{u}$$

$$e = \frac{3m}{M}$$

③ Find impulse of force b/w particle and rod.

Using IMT on particle.

$-J_1 = 0 - mu$

$$J_1 = mu$$

④ Find impulse by hinge at the time when particle strikes the rod.

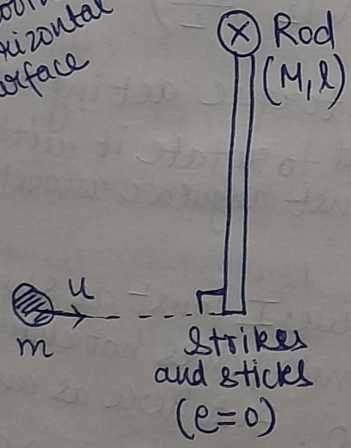
Using IMT on Rod $\Rightarrow$ $J_1 + J_2 = P_f - P_i$

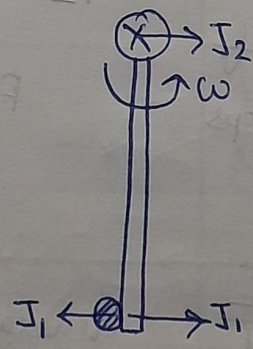
$mu + J_2 = MV_{com} - 0$

$mu + J_2 = M\left(\frac{3mu}{2M}\right) \Rightarrow J_2 = \frac{3mu}{2} - mu$

$$J_2 = \frac{mu}{2}$$

Q Smooth Horizontal Surface





Find ω of rod after collision

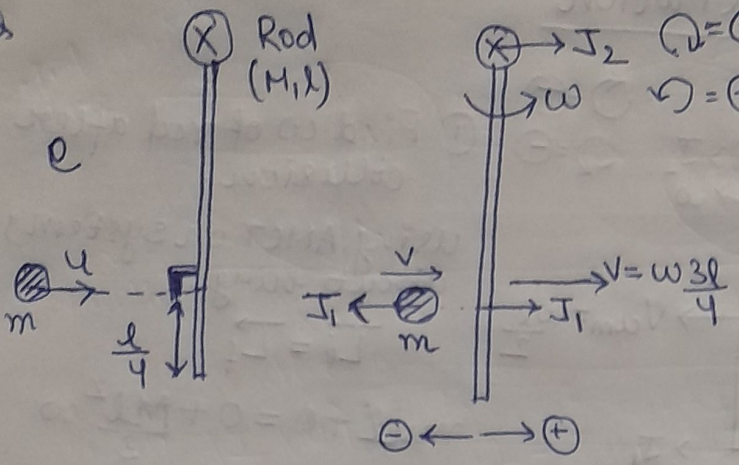
Using AMCT on system about hinge:

$\vec{L_i} = \vec{L_f}$ $(L = I\omega)$

$mul + 0 = \left(\frac{Ml^2}{3} + ml^2\right)\omega$

$$\omega = \frac{3mu}{(M+3m)l}$$

Q



⊖ = ⊕ ↻ = ⊕

Find velocity of particle and angular velocity of rod after collision.

Using AMCT about hinge:

$\vec{L_i} = \vec{L_f}$

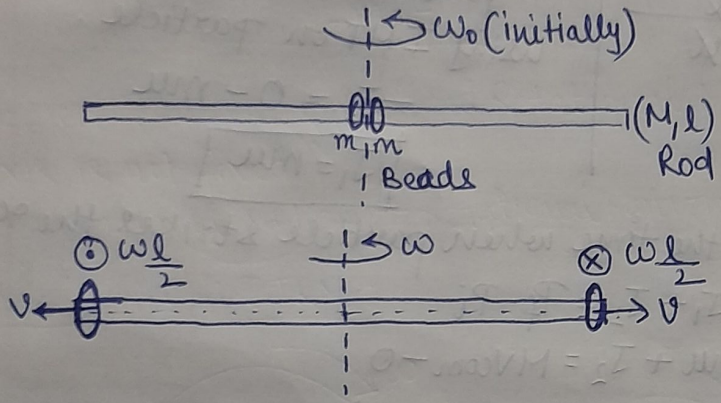
$$mu\left(\frac{3l}{4}\right) + 0 = mv\left(\frac{3l}{4}\right) + \frac{Ml^2}{3}\omega$$

$$\frac{3mu}{4} = \frac{3mv}{4} + \frac{Ml\omega}{3} \quad -(1)$$

$$e = \frac{v_{sep}}{v_{app}} = \frac{\frac{3\omega l}{4} - v}{u} \quad -(2)$$

We can solve (1) and (2) to get ω and v.

Q*



Find angular velocity of rod when bead reaches at ends of Rod.

Note: As particles move away $I_{system} \uparrow$, $\omega \downarrow$.

$\vec{L}$ remains constant.

Using AMCT about axis ⇒

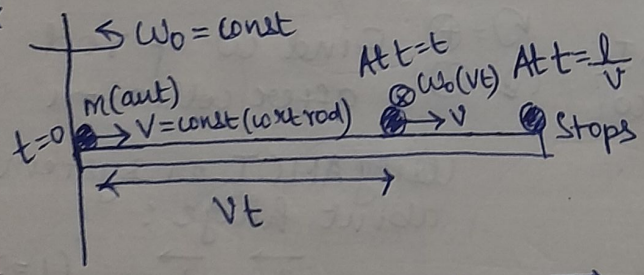
$\vec{L_i} = \vec{L_f}$

$$= \frac{Ml^2}{12}\omega_0 + 0 = \frac{Ml^2}{12}\omega + m\left(\omega\frac{l}{2}\right)\left(\frac{l}{2}\right) + m\left(\omega\frac{l}{2}\right)\left(\frac{l}{2}\right)$$

$$= \frac{Ml^2}{12}\omega_0 = \left(\frac{Ml^2}{12} + \frac{ml^2}{2}\right)\omega$$

$$\boxed{\omega = \frac{M\omega_0}{M+6m}}$$

Q**



Find torque acting on rod to rotate it with const angular velocity ω_0 →

#Concept: We know $\vec{\tau} = \frac{d\vec{L}}{dt}$

$L = I\omega$

$$\frac{dL}{dt} = I\frac{d\omega}{dt} + \omega\frac{dI}{dt},$$

have

Generally we I const as shape size geometry does not change so, $\frac{dI}{dt} = 0$. But here as ant moves $I \uparrow$ but ω = const ⇒ $\frac{d\omega}{dt} = 0$ here. Due to this $\vec{L} \uparrow$ and $\vec{\tau}$ is present.

→ → → →

L of system $= I\omega$

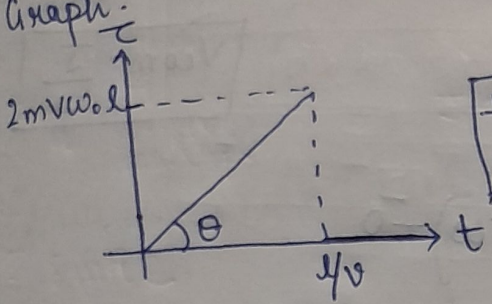
$$L = \left[\frac{Ml^2}{12} + m(vt)^2\right]\omega_0 = \left[\frac{Ml^2}{3} + mv^2t^2\right]\omega_0$$

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \left[0 + mv^2(2t)\right]\omega_0 \quad \therefore \boxed{\tau = 2mv^2\omega_0 t} \left(0 \le t \le \frac{l}{v}\right)$$

$\boxed{\tau = 0}$ $\left(t > \frac{l}{v}\right)$

↳ as ant stops moving so I is now const.

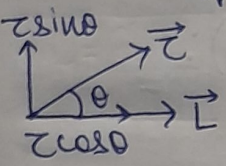
Graph:



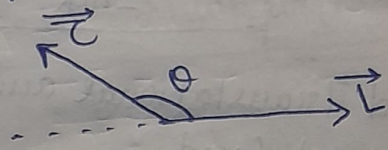
$\tan\theta = 2mv^2\omega_0$ (slope)

Note: ① $\vec{\tau} \parallel \vec{L} \Rightarrow |\vec{L}| \uparrow$ ② $\vec{\tau} \uparrow\downarrow \vec{L} \Rightarrow |\vec{L}| \downarrow$

③



④



Component $\Rightarrow |L| \downarrow$

$\tau\cos\theta = \tau_{tangential} = |L| \uparrow$

$\tau\sin\theta = \tau_{Normal} =$ Change direc^n of $\vec{L}$

Angular Impulse:

$AI = \vec{\tau}t$, if $\vec{\tau} = $ const

$AI = \int \vec{\tau}\,dt$, if $\vec{\tau} \neq$ const

→ Unit = N-m sec
= kg m²/sec

→ Dimension = $[ML^2T^{-1}]$

→ Axial Vector

→ Direction along $\vec{\tau}$

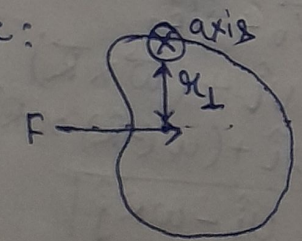
Angular Impulse-Angular Momentum theorem (AIAMT):

$$AI = \int \vec{\tau}\,dt = \int \frac{d\vec{L}}{dt}\,dt = \int_{\vec{L}_i}^{\vec{L}_f} d\vec{L}$$

$$\boxed{AI = \vec{L}_f - \vec{L}_i}$$

AIAMT states that AI is equal to change in angular momentum.

* Note:

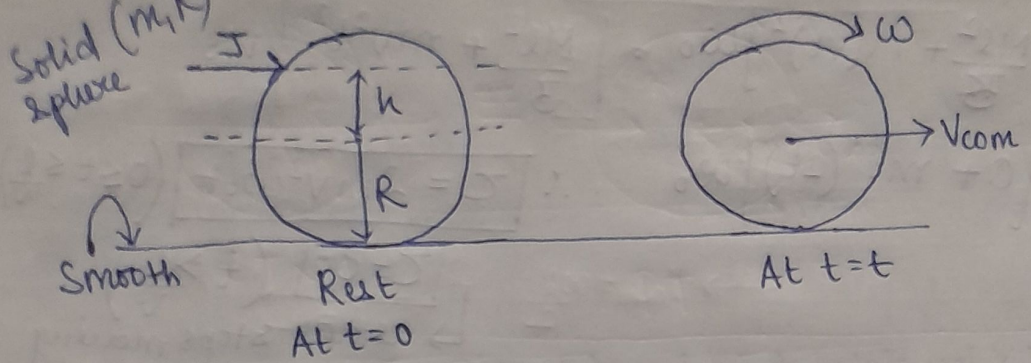


$J = \int F\,dt$

$AI = \int \vec{\tau}\,dt = \int F r_\perp\,dt = r_\perp \int F\,dt = r_\perp J$

$\therefore \boxed{AI = J r_\perp}$

Q



Find V_{com} and ω.

Using IMT:

$J = \Delta P$

$J = mV_{com} - 0$

$$V_{com} = \frac{J}{m}$$

Using AIAMT: $AI = J\perp_1$

$\Rightarrow AI = Jh$

Now, $AI = \Delta L \Rightarrow Jh = \left(\frac{2}{5}mR^2\omega\right) - 0$

$$\omega = \frac{5Jh}{2mR^2}$$

Combined Translational and Rotational Motion (CTRM):

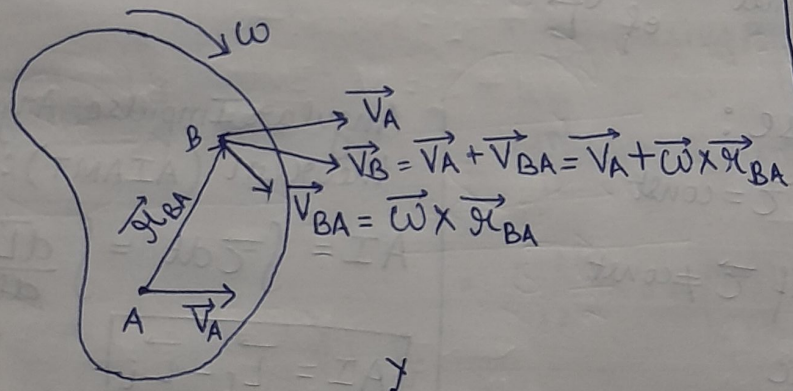
→ Axis is not fixed

→ Translational Motion + Rotational Motion

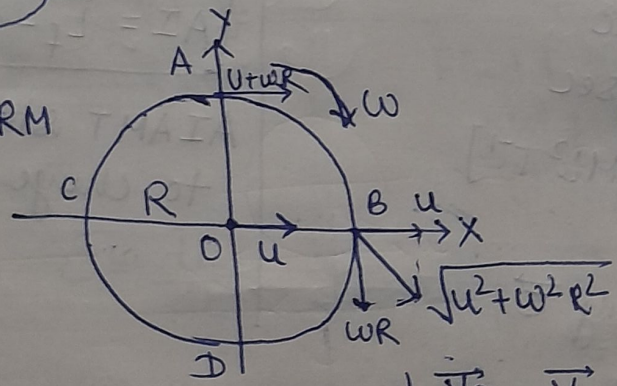
Kinematics of CTRM:

$$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$$

$$\vec{V}_B = \vec{V}_{BA} + \vec{V}_A$$



CTRM



$$\vec{\omega} = \omega(-\hat{k})$$

Method 1: Mathematical way

$\vec{V}_{AO} = \vec{V}_A - \vec{V}_O$

$\vec{V}_A = \vec{V}_{AO} + \vec{V}_O$

$= \vec{V}_O + \vec{V}_{AO}$

$= u\hat{i} + (\vec{\omega} \times \vec{R})$

$= u\hat{i} + (\omega(-\hat{k}) \times R\hat{j})$

$$\vec{V}_A = (U + \omega R)\hat{i}$$

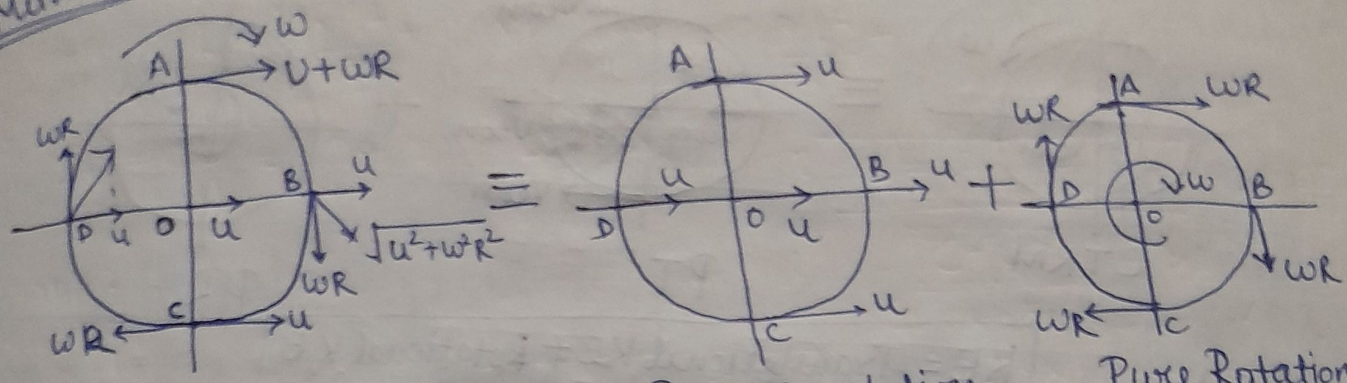
$\vec{V}_B = \vec{V}_O + \vec{V}_{BO}$

$= u\hat{i} + (\vec{\omega} \times \vec{R})$

$= u\hat{i} + (\omega(-\hat{k}) \times R\hat{i})$ ①

$$V_B = u\hat{i} - \omega R\hat{j}$$ ②

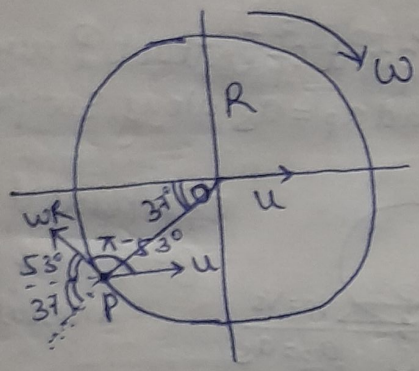
Method 2



CTRM Pure Translation Pure Rotation

∴ We can break CTRM as a combination of Pure translation and pure rotation.

Q

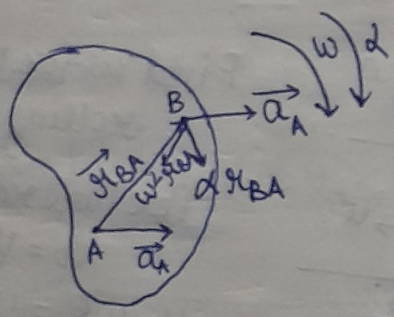


Find speed of point P

$$V_P = \sqrt{u^2+\omega^2R^2+2u\omega R\cos(\pi-53°)}$$

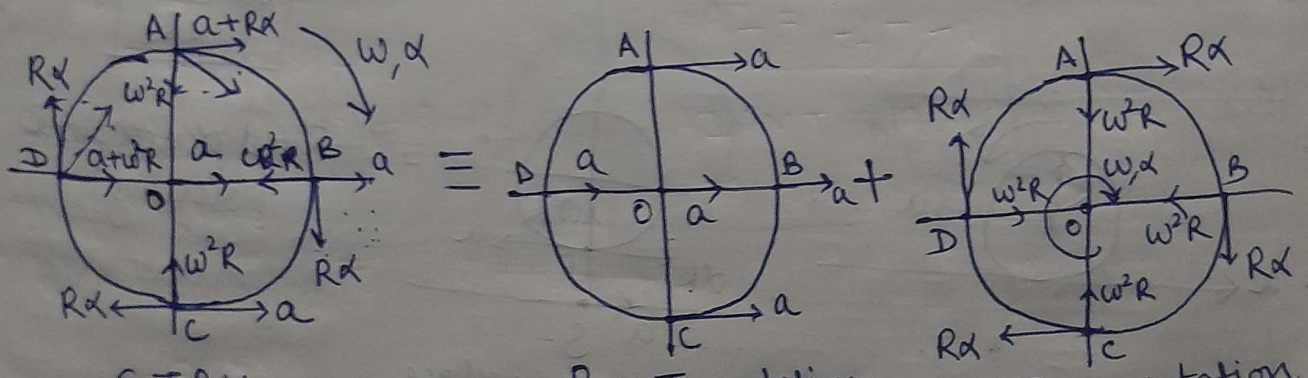
$$\boxed{V_P = \sqrt{u^2+\omega^2R^2-2u\omega R\cos 53°}}$$

Note:



$$\boxed{\vec{a}_B = \vec{a}_{BA} + \vec{a}_A}$$

Ex Find acc^n of A, B, C, D



CTRM Pure Translation Pure rotation

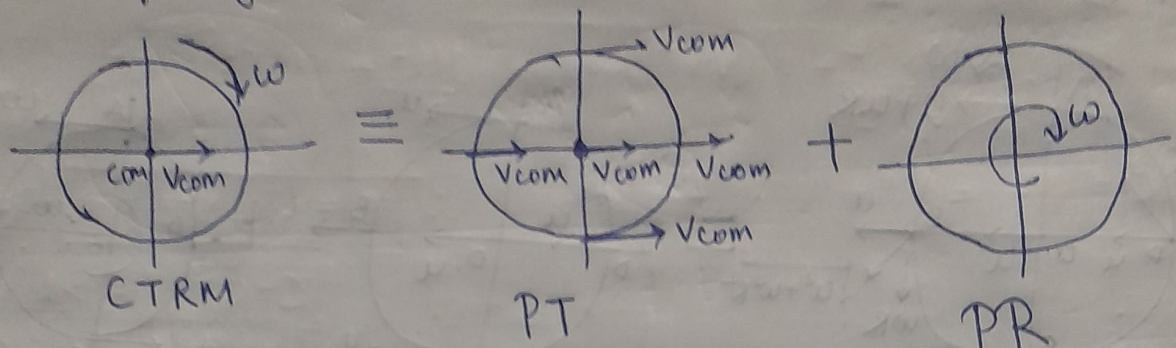
① $\vec{a}_A = (a+R\alpha)\hat{i} + \omega^2R(-\hat{j})$

② $\vec{a}_B = (a-\omega^2R)\hat{i} + R\alpha(-\hat{j})$

③ $\vec{a}_C = (a-R\alpha)\hat{i} + \omega^2R(\hat{j})$

④ $\vec{a}_D = (a+\omega^2R)\hat{i} + R\alpha(\hat{j})$

Note: KE of body in CTRM

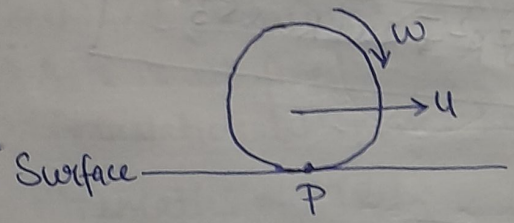


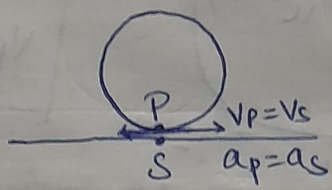
$$KE = \text{Translational KE} + \text{Rotational KE}$$

$$KE = \frac{1}{2} m V_{com}^2 + \frac{1}{2} I_{com} \omega^2$$

Pure Rolling / Rolling without slipping:

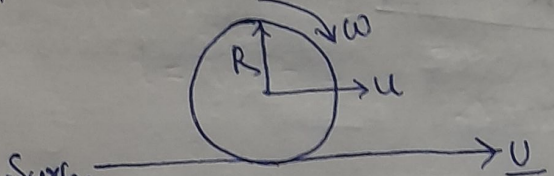
* There should be no relative motion b/w point of contact P and surface S, along surface ie; $V_P = V_S$ and $a_P = a_S$ along surface.

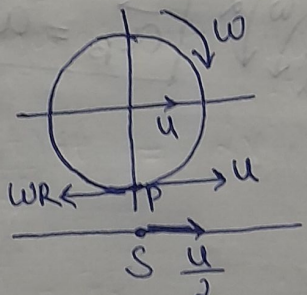




P = Point of contact
S = Point of surface in contact

Q





Find ω for pure rolling.

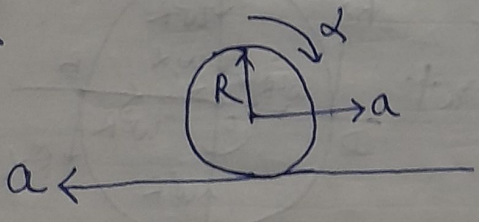
For Pure rolling:

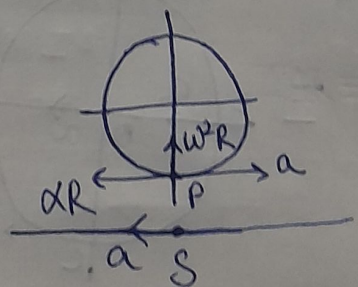
$V_P = V_S$

$u - \omega R = \frac{u}{2}$

$$\omega = \frac{u}{2R}$$

Q





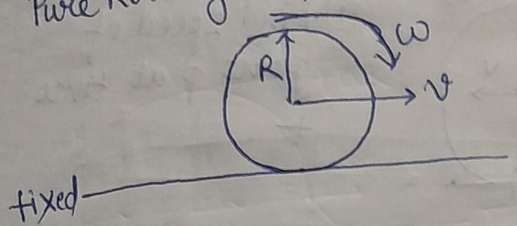
Find α for pure rolling

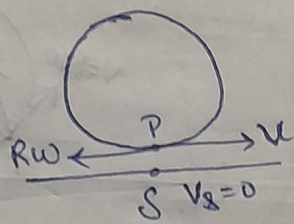
For pure rolling $\Rightarrow a_P = a_S$ along surface

$\alpha R - a = a$

$$\alpha = \frac{2a}{R}$$

Pure Rolling on fixed Surface:





For pure rolling:

$v_p = v_s$

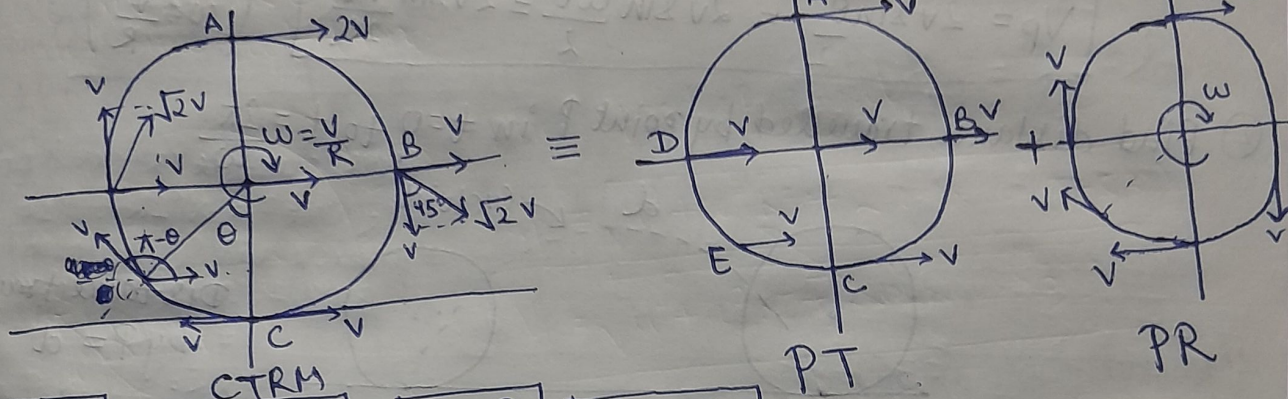
$v - R\omega = 0$

$\star \boxed{v = R\omega}$

$\star \boxed{\omega = \frac{v}{R}}$

Similarly, $\star \boxed{a = R\alpha}$ $\star \boxed{\alpha = \frac{a}{R}}$

Ex Find speed of given points A, B, C, D, E. Given pure rolling.



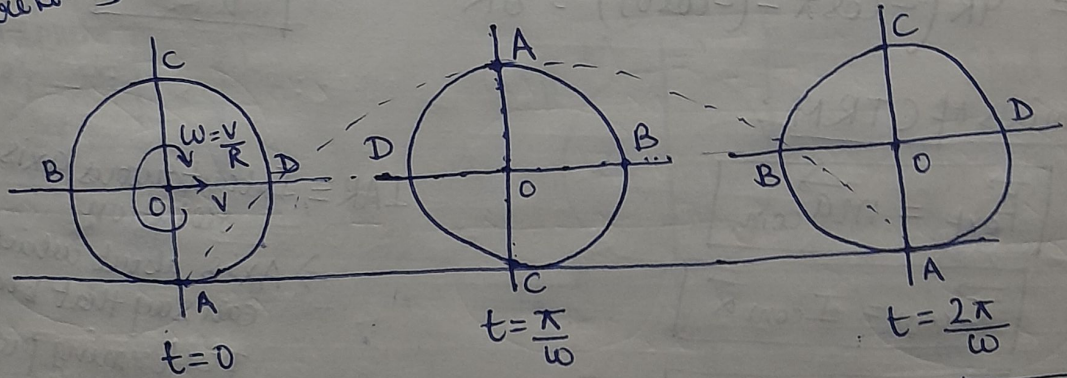
$\boxed{v_A = 2v}$ $\boxed{v_B = \sqrt{2}v}$ $\boxed{v_C = 0}$ $\boxed{v_D = \sqrt{2}v}$

$$v_E = \sqrt{v^2 + v^2 + 2v^2\cos(\pi-\theta)} = \sqrt{2v^2 + 2v^2(-\cos\theta)} = \sqrt{2v^2(1-\cos\theta)}$$

$$= \sqrt{2v^2\left(2\sin^2\frac{\theta}{2}\right)} = \sqrt{4v^2\sin^2\frac{\theta}{2}} = 2v\sin\frac{\theta}{2}$$

$\boxed{v_E = 2v\sin\frac{\theta}{2}}$

Note: Given Pure Rolling



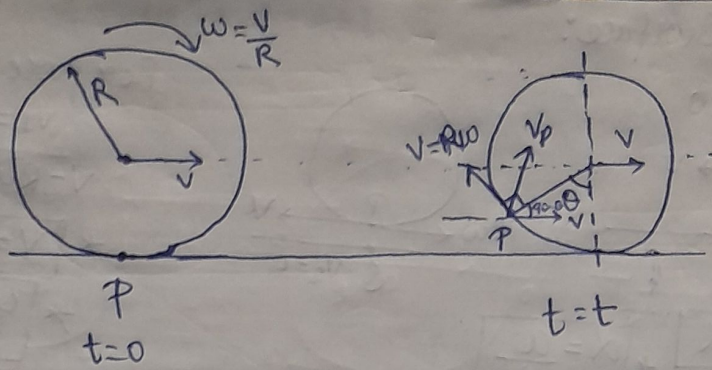
Displacement of O = $S_0 = vt = v \times \frac{2\pi}{\omega} = \omega R \times \frac{2\pi}{\omega} = 2\pi R$ $\boxed{S_0 = 2\pi R}$

Also, $\boxed{S_A = S_B = S_C = S_D = S_0 = 2\pi R}$

Distance travelled by O = $\boxed{d_0 = 2\pi R}$

But we can say that d_A, d_B, d_C, d_D may not be equal to $2\pi R$

Q

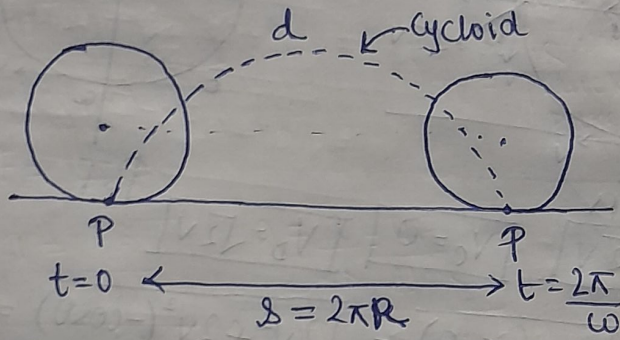


① Find speed of point P at time t.

$$V_p = \sqrt{V^2+V^2+2V^2\cos(\pi-\theta)} = \sqrt{2V^2-2V^2\cos\theta} = \sqrt{2V^2(1-\cos\theta)} = \sqrt{4V^2\sin^2\frac{\theta}{2}}$$

$$\boxed{V_p = 2V\sin\frac{\theta}{2} = 2V\sin\frac{\omega t}{2} = 2V\sin\frac{Vt}{2R}} \quad \left(\theta = \omega t,\ \omega = \frac{V}{R}\right)$$

② Find distance travelled by point P in $t=0$ to $t=\frac{2\pi}{\omega}$



Distance travelled by P = d

Since velocity of point P changes at every instance it moves,

$$d = \int_0^{2\pi/\omega} V_p\,dt = \int_0^{2\pi/\omega} 2V\sin\left(\frac{Vt}{2R}\right)dt = 2V\int_0^{2\pi/\omega}\sin\left(\frac{Vt}{2R}\right)dt = \frac{2V(2R)}{V}\left(-\cos\left(\frac{Vt}{2R}\right)\right)_0^{2\pi/\omega}$$

$$= 4R\left(-\cos\pi-(-\cos 0)\right) = 8R \qquad \therefore \boxed{d_p = 8R}$$

Dynamics of CTRM:

$$\boxed{\vec{F}_{ext} = m\vec{a}_{com}}$$

$$\boxed{\vec{\tau}_{com} = I_{com}\vec{\alpha}}$$

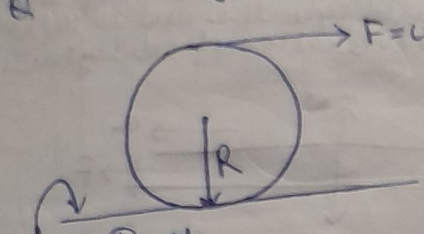
$$\boxed{\vec{\tau}_{IAR} = I_{IAR}\vec{\alpha}}$$

$$\boxed{KE = TKE + RKE = \frac{1}{2}mV_{com}^2 + \frac{1}{2}I_{com}\omega^2}$$

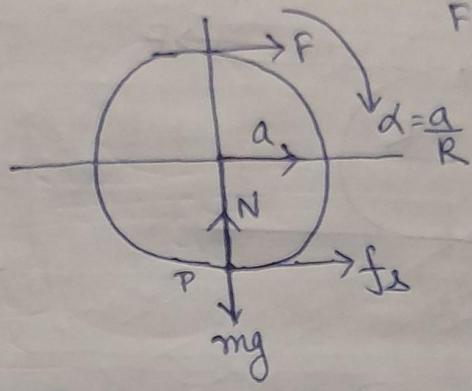
IAR ≡ Instantaneous axis of rotation.

⇒ Axis about which we can say that body is performing pure rotational motion.

Q Solid cylinder (m, R)



enough Rough
↳ μ↑ ⇒ No slipping
⇒ Pure Rolling
⇒ f_s ✓ ; f_k ✗



Find a_{com}, α of cylinder. Also find f_s on it

τ towards = +ve
τ opposite α = -ve

$\vec{F}_{ext} = m\vec{a}_{com}$

$\Rightarrow F + f_s = ma$ — ①

$\tau_{com} = I_{com}\alpha$

$FR - f_s R = \frac{mR^2}{2}\left(\frac{a}{R}\right)$

$F - f_s = \frac{ma}{2}$ — ②

From ① and ②

$F + F - \frac{ma}{2} = ma$

$2F = \frac{3ma}{2}$

$\boxed{a = \frac{4F}{3m}}$

$\boxed{\alpha = \frac{a}{R} = \frac{4F}{3mR}}$

$F + f_s = m\left(\frac{4F}{3m}\right)$

$\boxed{f_s = \frac{F}{3}}$

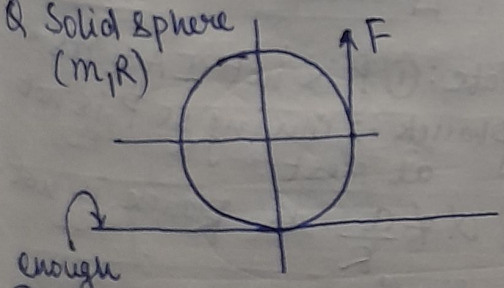
Alternate: Here we can say that P can act as IAR as its velocity is zero for an instant.

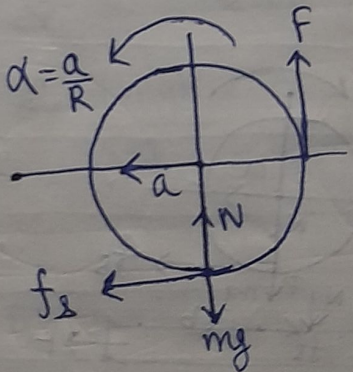
$\vec{\tau}_{IAR} = I_{IAR}\vec{\alpha} \Rightarrow F(2R) = \frac{3}{2}mR^2\left(\frac{a}{R}\right)$

$\boxed{a = \frac{4F}{3m}}$

* When body is performing pure rolling on fixed surface, then lowermost point (P) acts as IAR.

Q Solid sphere (m, R)





Find a_{com}, α. Also find f_s ($F < mg$)

$F_{ext} = ma_{com}$

$\boxed{f_s = ma}$ — ①

$\vec{\tau}_{com} = I_{com}\vec{\alpha}$

$= FR - f_s R = \frac{2}{5}mR^2\left(\frac{a}{R}\right)$

$= F - f_s = \frac{2}{5}ma$ — ②

Add ① and ②

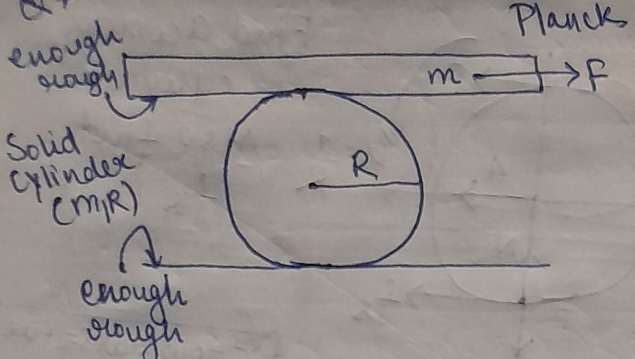
$F = \frac{7}{5}ma$

$\boxed{a = \frac{5F}{7m}}$

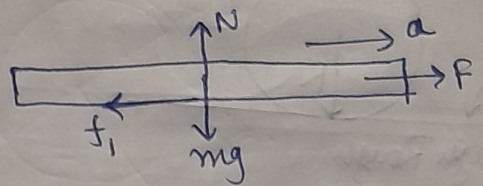
$\boxed{\alpha = \frac{5F}{7mR}}$

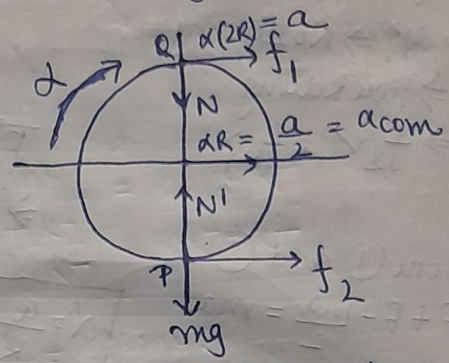
$\boxed{f_s = \frac{5F}{7}}$

Q*



Find acc^n of planck, acc^n of COM of solid cylinder and angular acceleration of solid cylinder





Here, surface is fixed so a_Q wrt $P = \alpha(2R)$. Also planck and SC are also in pure rolling so $a_{planck} = a_Q$

$a = 2\alpha R$

$\alpha R = \frac{a}{2} \Rightarrow a_{com}$

$F - f_1 = ma$ — ①

$f_1 + f_2 = \frac{ma}{2}$ — ②

$f_1 R - f_2 R = \frac{mR^2}{2}\left(\frac{a}{2R}\right)$

$f_1 - f_2 = \frac{ma}{4}$ — ③

Add ② and ③

$\boxed{f_1 = \frac{3ma}{8}}$

Put f_1 in ①

$F = \frac{11ma}{8}$

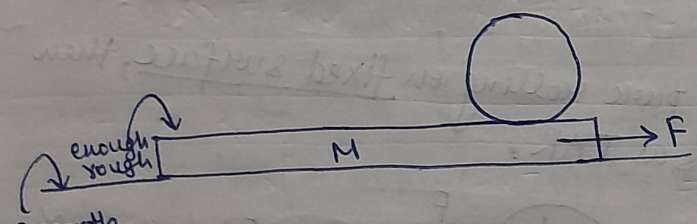
$\boxed{a = \frac{8F}{11m}}$

$\boxed{a_{com} = \frac{a}{2} = \frac{4F}{11m}}$

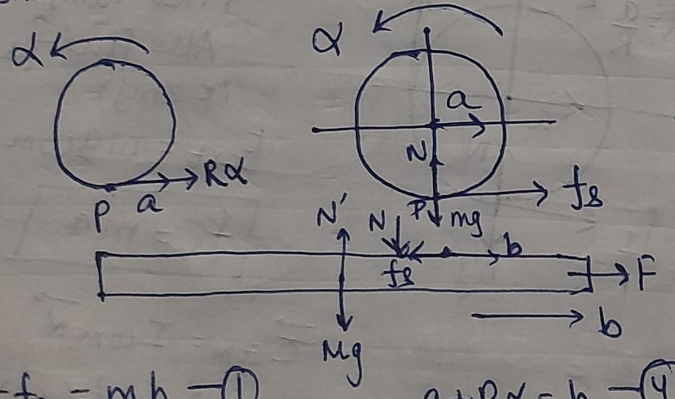
$\boxed{\alpha = \frac{a}{2R} = \frac{4F}{11mR}}$

Q*

Hollow sphere (m,R)



Find acc^n of planck, acc^n of COM of H.S., angular acc^n of H.S. and frictional force acting b/w shell and planck.



Note: ① P is not IAR as planck is moving so P is not at rest.

② $\alpha \neq \frac{a}{R}$ as planck is not fixed.

③ a_{com} of H.S $= a \neq b$ (acc^n of planck)

As in pure rolling only concept is $a_P = a_{planck}$

$F - f_s = mb$ — ①

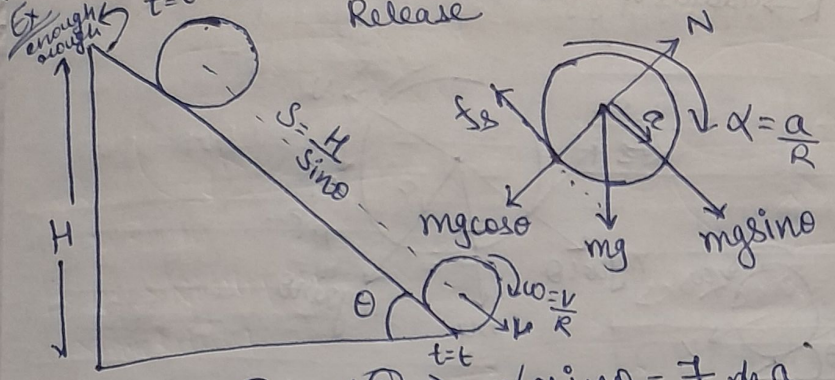
$f_s R = \frac{2}{3}mR^2\alpha$ — ②

$f_s = ma$ — ③

$a + R\alpha = b$ — ④

We can solve and get values respectively.

Solid Sphere (m, R)
Release



① Find acc^n of com of ball, α of ball and f_s acting on ball

$mg\sin\theta - f_s = ma$ — ①

$f_s R = \frac{2}{5}mR^2\left(\frac{a}{R}\right)$

$f_s = \frac{2}{5}ma$ — ②

Adding ① and ② ⇒ $mg\sin\theta = \frac{7}{5}ma$

$$\boxed{a = \frac{5g\sin\theta}{7}} \qquad \boxed{\alpha = \frac{5g\sin\theta}{7R}} \qquad \boxed{f_s = \frac{2mg\sin\theta}{7}}$$

② Find time taken by ball to reach the ground.

$S = ut + \frac{1}{2}at^2 \Rightarrow \frac{H}{\sin\theta} = \frac{1}{2}\left(\frac{5g\sin\theta}{7}\right)t^2 \Rightarrow \boxed{t = \sqrt{\frac{14H}{5g\sin^2\theta}}}$

③ Find v and ω when ball reaches ground.

$v^2 = u^2 + 2as \Rightarrow v^2 = 2\left(\frac{5g\sin\theta}{7}\right)\left(\frac{H}{\sin\theta}\right) \Rightarrow \boxed{v = \sqrt{\frac{10gH}{7}}}$

$\omega = \frac{v}{R} \Rightarrow \boxed{\omega = \sqrt{\frac{10gH}{7R^2}}}$ (we can also use TMEC to find v and ω)

④ Find KE of ball when it reaches ground.

$KE = \frac{1}{2}mv_{com}^2 + \frac{1}{2}I_{com}\omega^2$

$= \frac{1}{2}m\left(\frac{10gH}{7}\right) + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\left(\frac{10gH}{7R^2}\right)$

$= \frac{5mgH}{7} + \frac{2mgH}{7} = \frac{7mgH}{7}$

$$\boxed{KE = mgH}$$

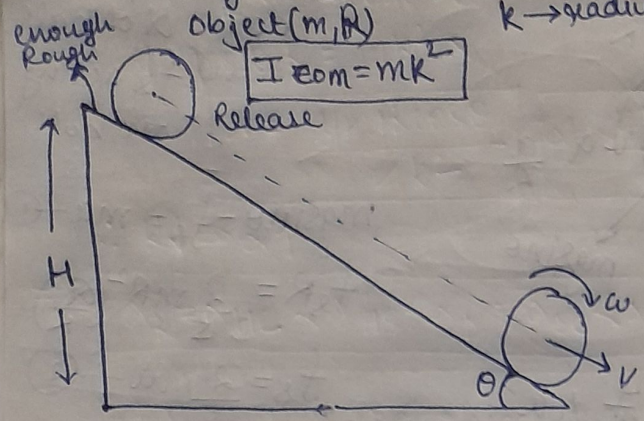
Alternate: TMEC ⇒ Loss in PE = Gain in KE

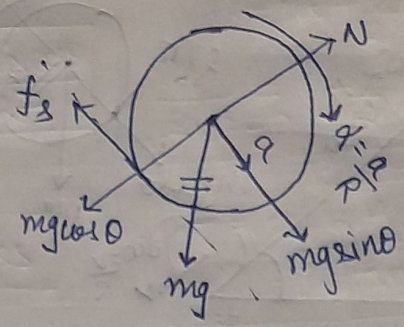
$$\boxed{KE = mgH}$$

Note: W.d by $f_s = 0$ as due to pure rolling velocity of point of contact along incline is 0 ⇒ Power of $f_s = 0 \Rightarrow w \cdot d = 0$
∴ No ext force ⇒ TMEC can be applied.

Pure Rolling on an Incline:

k → radius of gyration





$mg\sin\theta - f_s = ma$ — ①

$f_s R = mk^2\left(\frac{a}{R^2}\right)$ — ②

$f_s = mk^2\left(\frac{a}{R^2}\right)$ — ②

Add ① + ②

$mg\sin\theta = ma + \frac{mak^2}{R^2}$

$$a = \frac{g\sin\theta}{1+\frac{k^2}{R^2}} = \frac{g\sin\theta R^2}{R^2+k^2}$$

$$\alpha = \frac{a}{R} = \frac{g\sin\theta R}{R^2+k^2}$$

$$f_s = \frac{mg\sin\theta k^2}{R^2+k^2} = \frac{mg\sin\theta}{\frac{R^2}{k^2}+1}$$

using TMEC $\Rightarrow mgH = \frac{1}{2}mv^2 + \frac{1}{2}mk^2\left(\frac{v^2}{R^2}\right)$

$$gH = v^2\left(1+\frac{k^2}{R^2}\right)$$

$$V = \sqrt{\frac{2gH}{\left(1+\frac{k^2}{R^2}\right)}}$$

$$\omega = \frac{V}{R} = \sqrt{\frac{2gH}{R^2+k^2}}$$

★ Note: From the following results we observe:

① $K\downarrow \Rightarrow a\uparrow$ ② $K\downarrow \Rightarrow a\uparrow \Rightarrow v\uparrow$

③ $K\downarrow \Rightarrow V\uparrow \Rightarrow \omega\uparrow$ ④ $K\downarrow \Rightarrow a\uparrow \Rightarrow \alpha\uparrow$

⑤ $K\downarrow \Rightarrow f_s\downarrow$ ⑥ $K\uparrow \Rightarrow f_s\uparrow$

Object	Ring	Hollow Cylinder	Hollow Sphere	Disc	Solid Cylinder	Solid Sphere
$I_{com} \Rightarrow$	mR^2	mR^2	$\frac{2}{3}mR^2$	$\frac{mR^2}{2}$	$\frac{mR^2}{2}$	$\frac{2}{5}mR^2$
$K \Rightarrow$	R	R	$\sqrt{\frac{2}{3}}R = 0.8R$	$\frac{R}{\sqrt{2}} = 0.707R$	$\frac{R}{\sqrt{2}} = 0.707R$	$\sqrt{\frac{2}{5}}R = 0.63R$
$a \Rightarrow$	$a_R =$	$a_{HC} <$	$a_{HS} <$	$a_D =$	$a_{SC} <$	a_{SS}
$\alpha \Rightarrow$	$\alpha_R =$	$\alpha_{HC} <$	$\alpha_{HS} <$	$\alpha_D =$	$\alpha_{SC} <$	α_{SS}
$V \Rightarrow$	$V_R =$	$V_{HC} <$	$V_{HS} <$	$V_D =$	$V_{SC} <$	V_{SS}
$\omega \Rightarrow$	$\omega_R =$	$\omega_{HC} <$	$\omega_{HS} <$	$\omega_D =$	$\omega_{SC} <$	ω_{SS}
$f \Rightarrow$	$f_R =$	$f_{HC} >$	$f_{HS} >$	$f_D =$	$f_{SC} >$	f_{SS}

★ Solid sphere will reach the ground earliest (as $a\uparrow$) and ring and hollow cylinder will reach the ground at last.

★ KE of all the objects will be same as loss in PE is same for all when released from same pt. (Mechanical energy is conserved)

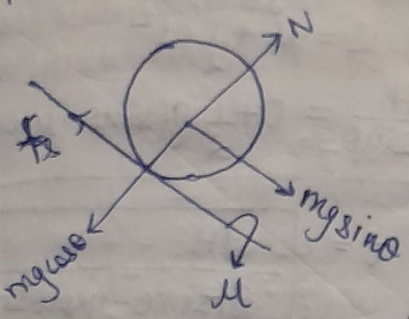
Coeffecient of friction required in this arrangement for rolling:

We know $f_s \le f_{lim} \Rightarrow f_s \le \mu N$

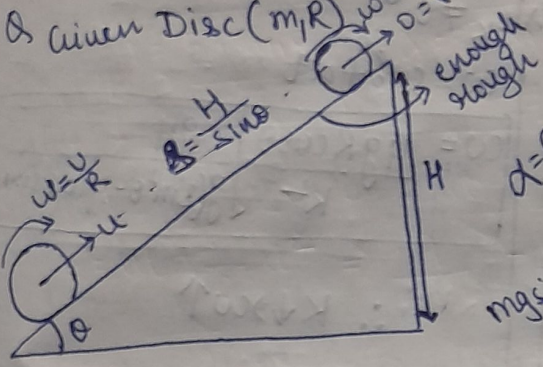
$$= \frac{mg\sin\theta\, K^2}{R^2+K^2} \le \mu mg\cos\theta$$

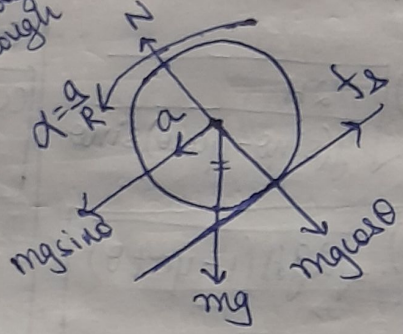
$$\boxed{\mu \ge \frac{\tan\theta\, K^2}{R^2+K^2}} \qquad \boxed{\mu_{min} = \frac{\tan\theta}{\left(\frac{R^2}{K^2}+1\right)}}$$

$$\boxed{① K\uparrow \Rightarrow \mu\uparrow \quad ② K\downarrow \mu\downarrow}$$



Q Given Disc(m,R) $\omega=0$, $v=V$





① Find a_{com} of disc and α of disc

② Find max^m height achieved by disc

① $mg\sin\theta - f_s = ma$ — ①

$+ f_s R = \frac{mR^2}{2}\left(\frac{a}{R}\right)$ — ②

$mg\sin\theta = \frac{3}{2}ma$

$$\boxed{a = \frac{2g\sin\theta}{3}}$$

$$\boxed{\alpha = \frac{2g\sin\theta}{3R}}$$

② Method 1:

$v^2 = u^2 + 2as$

$0 = u^2 - 2\left(\frac{2g\sin\theta}{3}\right)\frac{H}{\sin\theta}$

$$\boxed{H = \frac{3u^2}{4g}}$$

Method 2: using TMEC

$K_i + U_i = K_f + U_f$

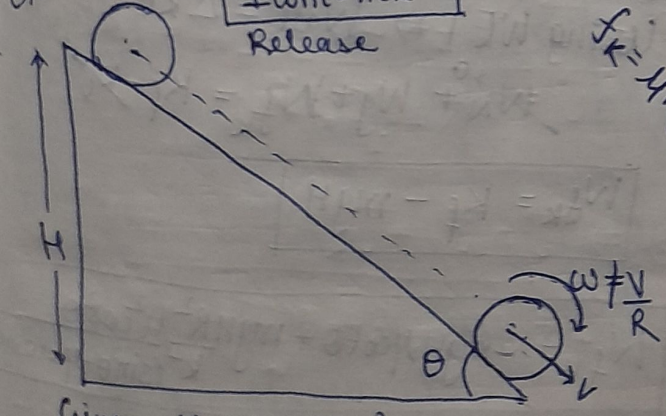
$\frac{1}{2}mu^2 + \frac{1}{2}\frac{mR^2}{2}\left(\frac{u^2}{R^2}\right) = mgH$

$$\boxed{H = \frac{3u^2}{4g}}$$

Ex ★★★ Object(m,R)

$\boxed{I_{com} = mK^2}$

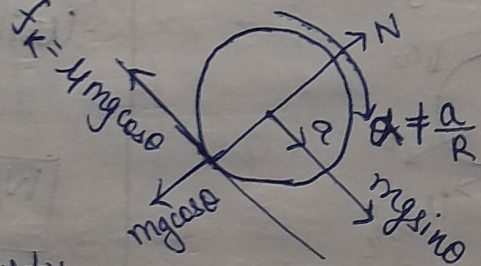
Release



Given $\mu < \frac{\tan\theta\, K^2}{R^2+K^2}$

⇒ Surface is not sufficiently rough

⇒ No pure rolling ⇒ Slipping ⇒ f_K ✓



① Find a_{com} and α of object

$mg\sin\theta - \mu mg\cos\theta = ma$

$$\boxed{a = g(\sin\theta - \mu\cos\theta)}$$

★ a is independent of K, this means all objects will have same acc^n ⇒ same velocity. Also, all objects reach ground in same time

$\frac{2gH}{R^2+K^2}$

Solid Sphere

$I = mR^2$

$K = 0.63R$

a_{SS}

x_{SS}

V_{SS}

ω_{SS}

f_{SS}

Now, $f_k R = I_{com}\alpha$

$\mu mg\cos\theta R = mk^2\alpha$

$\alpha = \frac{\mu mgR\cos\theta}{mk^2}$ $\quad$ $\alpha = \frac{\mu gR\cos\theta}{k^2}$

$\therefore K\downarrow \Rightarrow \alpha\uparrow$

$\Rightarrow \alpha_R = \alpha_{HC} < \alpha_{HS} < \alpha_D = \alpha_{SC} < \alpha_{SS}$

② Find time taken by object to reach ground.

$s = ut + \frac{1}{2}at^2$

$\frac{H}{\sin\theta} = 0 + \frac{1}{2}g(\sin\theta - \mu\cos\theta)t^2$

$t = \sqrt{\frac{2H}{g(\sin^2\theta - \mu\cos\theta\sin\theta)}}$

③ Find v_{com} and ω of object when it reaches ground.

Note: TMEC is not applicable here as $W_{f_k} \neq 0$

$v = u + at$

$v = 0 + g(\sin\theta - \mu\cos\theta)\sqrt{\frac{2H}{g(\sin\theta - \mu\cos\theta)\sin\theta}}$

$v = \sqrt{\frac{2gH(\sin\theta - \mu\cos\theta)}{\sin\theta}} = \sqrt{2gH(1 - \mu\cot\theta)}$

$\omega = \omega_0^{\,0} + \alpha t$

$\omega = \frac{\mu gR\cos\theta}{k^2}\sqrt{\frac{2H}{g(\sin^2\theta - \mu\cos\theta\sin\theta)}}$

$\therefore K\downarrow \Rightarrow \omega\uparrow$

④ Find KE of object when it reaches the ground.

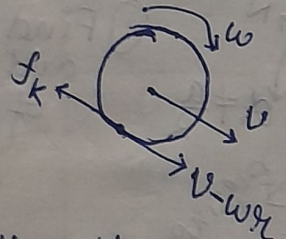
$KE = \frac{1}{2}mv^2 + \frac{1}{2}I_{com}\omega^2 = \frac{1}{2}m\left(2gH(1 - \mu\cot\theta)\right) + \frac{1}{2}mk^2\left(\frac{u^2g^2R^2\cos^2\theta(2H)}{k^4g^2(\sin^2\theta - \mu\cos\theta\sin\theta)}\right)$

$KE = mgH(1 - \mu\cot\theta) + mgH\frac{R^2}{k^2}\frac{u^2\cos^2\theta}{(\sin^2\theta - \mu\cos\theta\sin\theta)}$ $\quad (K\downarrow \Rightarrow KE\uparrow)$

⑤ Find work done by kinetic friction.

Method 1: Tough

At any time t



After time dt, $s = (v - \omega R)dt$

$W = \vec{F}\cdot\vec{s} = f_k(v - \omega R)dt\cos 180^\circ$

$W = \int_0^t -f_k(v - \omega R)dt$

write, v, ω & f_k as $f(t)$.

★ Method 2:

Using WET $\Rightarrow W_{all} = \Delta KE$

$W_N^{\,0} + W_g + W_{f_k} = K_f - K_i^{\,0}$

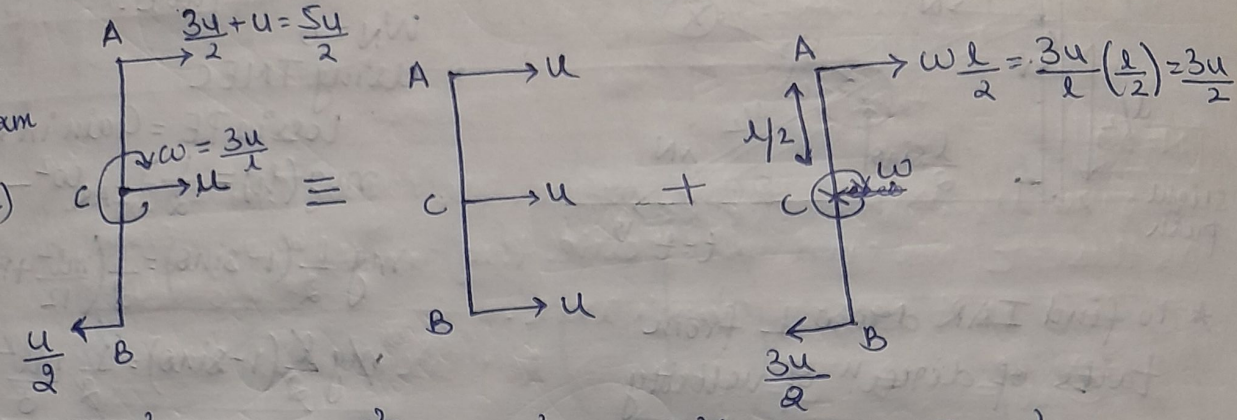
$W_{f_k} = K_f - mgH$

$W_{f_k} = -mgH\mu\cot\theta + mgH\frac{R^2}{k^2}\frac{u^2\cos^2\theta}{(\sin^2\theta - \mu\cos\theta\sin\theta)}$

Instantaneous Axis of Rotation (IAR):

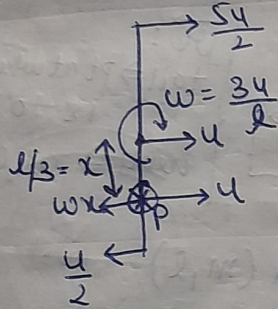
→ It is a axis about which an object performing CTRM, can be assumed to be performing pure rotational motion.

Ex uniform rod (m, l)



$$KE = \frac{1}{2} m v_{com}^2 + \frac{1}{2} I_{com} \omega^2 = \frac{1}{2} mu^2 + \frac{1}{2}\left(\frac{ml^2}{12}\right)\left(\frac{9u^2}{l}\right) = \frac{7mu^2}{8}$$

Now let us assume a point P on rod at x distance lower ~~from~~ C where velocity is zero. ⇒ P can act as IAR.



For zero velocity

$\omega x = u$

$\frac{3u}{l} x = u$

$$x = \frac{l}{3}$$

Now $KE = \frac{1}{2} I_{IAR} \omega^2$

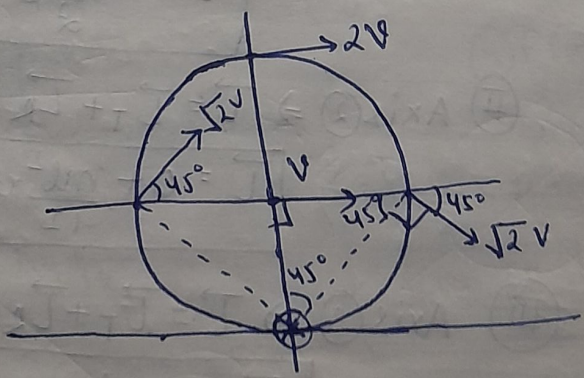
$$KE = \frac{1}{2}\left(\frac{ml^2}{12} + \frac{ml^2}{9}\right)\left(\frac{9u^2}{l}\right)$$

$$KE = \frac{7mu^2}{8}$$

From P we can say that Body is performing Pure rotation.

∴ P is IAR

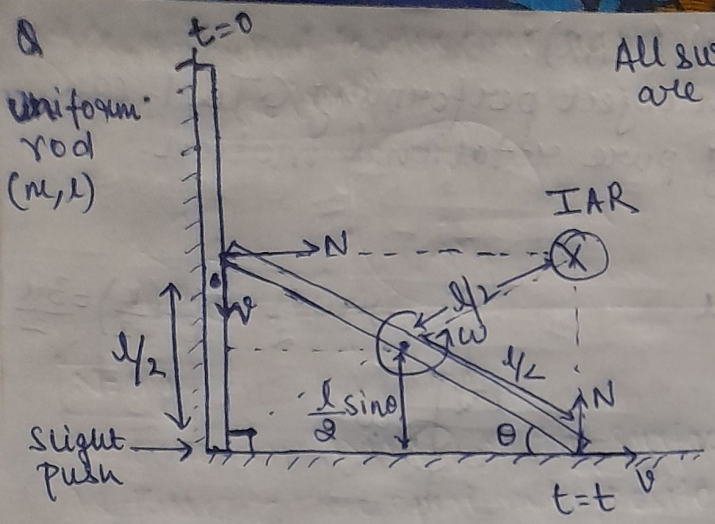
Ex Pure Rolling (fixed surface)



★ Here, IAR is intersection of perpendiculars drawn from direction of tail of velocity.

~~(For circular objects)~~

Q



All surfaces are smooth. Find angular velocity of rod when rod is making angle θ with the floor.

$W_N = 0$

⇒ Using TMEC

Loss in PE = Gain in KE

$mg(h) = \frac{1}{2} I_{IAR} \omega^2 - 0$

$mg\frac{l}{2}(1-\sin\theta) = \frac{1}{2}\left(\frac{ml^2}{12} + m\left(\frac{l}{2}\right)^2\right)\omega^2$

$mg\frac{l}{2}(1-\sin\theta) = \frac{1}{2}\frac{ml^2}{3}\omega^2$

$$\omega = \sqrt{\frac{3g(1-\sin\theta)}{l}}$$

★ To find IAR draw ⊥ from tails of direcⁿ of velocity

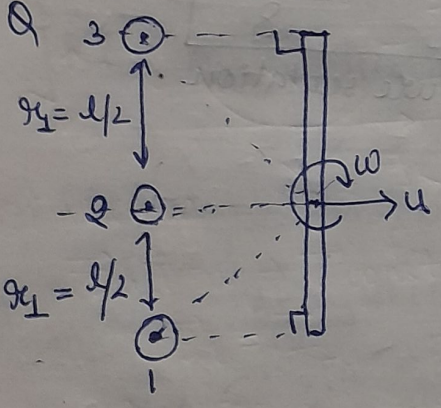
Angular Momentum of a body performing CTRM:

$$\vec{L} = \vec{L}_T + \vec{L}_{Rot}$$

$$\vec{L} = m(\vec{r} \times \vec{V}_{com}) + I_{com}\vec{\omega}$$

$\vec{r}$:
tail ⇒ pt at which $\vec{L}$ is to be found
head ⇒ at COM

Q



Given uniform rod (m, l)
Find $\vec{L}$ about all three axis.

(I) Axis ① ⇒ $\vec{L} = \vec{L}_T + \vec{L}_R$

$$\vec{L} = mu\frac{l}{2}\otimes + \frac{ml^2}{12}\omega\otimes$$

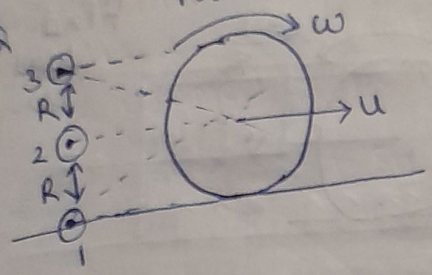
(II) Axis ② ⇒ $\vec{L} = \vec{L}_T + \vec{L}_R$

$$\vec{L} = 0 + \frac{ml^2}{12}\omega\otimes$$

(III) Axis ③ ⇒ $\vec{L} = \vec{L}_T + \vec{L}_R$

$$\vec{L} = mu\frac{l}{2}\odot + \frac{ml^2}{12}\omega\otimes$$

Hollow sphere



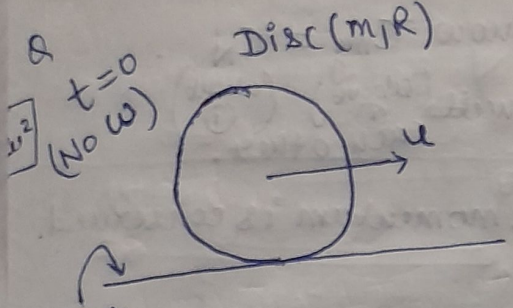
Find $\vec{L}$ of H.S. about all three axis

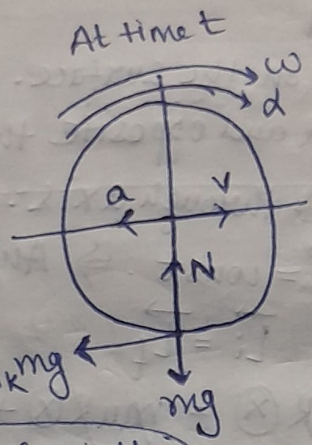
Axis ① $\Rightarrow \vec{L} = muR\otimes + \frac{2}{3}mR^2\omega\otimes$

Axis ② $\Rightarrow \vec{L} = 0 + \frac{2}{3}mR^2\omega\otimes$

Axis ③ $\Rightarrow \vec{L} = muR\odot + \frac{2}{3}mR^2\omega\otimes$

Q





Disc was moving with velocity u initially. After time t we can say that velocity be v. Due to τ of f_k there will be an α generated. Initially $\omega = 0$. But as time passes by $\omega\uparrow$, $V\downarrow$. when $V = \omega R$ we can say that body will start pure rolling. f_k will stop acting at this time.

① Find a_{com} and α

$f_k = ma$

$\mu_k mg = ma$

$\boxed{a = \mu_k g}$

$\vec{\tau}_{com} = I_{com}\vec{\alpha}$

$\mu_k mgR = \frac{mR^2}{2}\vec{\alpha}$

$\boxed{\alpha = \frac{2\mu_k g}{R}}$

② Find v_{com} and ω of disc at any time t.

$\boxed{V_{com} = u - \mu_k g t}$

$\boxed{\omega = \frac{2\mu_k g t}{R}}$ $\{\omega = \omega_0 + \alpha t\}$

③ Find the time when pure rolling starts.

For pure rolling, $V = \omega R$

$u - \mu_k g t = \frac{2\mu_k g t}{R}(R)$

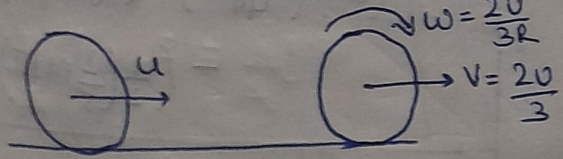
$\boxed{t = \frac{u}{3\mu_k g}}$

④ Find v_{com} and ω of disc when it starts pure rolling

$V = U - \mu_k g\left(\frac{u}{3\mu_k g}\right) \Rightarrow \boxed{V = \frac{2u}{3}}$

$\boxed{\omega = \frac{2u}{3R}}$

⑤ Find Work done by f_k.



After this pure rolling starts so Wd by $f_k = 0$

using WET $\Rightarrow W_{all} = \Delta KE$

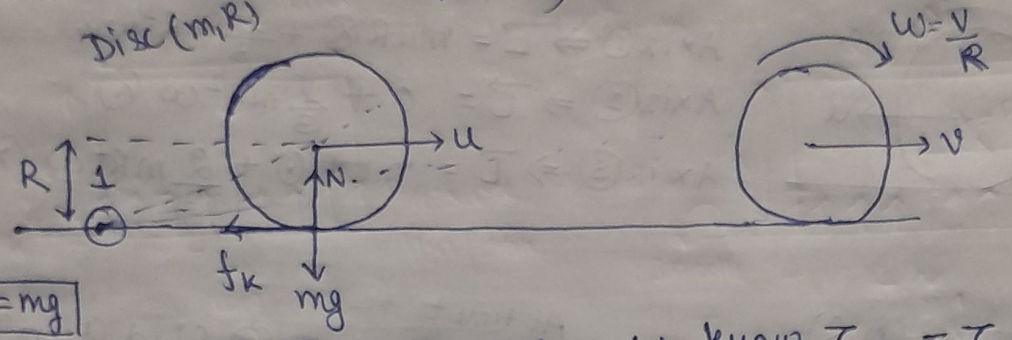
$W_N^{\,0} + W_{mg}^{\,0} + W_f = K_f - K_i$

$W_f = \left\{\frac{1}{2}m\left(\frac{4u^2}{9}\right) + \frac{1}{2}\left(\frac{mR^2}{2}\right)\left(\frac{4u^2}{9R^2}\right)\right\} - \left(\frac{1}{2}mu^2\right)$

$\boxed{W_f = -\frac{mu^2}{6}}$

Q* (Better method to solve prev question)

Disc (m, R)



Find v and ω.

$N = mg$

We take an axis ① on the surface. We know $\tau_{mg} = \tau_N$. As these τ are equal and opposite they cancell each other. (about ①) (about ①)

$\tau_{f_k} = 0$ as it passes through axis.

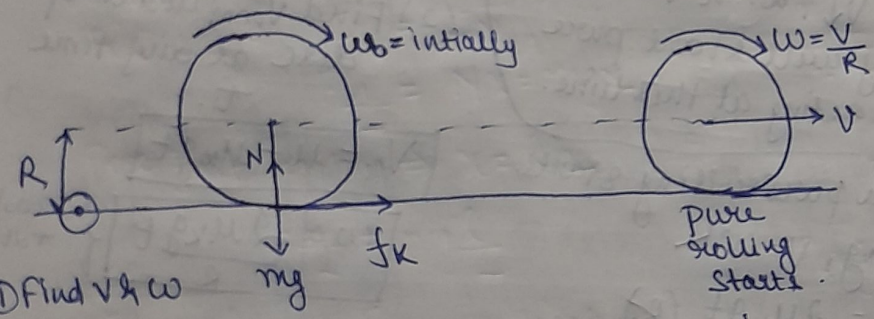
$\therefore \vec{\tau}_{net} = 0 \Rightarrow \vec{L} = \text{const} \Rightarrow$ Angular momentum is conserved.

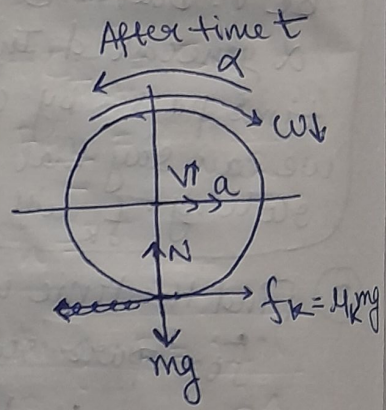
$$\vec{L_i} = \vec{L_f}$$

$$mvR\,\otimes = mvR\,\otimes + \frac{mR^2}{2}\left(\frac{v}{R}\right)\otimes$$

$$v = \frac{2u}{3} \qquad \omega = \frac{2v}{3R}$$

Q Ball (Solid sphere)





① Find v & ω

$\vec{\tau}_{net} = 0 \Rightarrow \vec{L}$ const

$$\vec{L_i} = \vec{L_f}$$

$$= \frac{2}{5}mR^2\omega_0\,\otimes = mvR\,\otimes + \frac{2}{5}mR^2\left(\frac{v}{R}\right)\otimes$$

$$= \frac{2}{5}\omega_0 R = \frac{7}{5}v$$

$$v = \frac{2\omega_0 R}{7} \qquad \omega = \frac{v}{R} = \frac{2\omega_0}{7}$$

② Find W.d by f_k.

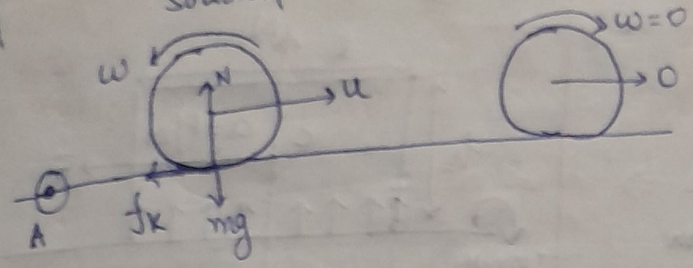
using WET $\Rightarrow W_{all} = \Delta KE$

$$W_g + W_N + W_{f_k} = K_f - K_i$$

$$W_{f_k} = \left(\frac{1}{2}m\left(\frac{4\omega_0^2R^2}{49}\right) + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\left(\frac{4\omega_0^2}{49}\right)\right) - \left(\frac{1}{2}\left(\frac{2}{5}mR^2\right)\omega_0^2\right)$$

$$W_{f_k} = \frac{14m\omega_0^2R^2}{49\times 5} - \frac{1}{5}m\omega_0^2R^2$$

Solid sphere



Find ω such that ball finally stops

About A $\vec{\tau}_{ext}=0 \Rightarrow \vec{L}=$ const

$\vec{L_i}=\vec{L_f}$

$= muR\otimes + \frac{2}{5}mR^2\omega\odot = 0+0$

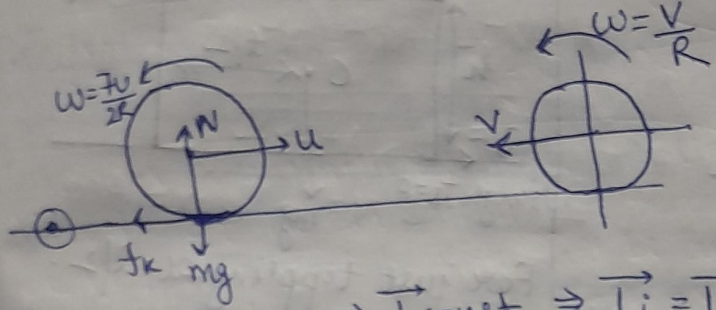
$\Rightarrow \vec{L_i}=0$

$\Rightarrow \cancel{m}uR = \frac{2}{5}\cancel{m}R^2\omega$

$$\boxed{\omega = \frac{5u}{2R}}$$

*If $\omega > \frac{5u}{2R}$, ball will roll back

Q



Find V and ω

⊙ → (+)

⊗ → (−)

About A $\vec{\tau}_{ext}=0 \Rightarrow \vec{\tau}$ const $\Rightarrow \vec{L_i}=\vec{L_f}$

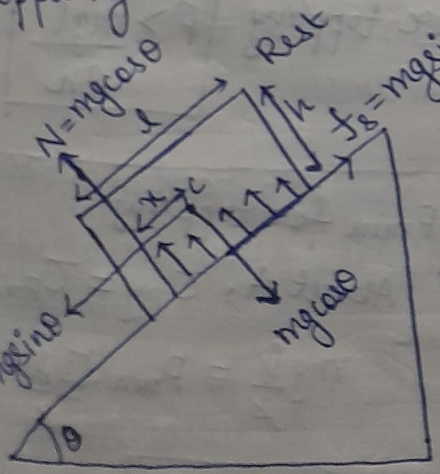
$$-muR + \frac{2}{5}mR^2\left(\frac{7u}{2R}\right) = mvR + \frac{2}{5}mR^2\left(\frac{V}{R}\right)$$

$$= \frac{2}{\cancel{5}}\cancel{m}uR = \frac{7}{\cancel{5}}\cancel{m}vR$$

$$\boxed{V=\frac{2v}{7}} \qquad \boxed{\omega = \frac{V}{R} = \frac{2u}{7R}}$$

Toppling:

Ex



Block is at rest. $\vec{\tau}_{ext}=0$

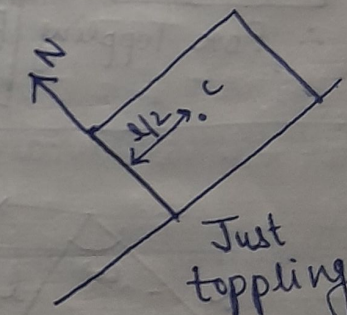
We have $\tau_f = mg\sin\theta\,\frac{h}{2}$

To balance this Normal will shift towards left.

As $\theta\uparrow \Rightarrow \tau_f\uparrow \Rightarrow \tau_N\uparrow$

For equilibrium

$\tau_{f_s} = \tau_N$



∴ For just toppling

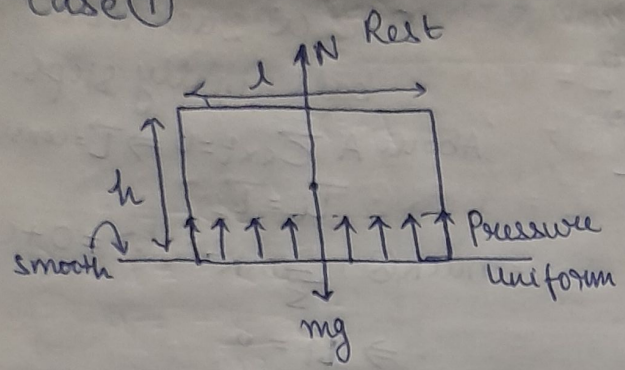
Normal passes through edge → $x = l/2$

$\tau_f = \tau_N$

$mg\sin\theta\,\frac{h}{2} = mg\cos\theta\,\frac{l}{2}$

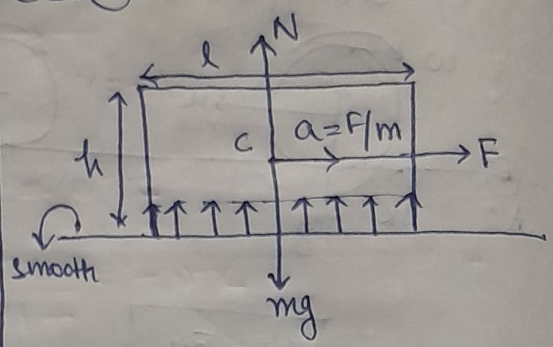
$$\boxed{\tan\theta = \frac{l}{h}}$$

Case ①



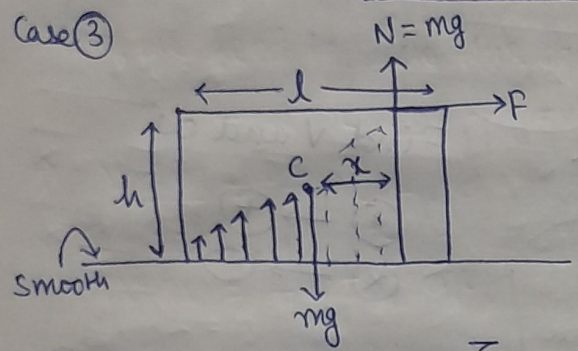
∴ No toppling in this case.
as $\vec{\tau}_{net}$ about C = 0

Case ②



∴ No toppling in this case.
as $\vec{\tau}_{net}$ about C = 0

Case ③



About C, we will have τ_F.
To balance this N will shift towards right.

For rest ⇒ $\tau_F = \tau_N$

$$F\frac{h}{2} = Nx = mgx$$

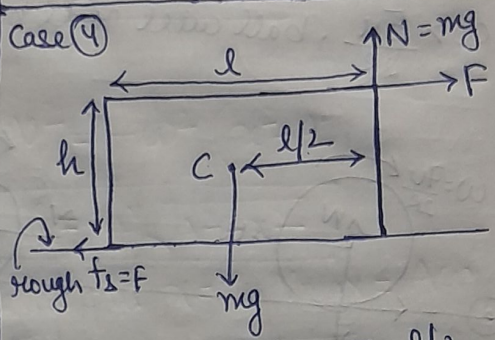
For just toppling ⇒ $x = l/2$

$$F\frac{h}{2} = mg\frac{l}{2}$$

$$\boxed{F = \frac{mgl}{h}} \Rightarrow F_{min} \text{ for toppling}$$

∴ For Toppling $\boxed{F > \frac{mgl}{h}}$

Case ④



For just toppling $x = l/2$.

$$\tau_F + \tau_{f_s} = \tau_N$$

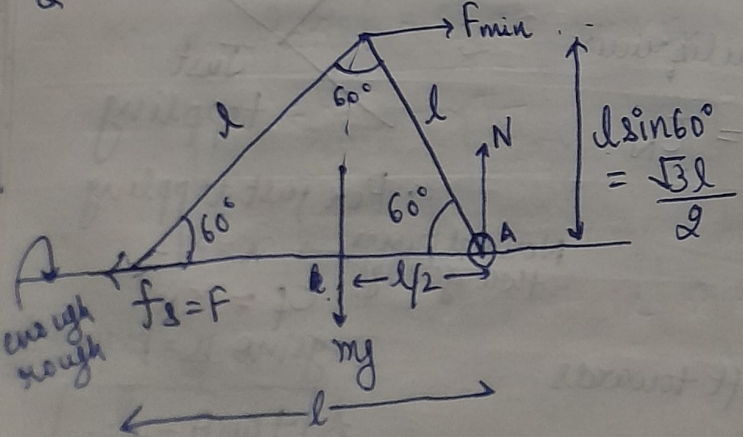
$$F\frac{h}{2} + F\frac{h}{2} = mg\frac{l}{2}$$

$$\Rightarrow Fh = mg\frac{l}{2}$$

∴ For just toppling $\boxed{F = \frac{mgl}{2h}}$

For toppling $\boxed{F > \frac{mgl}{2h}}$

Q



Find F_{min} for toppling

About A, $\tau_{mg} = \tau_{F_{min}}$

$$mg\frac{l}{2} = F_{min}\frac{\sqrt{3}l}{2}$$

$$\boxed{F_{min} = \frac{mg}{\sqrt{3}}}$$